

BEGINNING PYTHON 3

WITH CLAUDE 3

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BEGINNING PYTHON 3

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OSWALD CAMPESATO



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info@merclearning.com
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PREFACE

What Is the Value Proposition for This Book?

This book is a comprehensive guide designed to teach the fundamentals of Python programming while introducing the exciting possibilities of Generative AI. Whether you're a novice or a developer looking to integrate Claude 3 into your workflow, this book offers a clear, step-by-step path to mastering Python and leveraging AI-driven code generation.

We start with an introduction to some fundamental aspects of Python programming, which include various data types, number formatting, Unicode and UTF-8 handling, and text manipulation techniques. In addition, you will learn about loops, conditional logic, and reserved words in Python. You will also see how to handle user input, manage exceptions, and work with command-line arguments.

Next, the text transitions to the realm of Generative AI, discussing its distinction from Conversational AI. Popular platforms and models, including Claude 3, GPT-4, and their competitors, are presented to give readers an understanding of the current AI landscape. The book also sheds light on the capabilities of Claude 3, its strengths, weaknesses, and potential applications. In addition, you will learn how to generate a variety of Python 3 code samples via Claude 3 in chapter 6.

In essence, this book provides a modest bridge between the worlds of Python programming and AI, aiming to equip readers with the knowledge and skills to navigate both domains confidently.

The Target Audience

This book is intended primarily for people who want to learn both Python and how to use Claude 3 with Python. This book is also intended to reach an international audience of readers with highly diverse backgrounds in various age groups. In addition, this book uses standard English rather than colloquial

expressions that might be confusing to those readers. This book provides a comfortable and meaningful learning experience for the intended readers.

Do I Need to Learn the Theory Portions of This Book?

The answer depends on the extent to which you plan to become involved in working with Python and Claude 3, perhaps involving LLMs and generative AI. In general, it's probably worthwhile to learn the more theoretical aspects of LLMs that are discussed in this book.

Getting the Most from This Book

Some people learn well from prose, others learn well from sample code (and lots of it), which means that there's no single style that can be used for everyone.

Moreover, some programmers want to run the code first, see what it does, and then return to the code to delve into the details (and others use the opposite approach).

Consequently, there are various types of code samples in this book: some are short and some are long.

What Do I Need to Know for This Book?

Although this book is introductory in nature, some knowledge of Python 3.x with certainly be helpful for the code samples. Knowledge of other programming languages (such as Java) can also be helpful because of the exposure to programming concepts and constructs. The less technical knowledge that you have, the more diligence will be required in order to understand the various topics that are covered.

If you want to be sure that you can grasp the material in this book, glance through some of the code samples to get an idea of how much is familiar to you and how much is new for you.

Does This Book Contain Production-Level Code Samples?

This book contains basic Python code samples, and their primary purpose is to familiarize you with basic Python to help you understand the Python code generated via Claude 3. Moreover, clarity has higher priority than writing more compact code that is more difficult to understand (and possibly more prone to bugs). If you decide to use any of the code in this book, you ought to subject that code to the same rigorous analysis as the other parts of your code base.

What Are the Non-Technical Prerequisites for This Book?

Although the answer to this question is more difficult to quantify, it's very important to have a strong desire to learn about Generative AI, along with the motivation and discipline to read and understand the code samples. As a

reminder, even simple APIs can be a challenge to understand them the first time you encounter them, so be prepared to read the code samples several times.

How Do I Set Up a Command Shell?

If you are a Mac user, there are three ways to do so. The first method is to use Finder to navigate to Applications > Utilities and then double click on the Utilities application. Next, if you already have a command shell available, you can launch a new command shell by typing the following command:

```
open /Applications/Utilities/Terminal.app
```

A second method for Mac users is to open a new command shell on a MacBook from a command shell that is already visible simply by clicking `command+n` in that command shell, and your Mac will launch another command shell.

If you are a PC user, you can install Cygwin (open source <https://cygwin.com/>) that simulates bash commands or use another toolkit such as MKS (a commercial product). Please read the online documentation that describes the download and installation process. Note that custom aliases are not automatically set if they are defined in a file other than the main start-up file (such as `.bash_login`).

Companion Files

All the code samples and figures in this book may be obtained by writing to the publisher at info@merclearning.com.

What Are the “Next Steps” After Finishing This Book?

The answer to this question varies widely, mainly because the answer depends heavily on your objectives. If you are interested primarily in NLP, then you can learn about other LLMs (large language models).

If you are primarily interested in machine learning, there are some sub-fields of machine learning, such as deep learning and reinforcement learning (and deep reinforcement learning) that might appeal to you. Fortunately, there are many resources available, and you can perform an Internet search for those resources. One other point: the aspects of machine learning for you to learn depend on who you are: the needs of a machine learning engineer, data scientist, manager, student, or software developer are all different.

O. Campesato
December 2024

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INTRODUCTION TO PYTHON

This chapter contains an introduction to Python, with information about useful tools for installing Python modules, basic Python constructs, and how to work with some data types in Python.

The first part of this chapter covers how to install Python, some Python environment variables, and how to use the Python interpreter. Readers will see Python code samples and also how to save Python code in text files that they can launch from the command line. The second part of this chapter shows how to work with simple data types, such as numbers, fractions, and strings. The final part of this chapter discusses exceptions and how to use them in Python scripts.

Note: The Python files in this book are for Python 3.x.

TOOLS FOR PYTHON

The Anaconda Python distribution available for Windows, Linux, and Mac, and it's downloadable here: <http://continuum.io/downloads>

Anaconda is well-suited for modules such as `numpy` and `scipy`, and for Windows users, Anaconda appears to be a better alternative.

easy_install and pip

Both `easy_install` and `pip` are very easy to use when one needs to install Python modules. Whenever one needs to install a Python module (and there are many in this book), use either `easy_install` or `pip` with the following syntax:

```
easy_install <module-name>
pip install <module-name>
```

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Note: Python-based modules are easier to install, whereas modules with code written in C are usually faster but more difficult in terms of installation.

virtualenv

The `virtualenv` tool enables users to create isolated Python environments, and its home page is here: <http://www.virtualenv.org/en/latest/virtualenv.html>

`virtualenv` addresses the problem of preserving the correct dependencies and versions (and indirectly permissions) for different applications. Python novices might not need `virtualenv` right now, but keep this tool in mind.

IPython

Another very good tool is `IPython` (which won a Jolt award), and its home page is here:

<http://ipython.org/install.html>

Two very nice features of `IPython` are tab expansion and “?”, and an example of tab expansion is shown here:

```
python
Python 3.12.5 (v3.12.5:ff3bc82f7c9, Aug 7 2024, 05:32:06)
[Clang 13.0.0 (clang-1300.0.29.30)] on darwin
Type "help", "copyright", "credits" or "license" for more
information.
```

```
IPython 0.13.2 -- An enhanced Interactive Python.
?          -> Introduction and overview of IPython's
features.
%quickref  -> Quick reference.
help       -> Python's own help system.
object?    -> Details about 'object', use 'object??' for
extra details.
```

```
In [1]: di
%dirs   dict    dir      divmod
```

In the preceding session, type the characters `di`, and `iPython` will respond with the following line that contains all the functions that start with the letters `di`:

```
%dirs   dict    dir      divmod
```

Enter a question mark (“?”), and `ipython` provides textual assistance, the first part of which is here:

```
IPython -- An enhanced Interactive Python
=====
```

`IPython` offers a combination of convenient shell features, special commands and a history mechanism for both input (command history) and output (results caching, similar

to Mathematica). It is intended to be a fully compatible replacement for the standard Python interpreter, while offering vastly improved functionality and flexibility.

The next section shows how to check whether or not Python is installed, and also shows where Python can be downloaded.

PYTHON INSTALLATION

Before downloading anything, check if Python is already installed by typing the following command in a command shell:

```
python -V
```

The output for the Macbook used in this book is here:

```
Python 3.12.5
```

Note: Install Python 3.9 (or as close as possible to this version) so that the same version of Python that was used to test the Python files in this book is used.

To install Python, navigate to the Python home page and select the downloads link or navigate directly to this Web site:

<http://www.python.org/download/>

In addition, PythonWin is available for Windows, and its home page is here:

<http://www.cgl.ucsf.edu/Outreach/pc204/pythonwin.html>

Use any text editor that can create, edit, and save Python scripts and save them as plain text files (don't use Microsoft Word).

After Python is installed and configured, users are ready to work with the Python scripts in this book.

SETTING THE PATH ENVIRONMENT VARIABLE (WINDOWS ONLY)

The PATH environment variable specifies a list of directories that are searched whenever one specifies an executable program from the command line. A very good guide to setting up the environment so that the Python executable is always available in every command shell is to follow the instructions here:

<http://www.blog.pythonlibrary.org/2011/11/24/python-101-setting-up-python-on-windows/>

LAUNCHING PYTHON ON A MACHINE

There are three different ways to launch Python:

- Use the Python interactive interpreter.
- Launch Python scripts from the command line.
- Use an IDE.

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The next section shows how to launch the Python interpreter from the command line, and later in this chapter readers will learn how to launch Python scripts from the command line and they will also learn about Python IDEs.

Note: The emphasis in this book is to launch Python scripts from the command line or to enter code in the Python interpreter.

The Python Interactive Interpreter

Launch the Python interactive interpreter from the command line by opening a command shell and typing the following command:

```
python
```

Readers will see the following prompt (or something similar):

```
Python 3.12.5 (v3.12.5:ff3bc82f7c9, Aug 7 2024, 05:32:06)
[Clang 13.0.0 (clang-1300.0.29.30)] on darwin
Type "help", "copyright", "credits" or "license" for more
information.
>>>
```

Type the expression `2 + 7` at the prompt:

```
>>> 2 + 7
```

Python displays the following result:

```
9
>>>
```

Press `ctrl-d` to exit the Python shell.

Launch any Python script from the command line by preceding it with the word “python.” For example, if there is a Python script `myscript.py` that contains Python commands, launch the script as follows:

```
python myscript.py
```

As a simple illustration, suppose that the Python script `myscript.py` contains the following Python code:

```
print('Hello World from Python')
print('2 + 7 = ', 2+7)
```

When the user launches the preceding Python script, they will see the following output:

```
Hello World from Python
2 + 7 = 9
```

PYTHON IDENTIFIERS

A Python identifier is the name of a variable, function, class, module, or other Python object, and a valid identifier conforms to the following rules:

- It starts with a letter A to Z or a to z or an underscore (_).
- It contains zero or more letters, underscores, and digits (0 to 9).

Note: Python identifiers cannot contain characters such as @, \$, and %.

Python is a case-sensitive language, so `Abc` and `abc` are different identifiers in Python.

In addition, Python has the following naming convention:

- Class names start with an uppercase letter and all other identifiers with a lowercase letter
- An initial underscore is used for private identifiers.
- Two initial underscores are used for strongly private identifiers.

A Python identifier with two initial underscore and two trailing underscore characters indicates a language-defined special name.

LINES, INDENTATION, AND MULTILINES

Unlike other programming languages (such as Java or Objective-C), Python uses indentation instead of curly braces for code blocks. Indentation must be consistent in a code block, as shown here:

```
if True:
    print("ABC")
    print("DEF")
else:
    print("ABC")
    print("DEF")
```

Multi-line statements in Python can terminate with a new line or the backslash (“\”) character, as shown here:

```
total = x1 + \
        x2 + \
        x3
```

Obviously one can place `x1`, `x2`, and `x3` on the same line, so there is no reason to use three separate lines; however, this functionality is available in case one needs to add a set of variables that do not fit on a single line.

One can specify multiple statements in one line by using a semicolon (“;”) to separate each statement, as shown here:

```
a=10; b=5; print(a); print(a+b)
```

The output of the preceding code snippet is here:

```
10
15
```

Note: The use of semicolons and the continuation character are discouraged in Python.

QUOTATION AND COMMENTS IN PYTHON

Python allows single (‘), double (“) and triple (""" or ''') quotes for string literals, provided that they match at the beginning and the end of the string. One can use triple quotes for strings that span multiple lines. The following examples are legal Python strings:

```
word = 'word'
line = "This is a sentence."
para = """This is a paragraph. This paragraph contains
more than one sentence."""
```

A string literal that begins with the letter “r” (for “raw”) treats everything as a literal character and “escapes” the meaning of meta characters, as shown here:

```
a1 = r'\n'
a2 = r'\r'
a3 = r'\t'
print('a1:', a1, 'a2:', a2, 'a3:', a3)
```

The output of the preceding code block is here:

```
a1: \n a2: \r a3: \t
```

One can embed a single quote in a pair of double quotes (and vice versa) in order to display a single quote or a double quote. Another way to accomplish the same result is to precede a single or double quote with a backslash (“\”) character. The following code block illustrates these techniques:

```
b1 = ""
b2 = ""
b3 = '\ '
b4 = "\""
print('b1:', b1, 'b2:', b2)
print('b3:', b3, 'b4:', b4)
```

The output of the preceding code block is here:

```
b1: ' b2: "
b3: ' b4: "
```

A hash sign (#) that is not inside a string literal is the character that indicates the beginning of a comment. Moreover, all characters after the # and up

to the physical line end are part of the comment (and ignored by the Python interpreter). Consider the following code block:

```
#!/usr/bin/python
# First comment
print("Hello, Python!") # second comment
```

This will produce following result:

```
Hello, Python!
```

A comment may be on the same line after a statement or expression:

```
name = "Tom Jones" # This is also comment
```

Users can comment multiple lines as follows:

```
# This is comment one
# This is comment two
# This is comment three
```

A blank line in Python is a line containing only whitespace, a comment, or both.

SAVING CODE IN A MODULE

Earlier readers learned how to launch the Python interpreter from the command line and then enter Python commands. However, everything that is typed in the Python interpreter is only valid for the current session: If one exits the interpreter and then launches the interpreter again, previous definitions are no longer valid. Fortunately, Python enables users to store code in a text file, as discussed in the next section.

A *module* in Python is a text file that contains Python statements. In the previous section, users saw how the Python interpreter enables them to test code snippets whose definitions are valid for the current session. If one wants to retain the code snippets and other definitions, they should place them in a text file so they can execute that code outside of the Python interpreter.

The outermost statements in a Python are executed from top to bottom when the module is imported for the first time, which will then set up its variables and functions.

A Python module can be run directly from the command line, as shown here:

```
python first.py
```

As an illustration, place the following two statements in a text file called `first.py`:

```
x = 3
print(x)
```

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Now type the following command:

```
python first.py
```

The output from the preceding command is 3, which is the same as executing the preceding code from the Python interpreter.

When a Python module is run directly, the special variable `__name__` is set to `__main__`. Users will often see the following type of code in a Python module:

```
if __name__ == '__main__':  
    # do something here  
    print('Running directly')
```

The preceding code snippet enables Python to determine if a Python module was launched from the command line or imported into another Python module.

SOME STANDARD MODULES IN PYTHON

The Python Standard Library provides many modules that can simplify Python scripts. A list of the standard library modules is here:

<http://www.python.org/doc/>

Some of the most important Python modules include `cgi`, `math`, `os`, `pickle`, `random`, `re`, `socket`, `sys`, `time`, and `urllib`.

The code samples in this book use the modules `math`, `os`, `random`, `re`, `socket`, `sys`, `time`, and `urllib`. Users need to import these modules in order to use them in their code. For example, the following code block shows how to import four standard Python modules:

```
import datetime  
import re  
import sys  
import time
```

The code samples in this book import one or more of the preceding modules, as well as other Python modules.

THE `HELP()` AND `DIR()` FUNCTIONS

An Internet search for Python-related topics usually returns a number of links with useful information. Alternatively, users can check the official Python documentation site: *docs.python.org*

In addition, Python provides the `help()` and `dir()` functions that are accessible from the Python interpreter. The `help()` function displays documentation strings, whereas the `dir()` function displays defined symbols.

For example, if a user types `help(sys)`, they will see documentation for the `sys` module, whereas `dir(sys)` displays a list of the defined symbols.

Type the following command in the Python interpreter to display the string-related methods in Python:

```
>>> dir(str)
```

The preceding command generates the following output:

```
['_add_', '__class__', '__contains__', '__delattr__',
 '__doc__', '__eq__', '__format__', '__ge__', '__
getattribute__', '__getitem__', '__getnewargs__', '__
getslice__', '__gt__', '__hash__', '__init__', '__le__',
 '__len__', '__lt__', '__mod__', '__mul__', '__ne__',
 '__new__', '__reduce__', '__reduce_ex__', '__repr__',
 '__rmod__', '__rmul__', '__setattr__', '__sizeof__',
 '__str__', '__subclasshook__', '_formatter_field_name_
split', '_formatter_parser', 'capitalize', 'center',
'count', 'decode', 'encode', 'endswith', 'expandtabs',
'find', 'format', 'index', 'isalnum', 'isalpha', 'isdigit',
'islower', 'isspace', 'istitle', 'isupper', 'join',
'ljust', 'lower', 'lstrip', 'partition', 'replace',
'rfind', 'rindex', 'rjust', 'rpartition', 'rsplit',
rstrip', 'split', 'splitlines', 'startswith', 'strip',
'swapcase', 'title', 'translate', 'upper', 'zfill']
```

The preceding list gives users a consolidated “dump” of built-in functions (including some that are discussed later in this chapter). Although the `max()` function obviously returns the maximum value of its arguments, the purpose of other functions such as `filter()` or `map()` is not immediately apparent (unless one has used them in other programming languages). In any case, the preceding list provides a starting point for finding out more about various Python built-in functions that are not discussed in this chapter.

Note that while `dir()` does not list the names of built-in functions and variables, users can obtain this information from the standard module `__builtin__` that is automatically imported under the name `__builtins__`:

```
>>> dir(__builtins__)
```

The following command shows how to get more information about a function:

```
help(str.lower)
```

The output from the preceding command is here:

```
Help on method_descriptor:
```

```
lower(...)
```

```
    S.lower() -> string
```

```
    Return a copy of the string S converted to lowercase.
    (END)
```

Users should check the online documentation and also experiment with `help()` and `dir()` when they need additional information about a particular function or module.

COMPILE TIME AND RUNTIME CODE CHECKING

Python performs some compile-time checking, but most checks (including type, name, and so forth) are *deferred* until code execution. Consequently, if Python code references a user-defined function that does not exist, the code will compile successfully. In fact, the code will fail with an exception *only* when the code execution path references the nonexistent function.

As a simple example, consider the following Python function `myFunc` that references the nonexistent function called `DoesNotExist`:

```
def myFunc(x):
    if x == 3:
        print(DoesNotExist(x))
    else:
        print('x: ',x)
```

The preceding code will only fail when the `myFunc` function is passed the value 3, after which Python raises an error.

Chapter 2 explains how to define and invoke user-defined functions, along with an explanation of the difference between local versus global variables in Python.

Now that readers understand some basic concepts (such as how to use the Python interpreter) and how to launch custom Python modules, the next section discusses primitive data types in Python.

SIMPLE DATA TYPES IN PYTHON

Python supports primitive data types, such as numbers (integers, floating point numbers, and exponential numbers), strings, and dates. Python also supports more complex data types, such as lists (or arrays), tuples, and dictionaries, all of which are discussed in Chapter 3. The next several sections discuss some of the Python primitive data types, along with code snippets that show how to perform various operations on those data types.

WORKING WITH NUMBERS

Python provides arithmetic operations for manipulating numbers a straightforward manner that is similar to other programming languages. The following examples involve arithmetic operations on integers:

```
>>> 2+2
4
```

```
>>> 4/3
1
>>> 3*8
24
```

The following example assigns numbers to two variables and computes their product:

```
>>> x = 4
>>> y = 7
>>> x * y
28
```

The following examples demonstrate arithmetic operations involving integers:

```
>>> 2+2
4
>>> 4/3
1
>>> 3*8
24
```

Notice that division (“/”) of two integers is actually truncation in which only the integer result is retained. The following example converts a floating point number into exponential form:

```
>>> fnum = 0.00012345689000007
>>> "%.14e"%fnum
'1.23456890000070e-04'
```

Use the `int()` function and the `float()` function to convert strings to numbers:

```
word1 = "123"
word2 = "456.78"
var1 = int(word1)
var2 = float(word2)
print("var1: ",var1," var2: ",var2)
```

The output from the preceding code block is here:

```
var1: 123 var2: 456.78
```

Alternatively, one can use the `eval()` function:

```
word1 = "123"
word2 = "456.78"
var1 = eval(word1)
var2 = eval(word2)
print("var1: ",var1," var2: ",var2)
```

If one attempts to convert a string that is not a valid integer or a floating point number, Python raises an exception, so it's advisable to place a code in a `try/except` block (discussed later in this chapter).

Working with Other Bases

Numbers in Python are in base 10 (the default), but one can easily convert numbers to other bases. For example, the following code block initializes the variable `x` with the value 1234, and then displays that number in base 2, 8, and 16, respectively:

```
>>> x = 1234
>>> bin(x) '0b10011010010'
>>> oct(x) '0o2322'
>>> hex(x) '0x4d2' >>>
```

Use the `format()` function to suppress the `0b`, `0o`, or `0x` prefixes, as shown here:

```
>>> format(x, 'b') '10011010010'
>>> format(x, 'o') '2322'
>>> format(x, 'x') '4d2'
```

Negative integers are displayed with a negative sign:

```
>>> x = -1234
>>> format(x, 'b') '-10011010010'
>>> format(x, 'x') '-4d2'
```

The `chr()` Function

The Python `chr()` function takes a positive integer as a parameter and converts it to its corresponding alphabetic value (if one exists). The letters A through Z have decimal representation of 65 through 91 (which corresponds to hexadecimal 41 through 5b), and the lowercase letters a through z have decimal representation 97 through 122 (hexadecimal 61 through 7b).

Here is an example of using the `chr()` function to print uppercase A:

```
>>> x=chr(65)
>>> x
'A'
```

The following code block prints the ASCII values for a range of integers:

```
result = ""
for x in range(65,91):
    print(x, chr(x))
    result = result+chr(x)+' '
print("result: ",result)
```

Note: Python 2 uses ASCII strings whereas Python 3 uses UTF-8.

One can represent a range of characters with the following line:

```
for x in range(65,91):
```

However, the following equivalent code snippet is more intuitive:

```
for x in range(ord('A'), ord('Z')):
```

To display the result for lowercase letters, change the preceding range from (65, 91) to either of the following statements:

```
for x in range(65, 91):
for x in range(ord('a'), ord('z')):
```

The round() Function in Python

The Python `round()` function enables users to round decimal values to the nearest precision:

```
>>> round(1.23, 1)
1.2
>>> round(-3.42, 1)
-3.4
```

Formatting Numbers in Python

Python allows users to specify the number of decimal places of precision to use when printing decimal numbers, as shown here:

```
>>> x = 1.23456
>>> format(x, '0.2f')
'1.23'
>>> format(x, '0.3f')
'1.235'
>>> 'value is {:.3f}'.format(x) 'value is 1.235'
>>> from decimal import Decimal
>>> a = Decimal('4.2')
>>> b = Decimal('2.1')
>>> a + b
Decimal('6.3')
>>> print(a + b)
6.3
>>> (a + b) == Decimal('6.3')
True
>>> x = 1234.56789
>>> # Two decimal places of accuracy
>>> format(x, '0.2f')
'1234.57'
>>> # Right justified in 10 chars, one-digit accuracy
>>> format(x, '>10.1f')
' 1234.6'
>>> # Left justified
>>> format(x, '<10.1f') '1234.6 '
>>> # Centered
>>> format(x, '^10.1f') ' 1234.6 '
>>> # Inclusion of thousands separator
>>> format(x, ',')
'1,234.56789'
>>> format(x, '0,.1f')
'1,234.6'
```

WORKING WITH FRACTIONS

Python supports the `Fraction()` function (which is defined in the `fractions` module) that accepts two integers that represent the numerator and the denominator (which must be nonzero) of a fraction. Several examples of defining and manipulating fractions in Python are shown here:

```
>>> from fractions import Fraction
>>> a = Fraction(5, 4)
>>> b = Fraction(7, 16)
>>> print(a + b)
27/16
>>> print(a * b) 35/64
>>> # Getting numerator/denominator
>>> c = a * b
>>> c.numerator
35
>>> c.denominator
64
>>> # Converting to a float >>> float(c)
0.546875
>>> # Limiting the denominator of a value
>>> print(c.limit_denominator(8))
4
>>> # Converting a float to a fraction >>> x = 3.75
>>> y = Fraction(*x.as_integer_ratio())
>>> y
Fraction(15, 4)
```

Before delving into Python code samples that work with strings, the next section briefly discusses Unicode and UTF-8, both of which are character encodings.

UNICODE AND UTF-8

A Unicode string consists of a sequence of numbers that are between 0 and 0x10ffff, where each number represents a group of bytes. An encoding is the manner in which a Unicode string is translated into a sequence of bytes. Among the various encodings, UTF-8 (“Unicode transformation format”) is perhaps the most common, and it’s also the default encoding for many systems. The digit 8 in UTF-8 indicates that the encoding uses 8-bit numbers, whereas UTF-16 uses 16-bit numbers (but this encoding is less common).

The ASCII character set is a subset of UTF-8, so a valid ASCII string can be read as a UTF-8 string without any re-encoding required. In addition, a Unicode string can be converted into a UTF-8 string.

WORKING WITH UNICODE

Python supports Unicode, which means that users can render characters in different languages. Unicode data can be stored and manipulated in the same way as strings. Create a Unicode string by prepending the letter “u,” as shown here:

```
>>> u'Hello from Python!'
u'Hello from Python!'
```

Special characters can be included in a string by specifying their Unicode value. For example, the following Unicode string embeds a space (which has the Unicode value 0x0020) in a string:

```
>>> u'Hello\u0020from Python!'
u'Hello from Python!'
```

Listing 1.1 displays the contents of `unicode1.py` that illustrates how to display a string of characters in Japanese and another string of characters in Chinese (Mandarin).

LISTING 1.1: `unicode1.py`

```
chinese1 = u'\u5c07\u63a2\u8a0e HTML5 \u53ca\u5176\u4ed6'
hiragana = u'D3 \u306F \u304B\u3063\u3053\u3043\u3043 \u3067\u3059!'

print('Chinese:', chinese1)
print('Hiragana:', hiragana)
```

The output of Listing 1.2 is here:

```
Chinese: 將探討 HTML5 及其他
Hiragana: D3 は かつこいい です!
```

The next portion of this chapter shows how to “slice and dice” text strings with built-in Python functions.

WORKING WITH STRINGS

Users can concatenate two strings using the “+” operator. The following example prints a string and then concatenates two single-letter strings:

```
>>> 'abc'
'abc'
>>> 'a' + 'b'
'ab'
```

Use “+” or “*” to concatenate strings, as shown here:

```
>>> 'a' + 'a' + 'a'
'aaa'
```

```
>>> 'a' * 3
'aaa'
```

Assign strings to variables and print them using the `print()` command:

```
>>> print('abc')
abc
>>> x = 'abc'
>>> print(x)
abc
>>> y = 'def'
>>> print(x + y)
abcdef
```

“Unpack” the letters of a string and assign them to variables, as shown here:

```
>>> str = "World"
>>> x1,x2,x3,x4,x5 = str
>>> x1
'W'
>>> x2
'o'
>>> x3
'r'
>>> x4
'l'
>>> x5
'd'
```

The preceding code snippets show how easy it is to extract the letters in a text string, and in Chapter 3 users will learn how to “unpack” other Python data structures.

Extract substrings of a string as shown in the following examples:

```
>>> x = "abcdef"
>>> x[0]
'a'
>>> x[-1]
'f'
>>> x[1:3]
'bc'
>>> x[0:2] + x[5:]
'abf'
```

However, an error will occur if one attempts to “subtract” two strings:

```
>>> 'a' - 'b'
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: unsupported operand type(s) for -: 'str' and
'str'
```


The `try/except` construct in Python (discussed later in this chapter) enables one to handle the preceding type of exception more gracefully.

Comparing Strings

Use the methods `lower()` and `upper()` to convert a string to lowercase and uppercase, respectively, as shown here:

```
>>> 'Python'.lower()
'python'
>>> 'Python'.upper()
'PYTHON'
>>>
```

The methods `lower()` and `upper()` are useful for performing a case insensitive comparison of two ASCII strings. Listing 1.2 displays the contents of `compare.py` that uses the `lower()` function in order to compare two ASCII strings.

LISTING 1.2: `compare.py`

```
x = 'Abc'
y = 'abc'

if(x == y):
    print('x and y: identical')
elif (x.lower() == y.lower()):
    print('x and y: case insensitive match')
else:
    print('x and y: different')
```

Since `x` contains mixed case letters and `y` contains lowercase letters, Listing 1.2 displays the following output:

```
x and y: different
```

Formatting Strings in Python

Python provides the functions `string.lstring()`, `string.rstring()`, and `string.center()` for positioning a text string so that it is left-justified, right-justified, and centered, respectively. As seen in a previous section, Python also provides the `format()` method for advanced interpolation features.

Enter the following commands in the Python interpreter:

```
import string

str1 = 'this is a string'
print(string.ljust(str1, 10))
print(string.rjust(str1, 40))
print(string.center(str1, 40))
```

The output is shown here:

```
this is a string
           this is a string
      this is a string
```

The next portion of this chapter shows how to “slice and dice” text strings with built-in Python functions.

SLICING AND SPLICING STRINGS

Python enables users to extract substrings of a string (called “slicing”) using array notation. Slice notation is `start:stop:step`, where the start, stop, and step values are integers that specify the start value, end value, and the increment value. The interesting part about slicing in Python is that one can use the value `-1`, which operates from the right side instead of the left side of a string.

Some examples of slicing a string are here:

```
text1 = "this is a string"
print('First 7 characters:',text1[0:7])
print('Characters 2-4:',text1[2:4])
print('Right-most character:',text1[-1])
print('Right-most 2 characters:',text1[-3:-1])
```

The output from the preceding code block is here:

```
First 7 characters: this is
Characters 2-4: is
Right-most character: g
Right-most 2 characters: in
```

Later in this chapter readers will learn how to insert a string in the middle of another string.

Testing for Digits and Alphabetic Characters

Python enables users to examine each character in a string and then test whether that character is a bona fide digit or an alphabetic character. This section provides a precursor to regular expressions, after which you can perform an online search for articles that provide additional details for regular expressions.

Listing 1.3 displays the contents of `char_types.py` that illustrates how to determine if a string contains digits or characters. As you can see, the conditional logic in Listing 1.3 is self-explanatory.

LISTING 1.3: `char_types.py`

```
str1 = "4"
str2 = "4234"
```

```

str3 = "b"
str4 = "abc"
str5 = "alb2c3"

if(str1.isdigit()):
    print("this is a digit:",str1)

if(str2.isdigit()):
    print("this is a digit:",str2)

if(str3.isalpha()):
    print("this is alphabetic:",str3)

if(str4.isalpha()):
    print("this is alphabetic:",str4)

if(not str5.isalpha()):
    print("this is not pure alphabetic:",str5)

print("capitalized first letter:",str5.title())

```

Listing 1.3 initializes some variables, followed by two conditional tests that check whether or not `str1` and `str2` are digits using the `isdigit()` function. The next portion of Listing 1.3 checks if `str3`, `str4`, and `str5` are alphabetic strings using the `isalpha()` function. The output of Listing 1.3 is here:

```

this is a digit: 4
this is a digit: 4234
this is alphabetic: b
this is alphabetic: abc
this is not pure alphabetic: alb2c3
capitalized first letter: A1B2C3

```

SEARCH AND REPLACE A STRING IN OTHER STRINGS

Python provides methods for searching and also for replacing a string in a second text string. Listing 1.4 displays the contents of `find_pos1.py` that shows how to use the `find` function to search for the occurrence of one string in another string.

LISTING 1.4: find_pos1.py

```

item1 = 'abc'
item2 = 'Abc'
text = 'This is a text string with abc'

pos1 = text.find(item1)
pos2 = text.find(item2)

print('pos1=',pos1)
print('pos2=',pos2)

```

Listing 1.4 initializes the variables `item1`, `item2`, and `text`, and then searches for the index of the contents of `item1` and `item2` in the string `text`. The Python `find()` function returns the column number where the first successful match occurs; otherwise, the `find()` function returns a `-1` if a match is unsuccessful.

The output from launching Listing 1.4 is here:

```
pos1= 27
pos2= -1
```

In addition to the `find()` method, one can use the `in` operator when they want to test for the presence of an element, as shown here:

```
>>> lst = [1,2,3]
>>> 1 in lst
True
```

Listing 1.5 displays the contents of `replace1.py` that shows how to replace one string with another string.

LISTING 1.5: `replace1.py`

```
text = 'This is a text string with abc'
print('text:',text)
text = text.replace('is a', 'was a')
print('text:',text)
```

Listing 1.5 starts by initializing the variable `text` and then printing its contents. The next portion of Listing 1.5 replaces the occurrence of “is a” with “was a” in the string `text`, and then prints the modified string. The output from launching Listing 1.5 is here:

```
text: This is a text string with abc
text: This was a text string with abc
```

REMOVE LEADING AND TRAILING CHARACTERS

Python provides the functions `strip()`, `lstrip()`, and `rstrip()` to remove characters in a text string. Listing 1.6 displays the contents of `remove1.py` that shows how to search for a string.

LISTING 1.6: `remove1.py`

```
text = '    leading and trailing white space    '
print('text1:', 'x', text, 'y')

text = text.lstrip()
print('text2:', 'x', text, 'y')

text = text.rstrip()
print('text3:', 'x', text, 'y')
```

Listing 1.6 starts by concatenating the letter `x` and the contents of the variable `text`, and then printing the result. The second part of Listing 1.6 removes the leading white spaces in the string `text` and then appends the result to the letter `x`. The third part of Listing 1.6 removes the trailing white spaces in the string `text` (note that the leading white spaces have already been removed) and then appends the result to the letter `x`.

The output from launching Listing 1.6 is here:

```
text1: x    leading and trailing white space    y
text2: x leading and trailing white space      y
text3: x leading and trailing white space y
```

If one wants to remove extra white spaces inside a text string, they should use the `replace()` function as discussed in the previous section. The following example illustrates how this can be accomplished, which also contains the `re` module as a “preview” for what will be learned in Chapter 4:

```
import re
text = 'a  C b'
a = text.replace(' ', '')
b = re.sub('\s+', ' ', text)

print(a)
print(b)
```

The result is here:

```
aCb
a b
```

Chapter 2 shows how to use the `join()` function in order to remove extra white spaces in a text string.

PRINTING TEXT WITHOUT NEWLINE CHARACTERS

If one needs to suppress white space and a newline between objects output with multiple `print()` statements, they can use concatenation or the `write()` function.

The first technique is to concatenate the string representations of each object using the `str()` function prior to printing the result. For example, users should run the following statement in Python:

```
x = str(9)+str(0xff)+str(-3.1)
print('x: ',x)
```

The output is shown here:

```
x:  9255-3.1
```

The preceding line contains the concatenation of the numbers 9 and 255 (which is the decimal value of the hexadecimal number `0xff`) and `-3.1`.

Incidentally, one can use the `str()` function with modules and user-defined classes. An example involving the Python built-in module `sys` is here:

```
>>> import sys
>>> print(str(sys))
<module 'sys' (built-in)>
```

The following code snippet illustrates how to use the `write()` function to display a string:

```
import sys
write = sys.stdout.write
write('123')
write('123456789')
```

The output is here:

```
1233
1234567899
```

TEXT ALIGNMENT

Python provides the methods `ljust()`, `rjust()`, and `center()` for aligning text. The `ljust()` and `rjust()` functions left justify and right justify a text string, respectively, whereas the `center()` function will center a string. An example is shown in the following code block:

```
text = 'Hello World'
text.ljust(20)
'Hello World '
>>> text.rjust(20)
' Hello World'
>>> text.center(20)
' Hello World '
```

One can use the Python `format()` function to align text. Use the `<`, `>`, or `^` characters, along with a desired width, in order to right justify, left justify, and center the text, respectively. The following examples illustrate how to specify text justification:

```
>>> format(text, '>20')
'          Hello World'
>>>
>>> format(text, '<20')
'Hello World      '
>>>
>>> format(text, '^20')
'    Hello World    '
>>>
```

WORKING WITH DATES

Python provides a rich set of date-related functions. Listing 1.7 displays the contents of the Python script `date_time2.py` that displays various date-related values, such as the current date and time; the day of the week, month, and year; and the time in seconds since the epoch.

LISTING 1.7: `date_time2.py`

```
import time
import datetime

print("Time in seconds since the epoch: %s" %time.time())
print("Current date and time: " , datetime.datetime.now())
print("Or like this: " ,datetime.datetime.now().strftime
("%y-%m-%d-%H-%M"))

print("Current year: ", datetime.date.today().
strftime("%Y"))
print("Month of year: ", datetime.date.today().
strftime("%B"))
print("Week number of the year: ", datetime.date.today().
strftime("%W"))
print("Weekday of the week: ", datetime.date.today().
strftime("%w"))
print("Day of year: ", datetime.date.today().
strftime("%j"))
print("Day of the month : ", datetime.date.today().
strftime("%d"))
print("Day of week: ", datetime.date.today().
strftime("%A"))
```

Listing 1.8 displays the output generated by running the code in Listing 1.7.

LISTING 1.8: `datetime2.out`

```
Time in seconds since the epoch: 1375144195.66
Current date and time:  2013-07-29 17:29:55.664164
Or like this:  13-07-29-17-29
Current year:  2013
Month of year:  July
Week number of the year:  30
Weekday of the week:  1
Day of year:  210
Day of the month :  29
Day of week:  Monday
```

Python also enables users to perform arithmetic calculates with date-related values, as shown in the following code block:

```
>>> from datetime import timedelta
>>> a = timedelta(days=2, hours=6)
>>> b = timedelta(hours=4.5)
>>> c = a + b
>>> c.days
2
>>> c.seconds
37800
>>> c.seconds / 3600
10.5
>>> c.total_seconds() / 3600
58.5
```

Converting Strings to Dates

Listing 1.9 displays the contents of `string2date.py` that illustrates how to convert a string to a date, and also how to calculate the difference between two dates.

LISTING 1.9: `string2date.py`

```
from datetime import datetime

text = '2014-08-13'
y = datetime.strptime(text, '%Y-%m-%d')
z = datetime.now()
diff = z - y
print('Date difference:', diff)
```

The output from Listing 1.9 is shown here:

```
Date difference: -210 days, 18:58:40.197130
```

EXCEPTION HANDLING IN PYTHON

Unlike JavaScript, users cannot add a number and a string in Python. However, one can detect an illegal operation using the `try/except` construct in Python, which is similar to the `try/catch` construct in languages such as JavaScript and Java.

An example of a `try/except` block is here:

```
try:
    x = 4
    y = 'abc'
    z = x + y
except:
    print 'cannot add incompatible types:', x, y
```


When one runs the preceding code in Python, the `print()` statement in the `except` code block is executed because the variables `x` and `y` have incompatible types.

Earlier in the chapter readers also saw that subtracting two strings throws an exception:

```
>>> 'a' - 'b'
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: unsupported operand type(s) for -: 'str' and 'str'
```

A simple way to handle this situation is to use a `try/except` block:

```
>>> try:
...     print('a' - 'b')
... except TypeError:
...     print('TypeError exception while trying to subtract
two strings')
... except:
...     print('Exception while trying to subtract two
strings')
...
```

The output from the preceding code block is here:

```
TypeError exception while trying to subtract two strings
```

The preceding code block specifies the finer-grained exception called `TypeError`, followed by a generic `except` code block to handle all other exceptions that might occur during the execution of Python code. This style is similar to the exception handling in Java code.

Listing 1.10 displays the contents of `exception1.py` that illustrates how to handle various types of exceptions.

LISTING 1.10: exception1.py

```
import sys

try:
    f = open('myfile.txt')
    s = f.readline()
    i = int(s.strip())
except IOError as err:
    print("I/O error: {0}".format(err))
except ValueError:
    print("Could not convert data to an integer.")
except:
    print("Unexpected error:", sys.exc_info()[0])
    raise
```

Listing 1.10 contains a `try` block followed by three `except` statements. If an error occurs in the `try` block, the first `except` statement is compared with the type of exception that occurred. If there is a match, then the subsequent `print` statement is executed, and the program terminates. If not, a similar test is performed with the second `except` statement. If neither `except` statement matches the exception, the third `except` statement handles the exception, which involves printing a message and then “raising” an exception. Note that one can also specify multiple exception types in a single statement, as shown here:

```
except (NameError, RuntimeError, TypeError):
    print('One of three error types occurred')
```

The preceding code block is more compact, but `done` does not know which of the three error types occurred. Python allows users to define custom exceptions, but this topic is beyond the scope of this book.

HANDLING USER INPUT

Python enables users to read user input from the command line via the `input()` function or the `raw_input()` function. Typically, one assigns user input to a variable, which will contain all characters that users enter from the keyboard. User input terminates when users press the `<return>` key (which is included with the input characters). Listing 1.11 displays the contents of `user_input1.py` that prompts users for their name and then uses interpolation to display a response.

LISTING 1.11: `user_input1.py`

```
userInput = input("Enter your name: ")
print ("Hello %s, my name is Python" % userInput)
```

The output of Listing 1.11 is here (assume that the user entered the word Dave):

```
Hello Dave, my name is Python
```

The `print()` statement in Listing 1.11 uses string interpolation via `%s`, which substitutes the value of the variable after the `%` symbol. This functionality is obviously useful when one wants to specify something that is determined at run-time.

User input can cause exceptions (depending on the operations that the code performs), so it’s important to include exception-handling code.

Listing 1.12 displays the contents of `user_input2.py` that prompts users for a string and attempts to convert the string to a number in a `try/except` block.

LISTING 1.12: user_input2.py

```

userInput = input("Enter something: ")

try:
    x = 0 + eval(userInput)
    print('you entered the number:',userInput)
except:
    print(userInput,'is a string')

```

Listing 1.12 adds the number 0 to the result of converting a user's input to a number. If the conversion was successful, a message with the user's input is displayed. If the conversion failed, the `except` code block consists of a `print` statement that displays a message.

Note: This code sample uses the `eval()` function, which should be avoided so that the code does not evaluate arbitrary (and possibly destructive) commands.

Listing 1.13 displays the contents of `user_input3.py` that prompts users for two numbers and attempts to compute their sum in a pair of `try/except` blocks.

LISTING 1.13: user_input3.py

```

sum = 0

msg = 'Enter a number:'
val1 = input(msg)

try:
    sum = sum + eval(val1)
except:
    print(val1,'is a string')

msg = 'Enter a number:'
val2 = input(msg)

try:
    sum = sum + eval(val2)
except:
    print(val2,'is a string')

print('The sum of',val1,'and',val2,'is',sum)

```

Listing 1.13 contains two `try` blocks, each of which is followed by an `except` statement. The first `try` block attempts to add the first user-supplied number to the variable `sum`, and the second `try` block attempts to add the second user-supplied number to the previously entered number. An error message occurs if either input string is not a valid number; if both are valid numbers, a message is displayed containing the input numbers and their sum.

Be sure to read the caveat regarding the `eval()` function that is mentioned earlier in this chapter.

COMMAND-LINE ARGUMENTS

Python provides a `getopt` module to parse command-line options and arguments, and the Python `sys` module provides access to any command-line arguments via the `sys.argv`. This serves two purposes:

1. `sys.argv` is the list of command-line arguments.
2. `len(sys.argv)` is the number of command-line arguments.

Here `sys.argv[0]` is the program name, so if the Python program is called `test.py`, it matches the value of `sys.argv[0]`.

Now readers can provide input values for a Python program on the command line instead of providing input values by prompting users for their input.

As an example, consider the script `test.py` shown here:

```
#!/usr/bin/python
import sys
print('Number of arguments:', len(sys.argv), 'arguments')
print('Argument List:', str(sys.argv))
```

Run the previous as follows:

```
python test.py arg1 arg2 arg3
```

This will produce following result:

```
Number of arguments: 4 arguments.
Argument List: ['test.py', 'arg1', 'arg2', 'arg3']
```

The ability to specify input values from the command line provides useful functionality. For example, suppose that one has a custom Python class that contains the methods `add` and `subtract` to add and subtract a pair of numbers.

One can use command-line arguments in order to specify which method to execute on a pair of numbers, as shown here:

```
python MyClass add 3 5
python MyClass subtract 3 5
```

This functionality is very useful because users can programmatically execute different methods in a Python class, which means that they can write unit tests for their code as well.

Listing 1.14 displays the contents of `hello.py` that shows users how to use `sys.argv` to check the number of command line parameters.

LISTING 1.14: `hello.py`

```
import sys

def main():
    if len(sys.argv) >= 2:
        name = sys.argv[1]
    else:
        name = 'World'
    print('Hello', name)

# Standard boilerplate to invoke the main() function
if __name__ == '__main__':
    main()
```

Listing 1.14 defines the `main()` function that checks the number of command-line parameters: If this value is at least 2, then the variable `name` is assigned the value of the second parameter (the first parameter is `hello.py`), otherwise `name` is assigned the value `Hello`. The `print()` statement then prints the value of the variable `name`.

The final portion of Listing 1.14 uses conditional logic to determine whether or not to execute the `main()` function.

SUMMARY

This chapter explained how to work with numbers and perform arithmetic operations on numbers. Readers then learned how to work with strings and use string operations. They also learned how to use the `try/except` construct to handle exceptions that might occur in their Python code. The next chapter shows how to work with conditional statements, loops, and user-defined functions in Python.

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CONDITIONAL LOGIC, LOOPS, AND FUNCTIONS

This chapter introduces readers to various ways to perform conditional logic in Python, as well as control structures and user-defined functions in Python. Virtually every Python program that performs useful calculations requires some type of conditional logic or control structure (or both). Although the syntax for these Python features is slightly different from other languages, the functionality will be familiar.

The first part of this chapter contains code samples that illustrate how to handle `if-else` conditional logic in Python, as well as `if-elif-else` statements. The second part of this chapter discusses loops and `while` statements in Python. This section contains an assortment of examples (comparing strings, computing numbers raised to different exponents, and so forth) that illustrate various ways that one can use loops and `while` statements in Python.

The third part of this chapter contains examples that involve nested loops and recursion. The final part of this chapter introduces readers to user-defined Python functions.

PRECEDENCE OF OPERATORS IN PYTHON

When one has an expression involving numbers, they might remember that multiplication (“`*`”) and division (“`/`”) have higher precedence than addition (“`+`”) or subtraction (“`-`”). Exponentiation has even higher precedence than these four arithmetic operators.

Instead of relying on precedence rules, it’s simpler (as well as safer) to use parentheses. For example, `(x/y)+10` is clearer than `x/y+10`, even though they are equivalent expressions.

As another example, the following two arithmetic expressions are the equivalent, but the second is less error prone than the first:

$$x/y+3*z/8+x*y/z-3*x$$

$$(x/y) + (3*z) / 8 + (x*y) / z - (3*x)$$

In any case, the following Web site contains precedence rules for operators in Python:

http://www.mathcs.emory.edu/~valerie/courses/fall10/155/resources/op_precedence.html

PYTHON RESERVED WORDS

Every programming language has a set of reserved words: a set of words that cannot be used as identifiers, and Python is no exception. The Python reserved words are: `and`, `exec`, `not`, `assert`, `finally`, `or`, `break`, `for`, `pass`, `class`, `from`, `print`, `continue`, `global`, `raise`, `def`, `if`, `return`, `del`, `import`, `try`, `elif`, `in`, `while`, `else`, `is`, `with`, `except`, `lambda`, and `yield`.

If one inadvertently uses a reserved word as a variable, they will see an “invalid syntax” error message instead of a “reserved word” error message. For example, suppose they create a Python script `test1.py` with the following code:

```
break = 2
print('break =', break)
```

If they run the preceding Python code, they will see the following output:

```
File "test1.py", line 2
    break = 2
    ^
SyntaxError: invalid syntax
```

A quick inspection of the Python code reveals the fact that they are attempting to use the reserved word `break` as a variable.

WORKING WITH LOOPS IN PYTHON

Python supports `for` loops, `while` loops, and `range()` statements. The following subsections illustrate how one can use each of these constructs.

Python for Loops

Python supports the `for` loop whose syntax is slightly different from other languages (such as JavaScript and Java). The following code block shows

how to use a `for` loop in Python in order to iterate through the elements in a list:

```
>>> x = ['a', 'b', 'c']
>>> for w in x:
...     print(w)
...
a
b
c
```

The preceding code snippet prints three letters on three separate lines. Users can force the output to be displayed on the same line (which will “wrap” if they specify a large enough number of characters) by appending a comma “,” in the print statement, as shown here:

```
>>> x = ['a', 'b', 'c']
>>> for w in x:
...     print(w, end=' ')
...
a b c
```

One can use this type of code when they want to display the contents of a text file in a single line instead of multiple lines.

Python also provides the built-in `reversed()` function that reverses the direction of the loop, as shown here:

```
>>> a = [1, 2, 3, 4, 5]
>>> for x in reversed(a):
...     print(x)
...
5
4
3
2
1
```

Note that reversed iteration only works if the size of the current object can be determined or if the object implements a `__reversed__()` “magic” method.

A `for` Loop With `try/except` in Python

Listing 2.1 displays the contents of `string_to_nums.py` that illustrates how to calculate the sum of a set of integers that have been converted from strings.

LISTING 2.1: `string_to_nums.py`

```
line = '1 2 3 4 10e abc'
sum  = 0
invalidStr = ""
```

```

print('String of numbers:',line)

for str in line.split(" "):
    try:
        sum = sum + eval(str)
    except:
        invalidStr = invalidStr + str + ' '

print('sum:', sum)
if(invalidStr != ""):
    print('Invalid strings:',invalidStr)
else:
    print('All substrings are valid numbers')

```

Listing 2.1 initializes the variables `line`, `sum`, and `invalidStr`, and then displays the contents of `line`. The next portion of Listing 2.1 splits the contents of `line` into words, and then uses a `try` block in order to add the numeric value of each word to the variable `sum`. If an exception occurs, the contents of the current `str` is appended to the variable `invalidStr`.

When the loop has finished execution, Listing 2.1 displays the sum of the numeric words, followed by the list of words that are not numbers. The output from Listing 2.1 is here:

```

String of numbers: 1 2 3 4 10e abc
sum: 10
Invalid strings: 10e abc

```

Numeric Exponents in Python

Listing 2.2 displays the contents of `nth_exponent.py` that illustrates how to calculate intermediate powers of a set of integers.

LISTING 2.2: `nth_exponent.py`

```

maxPower = 4
maxCount = 4

def pwr(num):
    prod = 1
    for n in range(1,maxPower+1):
        prod = prod*num
        print(num,'to the power',n, 'equals',prod)
    print('-----')

for num in range(1,maxCount+1):
    pwr(num)

```

Listing 2.2 contains a function called `pwr()` that accepts a numeric value. This function contains a loop that prints the value of that number raised to the power `n`, where `n` ranges between 1 and `maxPower+1`.

The second part of Listing 2.2 contains a `for` loop that invokes the function `pwr()` with the numbers between 1 and `maxPower+1`. The output from Listing 2.2 is here:

```
1 to the power 1 equals 1
1 to the power 2 equals 1
1 to the power 3 equals 1
1 to the power 4 equals 1
-----
2 to the power 1 equals 2
2 to the power 2 equals 4
2 to the power 3 equals 8
2 to the power 4 equals 16
-----
3 to the power 1 equals 3
3 to the power 2 equals 9
3 to the power 3 equals 27
3 to the power 4 equals 81
-----
4 to the power 1 equals 4
4 to the power 2 equals 16
4 to the power 3 equals 64
4 to the power 4 equals 256
-----
```

NESTED LOOPS

Listing 2.3 displays the contents of `triangular1.py` that illustrates how to print a row of consecutive integers (starting from 1), where the length of each row is one greater than the previous row.

LISTING 2.3: `triangular1.py`

```
max = 8
for x in range(1,max+1):
    for y in range(1,x+1):
        print(y, ' ', end='')
    print()
```

Listing 2.3 initializes the variable `max` with the value 8, followed by an outer `for` loop whose loop variable `x` ranges from 1 to `max+1`. The inner loop has a loop variable `y` that ranges from 1 to `x+1`, and the inner loop prints the value of `y`. The output of Listing 2.4 is here:

```
1
1 2
1 2 3
1 2 3 4
```

```
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
```

THE `SPLIT()` FUNCTION WITH FOR LOOPS

Python supports various useful string-related functions, including the `split()` function and the `join()` function. The `split()` function is useful when one wants to tokenize (“split”) a line of text into words and then use a `for` loop to iterate through those words and process them accordingly.

The `join()` function does the opposite of `split()`: It “joins” two or more words into a single line. One can easily remove extra spaces in a sentence by using the `split()` function and then invoking the `join()` function, thereby creating a line of text with one white space between any two words.

USING THE `SPLIT()` FUNCTION TO COMPARE WORDS

Listing 2.4 displays the contents of `compare2.py` that illustrates how to use the `split` function to compare each word in a text string with another word.

LISTING 2.4: `compare2.py`

```
x = 'This is a string that contains abc and Abc'
y = 'abc'
identical = 0
casematch = 0

for w in x.split():
    if(w == y):
        identical = identical + 1
    elif (w.lower() == y.lower()):
        casematch = casematch + 1

if(identical > 0):
    print('found identical matches:', identical)

if(casematch > 0):
    print('found case matches:', casematch)

if(casematch == 0 and identical == 0):
    print('no matches found')
```

Listing 2.4 uses the `split()` function in order to compare each word in the string `x` with the word `abc`. If there is an exact match, the variable `identical` is incremented. If a match does not occur, a case-insensitive match of the current word is performed with the string `abc`, and the variable `casematch` is incremented if the match is successful.

The output from Listing 2.5 is here:

```
found identical matches: 1
found case matches: 1
```

USING THE `SPLIT()` FUNCTION TO PRINT JUSTIFIED TEXT

Listing 2.5 displays the contents of `fixed_column_count1.py` that illustrates how to print a set of words from a text string as justified text using a fixed number of columns.

LISTING 2.5: `fixed_column_count1.py`

```
import string

wordCount = 0
str1 = 'this is a string with a set of words in it'

print('Left-justified strings:')
print('-----')
for w in str1.split():
    print('%-10s' % w)
    wordCount = wordCount + 1
    if(wordCount % 2 == 0):
        print("")
print("\n")

print('Right-justified strings:')
print('-----')

wordCount = 0
for w in str1.split():
    print('%10s' % w)
    wordCount = wordCount + 1
    if(wordCount % 2 == 0):
        print()
```

Listing 2.5 initializes the variables `wordCount` and `str1`, followed by two `for` loops. The first `for` loop prints the words in `str1` in left-justified format, and the second `for` loop prints the words in `str1` in right-justified format. In both loops, a linefeed is printed after a pair of consecutive words is printed, which occurs whenever the variable `wordCount` is even. The output from Listing 2.5 is here:

```
Left-justified strings:
-----
this      is
a         string
with      a
set       of
words     in
it
```

Right-justified strings:

```
-----
      this          is
        a      string
      with          a
        set        of
      words        in
        it
```

USING THE `SPLIT()` FUNCTION TO PRINT FIXED WIDTH TEXT

Listing 2.6 displays the contents of `fixed_column_width1.py` that illustrates how to print a text string in a column of fixed width.

LISTING 2.6: `fixed_column_width1.py`

```
import string

left = 0
right = 0
columnWidth = 8

str1 = 'this is a string with a set of words in it and it
will be split into a fixed column width'
strLen = len(str1)

print('Left-justified column:')
print('-----')
rowCount = int(strLen/columnWidth)

for i in range(0,rowCount):
    left = i*columnWidth
    right = (i+1)*columnWidth-1
    word = str1[left:right]
    print("%-10s" % word)

# check for a 'partial row'
if(rowCount*columnWidth < strLen):
    left = rowCount*columnWidth-1;
    right = strLen
    word = str1[left:right]
    print("%-10s" % word)
```

Listing 2.6 initializes the integer variable `columnWidth` and the string variable `str1`. The variable `strLen` is the length of `str1`, and `rowCount` is `strLen` divided by `columnWidth`.

The next part of Listing 2.6 contains a loop that prints `rowCount` rows of characters, where each row contains `columnWidth` characters. The final

portion of Listing 2.6 prints any “leftover” characters that comprise a partial row.

The newspaper-style output (but without any partial whitespace formatting) from Listing 2.6 is here:

```
Left-justified column:
-----
this is
a strin
  with a
set of
ords in
it and
t will
e split
into a
ixed co
umn wid
th
```

USING THE `SPLIT()` FUNCTION TO COMPARE TEXT STRINGS

Listing 2.7 displays the contents of `compare_strings1.py` that illustrates how to determine whether or not the words in one text string are also words in a second text string.

LISTING 2.7: `compare_strings1.py`

```
text1 = 'a b c d'
text2 = 'a b c e d'

if(text2.find(text1) >= 0):
    print('text1 is a substring of text2')
else:
    print('text1 is not a substring of text2')

subStr = True
for w in text1.split():
    if(text2.find(w) == -1):
        subStr = False
        break

if(subStr == True):
    print('Every word in text1 is a word in text2')
else:
    print('Not every word in text1 is a word in text2')
```

Listing 2.7 initializes the string variables `text1` and `text2`, and uses conditional logic to determine whether or not `text1` is a substring of `text2` (and then prints a suitable message).

The next part of Listing 2.7 is a loop that iterates through the words in the string `text1` and checks if each of those words is also a word in the string `text2`. If a nonmatch occurs, the variable `subStr` is set to “False,” followed by the break statement that causes an early exit from the loop. The final portion of Listing 2.7 prints the appropriate message based on the value of `subStr`. The output from Listing 2.7 is here:

```
text1 is not a substring of text2
Every word in text1 is a word in text2
```

USING THE `SPLIT()` FUNCTION TO DISPLAY CHARACTERS IN A STRING

Listing 2.8 displays the contents of `string_chars1.py` that illustrates how to print the characters in a text string.

LISTING 2.8: `string_chars1.py`

```
text = 'abcdef'
for ch in text:
    print('char:',ch,'ord value:',ord(ch))
print
```

Listing 2.8 is straightforward: a `for` loop iterates through the characters in the string `text` and then prints the character and its `ord` value. The output from Listing 2.8 is here:

```
('char:', 'a', 'ord value:', 97)
('char:', 'b', 'ord value:', 98)
('char:', 'c', 'ord value:', 99)
('char:', 'd', 'ord value:', 100)
('char:', 'e', 'ord value:', 101)
('char:', 'f', 'ord value:', 102)
```

THE `JOIN()` FUNCTION

Another way to remove extraneous spaces is to use the `join()` function, as shown here:

```
text1 = '   there are       extra   spaces   '
print('text1:',text1)

text2 = ' '.join(text1.split())
print('text2:',text2)

text2 = 'XYZ'.join(text1.split())
print('text2:',text2)
```

The `split()` function “splits” a text string into a set of words, and also removes the extraneous white spaces. Next, the `join()` function “joins” together the words in the string `text1`, using a single white space as the

delimiter. The last code portion of the preceding code block uses the string `XYZ` as the delimiter instead of a single white space.

The output of the preceding code block is here:

```
text1:   there are      extra   spaces
text2: there are extra spaces
text2: thereXYZareXYZextraXYZspaces
```

PYTHON WHILE LOOPS

One can define a `while` loop to iterate through a set of numbers, as shown in the following examples:

```
>>> x = 0
>>> while x < 5:
...     print(x)
...     x = x + 1
...
0
1
2
3
4
5
```

Python uses indentation instead of curly braces that are used in other languages such as JavaScript and Java. Although Python lists are not discussed until Chapter 3, readers can probably understand the following simple code block that contains a variant of the preceding loop that they can use when working with lists:

```
lst = [1,2,3,4]

while lst:
    print('list:',lst)
    print('item:',lst.pop())
```

The preceding `while` loop terminates when the `lst` variable is empty, and there is no need to explicitly test for an empty list. The output from the preceding code is here:

```
list: [1, 2, 3, 4]
item: 4
list: [1, 2, 3]
item: 3
list: [1, 2]
item: 2
list: [1]
item: 1
```

This concludes the examples that use the `split()` function in order to process words and characters in a text string. The next part of this chapter shows examples of using conditional logic in Python code.

CONDITIONAL LOGIC IN PYTHON

Anyone who has written code in other programming languages, has undoubtedly seen `if/then/else` (or `if-elseif-else`) conditional statements. Although the syntax varies between languages, the logic is essentially the same. The following example shows how to use `if/elif` statements in Python:

```
>>> x = 25
>>> if x < 0:
...     print('negative')
... elif x < 25:
...     print('under 25')
... elif x == 25:
...     print('exactly 25')
... else:
...     print('over 25')
...
exactly 25
```

The preceding code block illustrates how to use multiple conditional statements, and the output is exactly what one would expect.

THE BREAK/CONTINUE/PASS STATEMENTS

The `break` statement in Python enables users to perform an “early exit” from a loop, whereas the `continue` statement essentially returns to the top of the loop and continues with the next value of the loop variable. The `pass` statement is essentially a “do nothing” statement.

Listing 2.9 displays the contents of `break_continue_pass.py` that illustrates the use of these three statements.

LISTING 2.9: `break_continue_pass.py`

```
print('first loop')
for x in range(1,4):
    if(x == 2):
        break
    print(x)

print('second loop')
for x in range(1,4):
    if(x == 2):
        continue
    print(x)
```

```

print('third loop')
for x in range(1,4):
    if(x == 2):
        pass
    print(x)

```

The output of Listing 2.9 is here:

```

first loop
1
second loop
1
3
third loop
1
2
3

```

COMPARISON AND BOOLEAN OPERATORS

Python supports a variety of Boolean operators, such as `in`, `not in`, `is`, `is not`, `and`, `or`, and `not`. The next several sections discuss these operators and provide some examples of how to use them.

The `in/not in/is/is not` Comparison Operators

The `in` and `not in` operators are used with sequences to check whether a value occurs or does not occur in a sequence. The operators `is` and `is not` determine whether or not two objects are the same object, which is important only matters for mutable objects such as lists. All comparison operators have the same priority, which is lower than that of all numerical operators. Comparisons can also be chained. For example, `a < b == c` tests whether `a` is less than `b` and moreover `b` equals `c`.

The `and`, `or`, and `not` Boolean Operators

The Boolean operators `and`, `or`, and `not` have lower priority than comparison operators. The Boolean `and` and `or` are binary operators whereas the Boolean `not` operator is a unary operator. Here are some examples:

`A and B` can only be true if both `A` and `B` are true

`A or B` is true if either `A` or `B` is true

`not (A)` is true if and only if `A` is false

One can also assign the result of a comparison or other Boolean expression to a variable, as shown here:

```

>>> string1, string2, string3 = '', 'b', 'cd'
>>> str4 = string1 or string2 or string3
>>> str4
'b'

```

The preceding code block initializes the variables `string1`, `string2`, and `string3`, where `string1` is an empty string. Next, `str4` is initialized via the `or` operator, and since the first nonnull value is `string2`, the value of `str4` is equal to `string2`.

LOCAL AND GLOBAL VARIABLES

Python variables can be local or global. A Python variable is local to a function if the following are true:

- a parameter of the function
- on the left-side of a statement in the function
- bound to a control structure (such as `for`, `with`, and `except`)

A variable that is referenced in a function but is not local (according to the previous list) is a nonlocal variable. Users can specify a variable as nonlocal with this snippet:

```
nonlocal z
```

A variable can be explicitly declared as global with this statement:

```
global z
```

The following code block illustrates the behavior of a global versus a local variable:

```
global z
z = 3

def changeVar(z):
    z = 4
    print('z in function:', z)

print('first global z:', z)

if __name__ == '__main__':
    changeVar(z)
    print('second global z:', z)
```

The output from the preceding code block is here:

```
first global z: 3
z in function: 4
second global z: 3
```

Uninitialized Variables and the Value None

Python distinguishes between an uninitialized variable and the value `None`. The former is a variable that has not been assigned a value, whereas the value `None` is a value that indicates “no value.” Collections and methods often return the value `None`, and one can test for the value `None` in conditional logic.

SCOPE OF VARIABLES

The accessibility or scope of a variable depends on where that variable has been defined. Python provides two scopes: global and local, with the added “twist” that global is actually module-level scope (i.e., the current file), and therefore one can have a variable with the same name in different files and they will be treated differently.

Local variables are straightforward: they are defined inside a function, and they can only be accessed inside the function where they are defined. Any variables that are not local variables have global scope, which means that those variables are “global” *only* with respect to the file where it has been defined, and they can be accessed anywhere in a file.

There are two scenarios to consider regarding variables. First, suppose two files (aka modules) `file1.py` and `file2.py` have a variable called `x`, and `file1.py` also imports `file2.py`. The question now is how to disambiguate between the `x` in the two different modules. As an example, suppose that `file2.py` contains the following two lines of code:

```
x = 3
print('unscoped x in file2:',x)
```

Suppose that `file1.py` contains the following code:

```
import file2 as file2

x = 5
print('unscoped x in file1:',x)
print('scoped x from file2:',file2.x)
```

Launch `file1.y` from the command line and the following output will appear:

```
unscoped x in file2: 3
unscoped x in file1: 5
scoped x from file2: 3
```

The second scenario involves a program contains a local variable and a global variable with the same name. According to the earlier rule, the local variable is used in the function where it is defined, and the global variable is used outside of that function.

The following code block illustrates the use of a global and local variable with the same name:

```
#!/usr/bin/python
# a global variable:
total = 0;

def sum(x1, x2):
    # this total is local:
    total = x1+x2;
```

```

    print("Local total : ", total)
    return total

# invoke the sum function
sum(2,3);
print("Global total : ", total)

```

When the above code is executed, it produces following result:

```

Local total :    5
Global total :    0

```

What about unscoped variables, such as specifying the variable `x` without a module prefix? The answer consists of the following sequence of steps that Python will perform:

1. Check the local scope for the name.
2. Ascend the enclosing scopes and check for the name.
3. Perform step #2 until the global scope (i.e., module level).
4. If `x` still hasn't been found, Python checks `__builtins__`.

```

Python 2.7.5 (default, Dec  2 2013, 18:34:31)
[GCC 4.2.1 Compatible Apple LLVM 4.2 (clang-425.0.28)] on
darwin
Type "help", "copyright", "credits" or "license" for more
information.
>>> x = 1
>>> g = globals()
>>> g
{'g': {...}, '__builtins__': <module '__builtin__' (built-
in)>, '__package__': None, 'x': 1, '__name__': '__main__',
'__doc__': None}
>>> g.pop('x')
1
>>> x
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'x' is not defined

```

Note: One can access the dicts that Python uses to track local and global scope by invoking `locals()` and `globals()` respectively.

PASS BY REFERENCE VERSUS VALUE

All parameters (arguments) in the Python language are passed by reference. Thus, if one changes what a parameter refers to within a function, the change is reflected in the calling function. For example:

```

def changeme(mylist):
    #This changes a passed list into this function
    mylist.append([1,2,3,4])

```

```

print("Values inside the function: ", mylist)
return

# Now you can call changeme function
mylist = [10,20,30]
changeme(mylist)
print("Values outside the function: ", mylist)

```

When maintaining reference of the passed object and appending values in the same object, the result is shown here:

```

Values inside the function:  [10, 20, 30, [1, 2, 3, 4]]
Values outside the function:  [10, 20, 30, [1, 2, 3, 4]]

```

The fact that values are passed by reference gives rise to the notion of mutability versus immutability that is discussed in Chapter 3.

ARGUMENTS AND PARAMETERS

Python differentiates between arguments to functions and parameter declarations in functions: a positional (mandatory) and keyword (optional/default value). This concept is important because Python has operators for packing and unpacking these kinds of arguments.

Python unpacks positional arguments from an iterable, as shown here:

```

>>> def foo(x, y):
...     return x - y
...
>>> data = 4,5
>>> foo(data) # only passed one arg
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: foo() takes exactly 2 arguments (1 given)
>>> foo(*data) # passed however many args are in tuple
-1

```

USING A WHILE LOOP TO FIND THE DIVISORS OF A NUMBER

Listing 2.10 displays the contents of `divisors1.py` that contains a `while` loop, conditional logic, and the `%` (modulus) operator in order to find the factors of any integer greater than 1.

LISTING 2.10: `divisors.py`

```

def divisors(num):
    div = 2

    while(num > 1):
        if(num % div == 0):
            print("divisor: ", div)

```

```

        num = num / div
    else:
        div = div + 1
    print("*** finished ***")

```

```
divisors(12)
```

Listing 2.10 defines a function `divisors()` that takes an integer value `num` and then initializes the variable `div` with the value 2. The `while` loop divides `num` by `div` and if the remainder is 0, it prints the value of `div` and then it divides `num` by `div`; if the value is not 0, then `div` is incremented by 1. This `while` loop continues as long as the value of `num` is greater than 1.

The output from Listing 2.10 passing in the value 12 to the function `divisors()` is here:

```

divisor:  2
divisor:  2
divisor:  3
** finished **

```

Listing 2.11 displays the contents of `divisors2.py` that contains a `while` loop, conditional logic, and the `%` (modulus) operator in order to find the factors of any integer greater than 1.

LISTING 2.11: `divisors2.py`

```

def divisors(num):
    primes = ""
    div = 2

    while(num > 1):
        if(num % div == 0):
            divList = divList + str(div) + ' '
            num = num / div
        else:
            div = div + 1
    return divList

result = divisors(12)
print('The divisors of',12,'are:',result)

```

Listing 2.11 is very similar to Listing 2.10: the main difference is that Listing 2.10 constructs the variable `divList` (which is a concatenated list of the divisors of a number) in the `while` loop, and then returns the value of `divList` when the `while` loop has completed. The output from Listing 2.11 is here:

```
The divisors of 12 are: 2 2 3
```


Using a while loop to Find Prime Numbers

Listing 2.12 displays the contents of `divisors3.py` that contains a `while` loop, conditional logic, and the `%` (modulus) operator in order to count the number of prime factors of any integer greater than 1. If there is only one divisor for a number, then that number is a prime number.

LISTING 2.12: `divisors3.py`

```
def divisors(num):
    count = 1
    div = 2
    while(div < num):
        if(num % div == 0):
            count = count + 1
        div = div + 1
    return count

result = divisors(12)

if(result == 1):
    print('12 is prime')
else:
    print('12 is not prime')
```

USER-DEFINED FUNCTIONS IN PYTHON

Python provides built-in functions and also enables users to define their own functions. One can define functions to provide the required functionality. Here are simple rules to define a function in Python:

- Function blocks begin with the keyword `def` followed by the function name and parentheses.
- Any input arguments should be placed within these parentheses.
- The first statement of a function can be an optional statement - the documentation string of the function or docstring.
- The code block within every function starts with a colon (`:`) and is indented.
- The statement `return [expression]` exits a function, optionally passing back an expression to the caller. A return statement with no arguments is the same as `return None`.
- If a function does not specify return statement, the function automatically returns `None`, which is a special type of value in Python.

A very simple custom Python function is here:

```
>>> def func():
...     print 3
...
>>> func()
3
```

The preceding function is trivial, but it does illustrate the syntax for defining custom functions in Python. The following example is slightly more useful:

```
>>> def func(x):
...     for i in range(0,x):
...         print(i)
...
>>> func(5)
0
1
2
3
4
```

SPECIFYING DEFAULT VALUES IN A FUNCTION

Listing 2.13 displays the contents of `default_values.py` that illustrates how to specify default values in a function.

LISTING 2.13: `default_values.py`

```
def numberFunc(a, b=10):
    print (a,b)

def stringFunc(a, b='xyz'):
    print (a,b)

def collectionFunc(a, b=None):
    if(b is None):
        print('No value assigned to b')

numberFunc(3)
stringFunc('one')
collectionFunc([1,2,3])
```

Listing 2.13 defines three functions, followed by an invocation of each of those functions. The functions `numberFunc()` and `stringFunc()` print a list contain the values of their two parameters, and `collectionFunc()` displays a message if the second parameter is `None`. The output from Listing 2.13 is here:

```
(3, 10)
('one', 'xyz')
No value assigned to b
```

Returning Multiple Values From a Function

This task is accomplished by the code in Listing 2.14, which displays the contents of `multiple_values.py`.

LISTING 2.14: multiple_values.py

```
def MultipleValues():
    return 'a', 'b', 'c'

x, y, z = MultipleValues()

print('x:',x)
print('y:',y)
print('z:',z)
```

The output from Listing 2.14 is here:

```
x: a
y: b
z: c
```

FUNCTIONS WITH A VARIABLE NUMBER OF ARGUMENTS

Python enables users to define functions with a variable number of arguments. This functionality is useful in many situations, such as computing the sum, average, or product of a set of numbers. For example, the following code block computes the sum of two numbers:

```
def sum(a, b):
    return a + b

values = (1, 2)
s1 = sum(*values)
print('s1 = ', s1)
```

The output of the preceding code block is here:

```
s1 = 3
```

The `sum` function in the preceding code block can only be used for two numeric values.

Listing 2.15 displays the contents of `variable_sum1.py` that illustrates how to compute the sum of a variable number of numbers.

LISTING 2.15: variable_sum1.py

```
def sum(*values):
    sum = 0
    for x in values:
        sum = sum + x
    return sum

values1 = (1, 2)
s1 = sum(*values1)
print('s1 = ',s1)
```

```
values2 = (1, 2, 3, 4)
s2 = sum(*values2)
print('s2 = ', s2)
```

Listing 2.15 defines the function `sum` whose parameter values can be an arbitrary list of numbers. The next portion of this function initializes `sum` to 0, and then a `for` loop iterates through values and adds each of its elements to the variable `sum`. The last line in the function `sum()` returns the value of the variable `sum`. The output from Listing 2.15 is here:

```
s1 = 3
s2 = 10
```

LAMBDA EXPRESSIONS

Listing 2.16 displays the contents of `lambda1.py` that illustrates how to create a simple lambda expression in Python.

LISTING 2.16: `lambda1.py`

```
add = lambda x, y: x + y

x1 = add(5, 7)
x2 = add('Hello', 'Python')

print(x1)
print(x2)
```

Listing 2.16 defines the lambda expression `add` that accepts two input parameters and then returns their sum (for numbers) or their concatenation (for strings).

The output from Listing 2.16 is here:

```
12
HelloPython
```

RECURSION

Recursion is a powerful technique that can provide an elegant solution to various problems. The following subsections contain examples of using recursion to calculate some well-known numbers.

Calculating Factorial Values

The factorial value of a positive integer `n` is the product of all the integers between 1 and `n`. The symbol for factorial is the exclamation point (“!”) and some sample factorial values are here:

```
1! = 1
2! = 2
3! = 6
```

4! = 20
5! = 120

The formula for the factorial value of a number is succinctly defined as follows:

```
Factorial(n) = n*Factorial(n-1) for n > 1 and
Factorial(1) = 1
```

Listing 2.17 displays the contents of `factorial.py` that illustrates how to use recursion in order to calculate the factorial value of a positive integer.

LISTING 2.17: factorial.py

```
def factorial(num):
    if (num > 1):
        return num * factorial(num-1)
    else:
        return 1

result = factorial(5)
print('The factorial of 5 =', result)
```

Listing 2.17 contains the function `factorial` that implements the recursive definition of the factorial value of a number. The output from Listing 2.17 is here:

The factorial of 5 = 120

In addition to a recursive solution, there is also an iterative solution for calculating the factorial value of a number. Listing 2.18 displays the contents of `factorial2.py` that illustrates how to use the `range()` function in order to calculate the factorial value of a positive integer.

LISTING 2.18: factorial2.py

```
def factorial2(num):
    prod = 1
    for x in range(1,num+1):
        prod = prod * x
    return prod

result = factorial2(5)
print 'The factorial of 5 =', result
```

Listing 2.18 defines the function `factorial2()` with a parameter `num`, followed by the variable `prod` which has an initial value of 1. The next part of `factorial2()` is a for loop whose loop variable `x` ranges between 1 and `num+1`, and each iteration through that loop multiplies the value of `prod` with

the value of `x`, thereby computing the factorial value of `num`. The output from Listing 2.18 is here:

```
The factorial of 5 = 120
```

Calculating Fibonacci Numbers

The set of Fibonacci numbers represent some interesting patterns (such as the pattern of a sunflower) in nature, and its recursive definition is here:

```
Fib(0) = 0
Fib(1) = 1
Fib(n) = Fib(n-1) + Fib(n-2) for n >= 2
```

Listing 2.19 displays the contents of `fib.py` that illustrates how to calculate Fibonacci numbers.

LISTING 2.19: `fib.py`

```
def fib(num):
    if (num == 0):
        return 1
    elif (num == 1):
        return 1
    else:
        return fib(num-1) + fib(num-2)

result = fib(10)
print('Fibonacci value of 5 =', result)
```

Listing 2.19 defines the `fib()` function with the parameter `num`. If `num` equals 0 or 1 then `fib()` returns `num`; otherwise, `fib()` returns the result of adding `fib(num-1)` and `fib(num-2)`. The output from Listing 2.19 is here:

```
Fibonacci value of 10 = 89
```

Calculating the GCD of Two Numbers

The GCD (greatest common divisor) of two positive integers is the largest integer that divides both integers with a remainder of 0. Some values are shown here:

```
gcd(6,2)    = 2
gcd(10,4)   = 2
gcd(24,16)  = 8
```

Listing 2.20 uses recursion and Euclid's algorithm in order to find the GCD of two positive integers.

LISTING 2.20: gcd.py

```
def gcd(num1, num2):
    if(num1 % num2 == 0):
        return num2
    elif (num1 < num2):
        print("switching ", num1, " and ", num2)
        return gcd(num2, num1)
    else:
        print("reducing", num1, " and ", num2)
        return gcd(num1-num2, num2)

result = gcd(24, 10)
print("GCD of", 24, "and", 10, "=", result)
```

Listing 2.20 defines the function `gcd()` with the parameters `num1` and `num2`. If `num1` is divisible by `num2`, the function returns `num2`. If `num1` is less than `num2`, then the GCD is invoked by switching the order of `num1` and `num2`. In all other cases, `gcd()` returns the result of computing `gcd()` with the values `num1-num2` and `num2`. The output from Listing 2.20 is here:

```
reducing 24 and 10
reducing 14 and 10
switching 4 and 10
reducing 10 and 4
reducing 6 and 4
switching 2 and 4
GCD of 24 and 10 = 2
```

Calculating the LCM of Two Numbers

The LCM (lowest common multiple) of two positive integers is the smallest integer that is a multiple of those two integers. Some values are shown here:

```
lcm(6,2)    = 2
lcm(10,4)   = 20
lcm(24,16)  = 48
```

In general, if `x` and `y` are two positive integers, users can calculate their LCM as follows:

```
lcm(x,y) = x*y/gcd(x,y)
```

Listing 2.21 displays the contents of `lcm.py` that invokes the `gcd()` function that is defined in the previous section in order to calculate the LCM of two positive integers.

LISTING 2.21: lcm.py

```
def gcd(num1, num2):
    if(num1 % num2 == 0):
        return num2
```

```

elif (num1 < num2):
    #print("switching ", num1, " and ", num2)
    return gcd(num2, num1)
else:
    #print("reducing", num1, " and ", num2)
    return gcd(num1-num2, num2)

def lcm(num1, num2):
    gcd1 = gcd(num1, num2)
    lcm1 = num1*num2/gcd1
    return lcm1

result = lcm(24, 10)
print("The LCM of", 24, "and", 10, "=", result)

```

Listing 2.21 defines the function `gcd()` that was discussed in the previous section, followed by the function `lcm()` that takes the parameters `num1` and `num2`. The first line in `lcm()` computes `gcd1`, which is the `gcd()` of `num1` and `num2`. The second line in `lcm()` computes `lcm1`, which is `num1` multiplied by `num2` divided by `gcd1`. The third line in `lcm()` returns the value of `lcm1`. The output of Listing 2.21 is here:

```
The LCM of 24 and 10 = 120.0
```

SUMMARY

This chapter explained how to use condition logic, such as `if/elif` statement. Readers also learned how to work with loops in Python, including `for` loops and `while` loops. They learned how to compute various values, such as the GCD (greatest common divisor) and LCM (lowest common multiple) of a pair of numbers, and also how to determine whether or not a positive integer is prime.

PYTHON DATA STRUCTURES

Chapters 1 and 2 taught readers how to work with numbers and strings, as well as control structures in Python. In this chapter, readers will learn about Python collections, such as lists (or arrays), sets, tuples, and dictionaries. They will see many short code blocks that will help them rapidly learn how to work with these data structures in Python. After readers have finished this chapter, they will be in a better position to create more complex Python modules using one or more of these data structures.

The first part of this chapter discusses Python lists and shows code samples that illustrate various methods that are available for manipulating lists. The second part of this chapter discusses Python sets and how they differ from Python lists. The third part of this chapter discusses Python tuples, and the final part of this chapter discusses Python dictionaries.

Another detail to keep in mind is that this chapter contains Python code samples for counting uppercase and lowercase letters in a string, as well as code for a stack and a queue, all of which are generated by Claude 3.

One can use a different LLM (e.g., ChatGPT, Gemini, and so forth) if they wish, which is to say that that access to Claude 3 is not an absolute requirement. Note that although other LLMs will generate similar Python code, one might see differences in the generated code when compared with the Python code that was generated by Claude 3.

Incidentally, Chapter 6 is devoted to generating Python code via Claude 3.

WORKING WITH LISTS

Python supports a list data type, along with a rich set of list-related functions. Since lists are not typed, users can create a list of different data types, as well as multidimensional lists. The next several sections show how to manipulate list structures in Python.

Lists and Basic Operations

A Python list consists of comma-separated values enclosed in a pair of square brackets. The following examples illustrate the syntax for defining a list in Python, and also how to perform various operations on a Python list:

```
>>> list = [1, 2, 3, 4, 5]
>>> list
[1, 2, 3, 4, 5]
>>> list[2]
3
>>> list2 = list + [1, 2, 3, 4, 5]
>>> list2
[1, 2, 3, 4, 5, 1, 2, 3, 4, 5]
>>> list2.append(6)
>>> list2
[1, 2, 3, 4, 5, 1, 2, 3, 4, 5, 6]
>>> len(list)
5
>>> x = ['a', 'b', 'c']
>>> y = [1, 2, 3]
>>> z = [x, y]
>>> z[0]
['a', 'b', 'c']
>>> len(x)
3
```

One can assign multiple variables to a list, provided that the number and type of the variables match the structure. Here is an example:

```
>>> point = [7,8]
>>> x,y = point
>>> x
7
>>> y
8
```

The following example shows how to assign values to variables from a more complex data structure:

```
>>> line = ['a', 10, 20, (2014,01,31)]
>>> x1,x2,x3,date1 = line
>>> x1
'a'
>>> x2
10
>>> x3
20
>>> date1
(2014, 1, 31)
```

If one wants to access the year/month/date components of the `date1` element in the preceding code block, they can do so with the following code block:

```

>>> line = ['a', 10, 20, (2014,01,31)]
>>> x1,x2,x3,(year,month,day) = line
>>> x1
'a'
>>> x2
10
>>> x3
20
>>> year
2014
>>> month
1
>>> day
31

```

If the number and/or structure of the variables do not match the data, an error message is displayed, as shown here:

```

>>> point = (1,2)
>>> x,y,z = point
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: need more than 2 values to unpack

```

If the number of variables that you specify is less than the number of data items, an error message will appear, as shown here:

```

>>> line = ['a', 10, 20, (2014,01,31)]
>>> x1,x2 = line
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: too many values to unpack

```

Reversing and Sorting a List

The Python `reverse()` method reverses the contents of a list, as shown here:

```

>>> a = [4, 1, 2, 3]
>>> a.reverse()
[3, 2, 1, 4]

```

The Python `sort()` method sorts a list:

```

>>> a = [4, 1, 2, 3]
>>> a.sort()
[1, 2, 3, 4]

```

One can sort a list and then reverse its contents, as shown here:

```

>>> a = [4, 1, 2, 3]
>>> a.reverse(a.sort())
[4, 3, 2, 1]

```

Another way to reverse a list:

```
>>> L = [0,10,20,40]
>>> L[::-1]
[40, 20, 10, 0]
```

Keep in mind is that `reversed(array)` is an iterable and not a list. Although users can convert the reversed array to a list with this code snippet:

```
list(reversed(array)) or L[::-1]
```

Listing 3.1 contains a Python list consisting of four strings, followed by three so-called “list comprehensions” that are lists of strings.

LISTING 3.1: uppercase1.py

```
list1 = ['a', 'list', 'of', 'words']
list2 = [s.upper() for s in list1]
list3 = [s for s in list1 if len(s) <=2 ]
list4 = [s for s in list1 if 'w' in s ]

print('list1:',list1)
print('list2:',list2)
print('list3:',list3)
print('list4:',list4)
```

The output from launching the code in Listing 3.1 is here:

```
list1: ['a', 'list', 'of', 'words']
list2: ['A', 'LIST', 'OF', 'WORDS']
list3: ['a', 'of']
list4: ['words']
```

Lists and Arithmetic Operations

The minimum value of a list of numbers is the first number of in the sorted list of numbers. If one reverses the sorted list, the first number is the maximum value. There are several ways to reverse a list, starting with the technique shown in the following code:

```
x = [3,1,2,4]
maxList = x.sort()
minList = x.sort(x.reverse())

min1 = min(x)
max1 = max(x)
print min1
print max1
```

The output of the preceding code block is here:

```
1
4
```

A second (and better) way to sort a list is shown here:

```
minList = x.sort(reverse=True)
```

A third way to sort a list involves the built-in functional version of the `sort()` method, as shown here:

```
sorted(x, reverse=True)
```

The preceding code snippet is useful when one does not want to modify the original order of the list or they want to compose multiple list operations on a single line.

Lists and Filter-Related Operations

Python enables users to filter a list via list comprehensions (introduced earlier) as here:

```
mylist = [1, -2, 3, -5, 6, -7, 8]
pos = [n for n in mylist if n > 0]
neg = [n for n in mylist if n < 0]

print pos
print neg
```

Users can also specify `if/else` logic in a filter, as shown here:

```
mylist = [1, -2, 3, -5, 6, -7, 8]
negativeList = [n if n < 0 else 0 for n in mylist]
positiveList = [n if n > 0 else 0 for n in mylist]

print positiveList
print negativeList
```

The output of the preceding code block is here:

```
[1, 3, 6, 8]
[-2, -5, -7]
[1, 0, 3, 0, 6, 0, 8]
[0, -2, 0, -5, 0, -7, 0]
```

SORTING LISTS OF NUMBERS AND STRINGS

Listing 3.2 displays the contents of the Python script `sorted1.py` that determines whether or not two lists are sorted.

LISTING 3.2: `sorted1.py`

```
list1 = [1,2,3,4,5]
list2 = [2,1,3,4,5]
```

```

sort1 = sorted(list1)
sort2 = sorted(list2)

if(list1 == sort1):
    print(list1,'is sorted')
else:
    print(list1,'is not sorted')

if(list2 == sort2):
    print(list2,'is sorted')
else:
    print(list2,'is not sorted')

```

Listing 3.2 initializes the lists `list1` and `list2`, and the sorted lists `sort1` and `sort2` based on the lists `list1` and `list2`, respectively. If `list1` equals `sort1` then `list1` is already sorted; similarly, if `list2` equals `sort2` then `list2` is already sorted.

The output from Listing 3.2 is here:

```

[1, 2, 3, 4, 5] is sorted
[2, 1, 3, 4, 5] is not sorted

```

Note that if one sorts a list of character strings the output is case sensitive, and that uppercase letters appear before lowercase letters. This is due to the fact that the collating sequence for ASCII places uppercase letter (decimal 65 through decimal 90) before lowercase letters (decimal 97 through decimal 127). The following example provides an illustration:

```

>>> list1 = ['a', 'A', 'b', 'B', 'Z']
>>> print sorted(list1)
['A', 'B', 'Z', 'a', 'b']

```

One can also specify the reverse option so that the list is sorted in reverse order:

```

>>> list1 = ['a', 'A', 'b', 'B', 'Z']
>>> print sorted(list1, reverse=True)
['b', 'a', 'Z', 'B', 'A']

```

It's possible to sort a list based on the length of the items in the list:

```

>>> list1 = ['a', 'AA', 'bbb', 'BBBBB', 'ZZZZZZ']
>>> print sorted(list1, key=len)
['a', 'AA', 'bbb', 'BBBBB', 'ZZZZZZ']
>>> print sorted(list1, key=len, reverse=True)
['ZZZZZZ', 'BBBBB', 'bbb', 'AA', 'a']

```

Users can specify `str.lower` if you they treat uppercase letters as though they are lowercase letters during the sorting operation, as shown here:

```

>>> print sorted(list1, key=str.lower)
['a', 'AA', 'bbb', 'BBBBB', 'ZZZZZZ']

```

EXPRESSIONS IN LISTS

The following construct is similar to a for loop but without the colon “:” character that appears at the end of a loop construct. Consider the following example:

```
nums = [1, 2, 3, 4]
cubes = [ n*n*n for n in nums ]

print 'nums: ',nums
print 'cubes:',cubes
```

The output from the preceding code block is here:

```
nums:  [1, 2, 3, 4]
cubes: [1, 8, 27, 64]
```

CONCATENATING A LIST OF WORDS

Python provides the `join()` method for concatenating text strings, as shown here:

```
>>> parts = ['Is', 'SF', 'In', 'California?']
>>> ' '.join(parts)
'Is SF In California?'
>>> ','.join(parts)
'Is,SF,In,California?'
>>> ''.join(parts) 'IsSFInCalifornia?'
```

There are several ways to concatenate a set of strings and then print the result. The following is the most inefficient way to do so:

```
print "This" + " is" + " a" + " sentence"
```

Either of the following is preferred:

```
print "%s %s %s %s" % ("This", "is", "a", "sentence")
print " ".join(["This","is","a","sentence"])
```

THE BUBBLESORT IN PYTHON

The previous sections contains examples that illustrate how to sort a list of numbers using the `sort()` function. Sometimes one needs to implement different types of sorts in Python. Listing 3.3 displays the contents of `bubble_sort.py` that illustrates how to implement the bubble sort in Python.

LISTING 3.3: bubble_sort.py

```
list1 = [1, 5, 3, 4]

print("Initial list:",list1)
```

```

for i in range(0, len(list1)-1):
    for j in range(i+1, len(list1)):
        if(list1[i] > list1[j]):
            temp = list1[i]
            list1[i] = list1[j]
            list1[j] = temp

```

```
print("Sorted list: ", list1)
```

The output from Listing 3.3 is here:

```

Initial list: [1, 5, 3, 4]
Sorted list:  [1, 3, 4, 5]

```

THE PYTHON RANGE() FUNCTION

In this section readers will learn about the Python `range()` function that they can use to iterate through a list, as shown here:

```

>>> for i in range(0,5):
...     print i
...
0
1
2
3
4

```

One can use a `for` loop to iterate through a list of strings, as shown here:

```

>>> x
['a', 'b', 'c']
>>> for w in x:
...     print w
...
a
b
c

```

One can use a `for` loop to iterate through a list of strings and provide additional details, as shown here:

```

>>> x
['a', 'b', 'c']
>>> for w in x:
...     print len(w), w
...
1 a
1 b
1 c

```

The preceding output displays the length of each word in the list `x`, followed by the word itself.

Counting Digits, Uppercase, and Lowercase Letters

Listing 3.4 displays the contents of the Python file `count_char_types.py` that counts the occurrences of digits and letters in a string.

LISTING 3.4: `count_char_types.py`

```
str1 = "abc4234AFde"
digitCount = 0
alphaCount = 0
upperCount = 0
lowerCount = 0

for i in range(0,len(str1)):
    char = str1[i]
    if(char.isdigit()):
        #print("this is a digit:",char)
        digitCount += 1
    elif(char.isalpha()):
        #print("this is alphabetic:",char)
        alphaCount += 1
        if(char.upper() == char):
            upperCount += 1
        else:
            lowerCount += 1

print('Original String:  ',str1)
print('Number of digits:  ',digitCount)
print('Total alphanumeric:',alphaCount)
print('Upper Case Count:  ',upperCount)
print('Lower Case Count:  ',lowerCount)
```

Listing 3.4 initializes counter-related variables, followed by a loop (with loop variable `i`) that iterates from 0 to the length of the string `str1`. The string variable `char` is initialized with the letter at index `i` of the string `str1`. The next portion of the loop uses conditional logic to determine whether `char` is a digit or an alphabetic character; in the latter case, the code checks whether or not the character is uppercase or lowercase. In all cases, the values of the appropriate counter-related variables are incremented.

The output of Listing 3.4 is here:

```
Original String:  abc4234AFde
Number of digits:  4
Total alphanumeric: 7
Upper Case Count:  2
Lower Case Count:  5
```

Counting Digits and Other Elements With Claude 3

The following prompt was given to Claude 3.

Please generate Python code that counts the number of digits, upper case letters, and lower case letters in a randomly generated string, and provide an explanation of the Python code.

Claude 3:

LISTING 3.5: `claude3_count_types.py`

```
import random
import string

def generate_random_string(length):
    """
    Generates a random string of specified length.

    Args:
        length: The length of the string to generate.

    Returns:
        A random string of the specified length.
    """
    letters = string.ascii_letters + string.digits
    return ''.join(random.choice(letters) for _ in
range(length))

def count_digits_uppercase_lowercase(text):
    """
    Counts the number of digits, uppercase letters, and
lowercase letters in a string.

    Args:
        text: The string to analyze.

    Returns:
        A dictionary containing the counts of digits,
uppercase letters, and lowercase letters.
    """
    digit_count = 0
    uppercase_count = 0
    lowercase_count = 0
    for char in text:
        if char.isdigit():
            digit_count += 1
        elif char.isupper():
            uppercase_count += 1
```

```

        elif char.islower():
            lowercase_count += 1
    return {
        "digits": digit_count,
        "uppercase": uppercase_count,
        "lowercase": lowercase_count
    }

# Generate a random string
random_string = generate_random_string(10)

# Count the digits, uppercase letters, and lowercase
letters
counts = count_digits_uppercase_lowercase(random_string)

# Print the results
print("Random string:", random_string)
print("Digits:", counts["digits"])
print("Uppercase letters:", counts["uppercase"])
print("Lowercase letters:", counts["lowercase"])

```

Explanation:**Import Libraries:**

random library is used to generate random characters.

string library provides access to predefined sets of characters like alphanumeric characters.

Functions:

generate_random_string:

This function takes a length as input and generates a random string of that length.

It uses `string.ascii_letters` and `string.digits` to get all uppercase, lowercase letters, and digits.

It then uses a loop and `random.choice` to randomly select characters and join them into a string.

count_digits_uppercase_lowercase:

This function takes a text as input and returns a dictionary with counts for digits, uppercase, and lowercase letters.

It uses three counters initialized to 0: `digit_count`, `uppercase_count`, and `lowercase_count`.

It iterates through each character in the text string.

For each character, it uses `char.isdigit()`, `char.isupper()`, and `char.islower()` methods to check its type and increment the corresponding counter.

Finally, it returns a dictionary with the counts for each character type.

Code Execution:

`generate_random_string` is called with a length of 10 to create a random string.

`count_digits_uppercase_lowercase` is called with the generated string to count characters.

Finally, the results are printed, including the original string and counts for each character type.

Launch the code in Listing 3.5 and you will see the following output:

```
Random string: 4jv9UqpFYi
Digits: 2
Uppercase letters: 3
Lowercase letters: 5
```

ARRAYS AND THE `APPEND()` FUNCTION

Although Python does have an array type (`import array`), which is essentially a heterogeneous list, the array type has no advantages over the list type other than a slight saving in memory use. Users can also define heterogeneous arrays:

```
a = [10, 'hello', [5, '77']]
```

They can append a new element to an element inside a list:

```
>>> a = [10, 'hello', [5, '77']]
>>> a[2].append('abc')
>>> a
[10, 'hello', [5, '77', 'abc']]
```

They can assign simple variables to the elements of a list, as shown here:

```
myList = [ 'a', 'b', 91.1, (2014, 01, 31) ]
x1, x2, x3, x4 = myList
print 'x1:', x1
print 'x2:', x2
print 'x3:', x3
print 'x4:', x4
```

The output of the preceding code block is here:

```
x1: a
x2: b
x3: 91.1
x4: (2014, 1, 31)
```

The Python `split()` function is more convenient (especially when the number of elements is unknown or variable) than the preceding sample, and you will see examples of the `split()` function in the next section.

WORKING WITH LISTS AND THE `SPLIT()` FUNCTION

One can use the Python `split()` function to split the words in a text string and populate a list with those words. An example is here:

```
>>> x = "this is a string"
>>> list = x.split()
>>> list
['this', 'is', 'a', 'string']
```

A simple way to print the list of words in a text string is shown here:

```
>>> x = "this is a string"
>>> for w in x.split():
...     print w
...
this
is
a
string
```

Users can search for a word in a string as follows:

```
>>> x = "this is a string"
>>> for w in x.split():
...     if(w == 'this'):
...         print "x contains this"
...
x contains this
...
```

COUNTING WORDS IN A LIST

Python provides the `Counter` class that enables you to count the words in a list. Listing 3.6 displays the contents of `count_word2.py` that displays the top three words with greatest frequency.

LISTING 3.6: `count_word2.py`

```
from collections import Counter

mywords = ['a', 'b', 'a', 'b', 'c', 'a', 'd', 'e', 'f',
           'b']

word_counts = Counter(mywords)
topThree = word_counts.most_common(3)
print(topThree)
```

Listing 3.6 initializes the variable `mywords` with a set of characters and then initializes the variable `word_counts` by passing `mywords` as an argument to `Counter`. The variable `topThree` is an array containing the three most common characters (and their frequency) that appear in `mywords`. The output from Listing 3.6 is here:

```
[('a', 3), ('b', 3), ('c', 1)]
```

ITERATING THROUGH PAIRS OF LISTS

Python supports operations on pairs of lists, which means that users can perform vector-like operations. The following snippet multiplies every list element by 3:

```
>>> list1 = [1, 2, 3]
>>> [3*x for x in list1]
[3, 6, 9]
```

Create a new list with pairs of elements consisting of the original element and the original element multiplied by 3:

```
>>> list1 = [1, 2, 3]
>>> [[x, 3*x] for x in list1]
[[1, 3], [2, 6], [3, 9]]
```

Compute the product of every pair of numbers from two lists:

```
>>> list1 = [1, 2, 3]
>>> list2 = [5, 6, 7]
>>> [a*b for a in list1 for b in list2]
[5, 6, 7, 10, 12, 14, 15, 18, 21]
```

Calculate the sum of every pair of numbers from two lists:

```
>>> list1 = [1, 2, 3]
>>> list2 = [5, 6, 7]
>>> [a+b for a in list1 for b in list2]
[6, 7, 8, 7, 8, 9, 8, 9, 10]
```

Calculate the pair-wise product of two lists:

```
>>> [list1[i]*list2[i] for i in range(len(list1))]
[8, 12, -54]
```

OTHER LIST-RELATED FUNCTIONS

Python provides additional functions that can be used with lists, such as `append()`, `insert()`, `delete()`, `pop()`, and `extend()`. Python also supports the functions `index()`, `count()`, `sort()`, and `reverse()`. Examples of these functions are illustrated in the following code block.

Define a Python list (notice that duplicates are allowed):

```
>>> a = [1, 2, 3, 2, 4, 2, 5]
```

Display the number of occurrences of 1 and 2:

```
>>> print a.count(1), a.count(2)
1 3
```

Insert -8 in position 3:

```
>>> a.insert(3,-8)
>>> a
[1, 2, 3, -8, 2, 4, 2, 5]
```

Remove occurrences of 3:

```
>>> a.remove(3)
>>> a
[1, 2, -8, 2, 4, 2, 5]
```

Remove occurrences of 1:

```
>>> a.remove(1)
>>> a
[2, -8, 2, 4, 2, 5]
```

Append 19 to the list:

```
>>> a.append(19)
>>> a
[2, -8, 2, 4, 2, 5, 19]
```

Print the index of 19 in the list:

```
>>> a.index(19)
6
```

Reverse the list:

```
>>> a.reverse()
>>> a
[19, 5, 2, 4, 2, -8, 2]
```

Sort the list:

```
>>> a.sort()
>>> a
[-8, 2, 2, 2, 4, 5, 19]
```

Extend list a with list b:

```
>>> b = [100,200,300]
>>> a.extend(b)
>>> a
[-8, 2, 2, 2, 4, 5, 19, 100, 200, 300]
```

Remove the first occurrence of 2:

```
>>> a.pop(2)
2
>>> a
[-8, 2, 2, 4, 5, 19, 100, 200, 300]
```

Remove the last item of the list:

```
>>> a.pop()
300
>>> a
[-8, 2, 2, 4, 5, 19, 100, 200]
```

CREATE A STACK USING CLAUDE 3

The following prompt was given to Claude 3.

Please generate a Python code sample that implements a stack for floating point number, along with an explanation of the code.

Claude 3:

LISTING 3.7: `claude3_stack.py`

```
import random

class RandomNumberStack:
    def __init__(self, size):
        self.stack = []
        self.size = size

    def push(self, number):
        if len(self.stack) >= self.size:
            raise OverflowError("Stack is full!")
        self.stack.append(number)

    def pop(self):
        if len(self.stack) == 0:
            raise ValueError("Stack is empty!")
        return self.stack.pop()

    def peek(self):
        if len(self.stack) == 0:
            raise ValueError("Stack is empty!")
        return self.stack[-1]

    def is_empty(self):
        return len(self.stack) == 0

    #def size(self):
    def stack_size(self):
        return len(self.stack)

    def generate_random_numbers(self, n):
        for _ in range(n):
            self.push(random.randint(1, 100))
```



```

def __str__(self):
    return str(self.stack)

# Example usage
stack = RandomNumberStack(5)

# Generate 3 random numbers and push them to the stack
stack.generate_random_numbers(3)

print("Stack after pushing:", stack)

# Pop and peek at the top element
popped_element = stack.pop()
peeked_element = stack.peek()

print("Popped element:", popped_element)
print("Peeked element:", peeked_element)

print("Stack after popping:", stack)
print("Is stack empty:", stack.is_empty())
#print("Stack size:", stack.size())
print("Stack size:", stack.stack_size())

```

Explanation:**RandomNumberStack class:**

This class defines a stack data structure that can hold a random set of numbers.

The `__init__` method initializes the stack with a specified size.

The push method adds an element to the top of the stack, checking for overflow.

The pop method removes and returns the element from the top of the stack, checking for underflow.

The peek method returns the element at the top of the stack without removing it.

The `is_empty` method returns True if the stack is empty and False otherwise.

The size method returns the number of elements in the stack.

The `generate_random_numbers` method pushes a specified number of random integers to the stack.

The `__str__` method returns a string representation of the stack.

Example usage:

This code creates an instance of the `RandomNumberStack` class with a size of 5.

It then generates three random numbers and pushes them to the stack.

The script pops one element and peeks at the top element, demonstrating the pop and peek methods.

Finally, it prints the remaining contents of the stack, checks if it's empty, and checks its size.

Launch the code in Listing 3.7 and you will see the following output:

```
Stack after pushing: [21, 12, 34]
Popped element: 34
Peeked element: 12
Stack after popping: [21, 12]
Is stack empty: False
Stack size: 2
```

The next section shows how to work with vectors in Python.

WORKING WITH VECTORS

A vector is a one-dimensional array of values, and you can perform vector-based operations, such as addition, subtraction, and inner product. Listing 3.8 displays the contents of `my_vectors.py` that illustrates how to perform vector-based operations.

LISTING 3.8: `my_vectors.py`

```
v1 = [1,2,3]
v2 = [1,2,3]
v3 = [5,5,5]

s1 = [0,0,0]
d1 = [0,0,0]
p1 = 0

print("Initial Vectors")
print('v1:',v1)
print('v2:',v2)
print('v3:',v3)

for i in range(len(v1)):
    d1[i] = v3[i] - v2[i]
    s1[i] = v3[i] + v2[i]
    p1    = v3[i] * v2[i] + p1

print("After operations")
print('d1:',d1)
```

```
print('s1:',s1)
print('p1:',p1)
```

Listing 3.8 starts with the definition of three lists in Python, each of which represents a vector. The lists `d1` and `s1` represent the difference of `v2` and the sum `v2`, respectively. The number `p1` represents the “inner product” (also called the “dot product”) of `v3` and `v2`. The output from Listing 3.6 is here:

```
Initial Vectors
v1: [1, 2, 3]
v2: [1, 2, 3]
v3: [5, 5, 5]
After operations
d1: [4, 3, 2]
s1: [6, 7, 8]
p1: 30
```

WORKING WITH MATRICES

A two-dimensional matrix is a two-dimensional array of values, and one can easily create such a matrix. For example, the following code block illustrates how to access different elements in a 2D matrix:

```
mm = [["a", "b", "c"], ["d", "e", "f"], ["g", "h", "i"]];
print('mm: ',mm)
print('mm[0]: ',mm[0])
print('mm[0][1]: ',mm[0][1])
```

The output from the preceding code block is here:

```
mm:          [['a', 'b', 'c'], ['d', 'e', 'f'], ['g', 'h', 'i']]
mm[0]:       ['a', 'b', 'c']
mm[0][1]: b
```

Listing 3.9 displays the contents of `my2D_matrix.py` that illustrates how to create and populate 2 two-dimensional matrices.

LISTING 3.9: my2D_matrix.py

```
rows = 3
cols = 3

my2DMatrix = [[0 for i in range(rows)] for j in
range(rows)]
print('Before:',my2DMatrix)

for row in range(rows):
    for col in range(cols):
        my2DMatrix[row][col] = row*row+col*col
print('After: ',my2DMatrix)
```

Listing 3.9 initializes the variables `rows` and `cols` and then uses them to create the `rows x cols` matrix `my2DMatrix` whose values are initially 0. The next part of Listing 3.9 contains a nested loop that initializes the element of `my2DMatrix` whose position is `(row, col)` with the value `row*row+col*col`. The last line of code in Listing 3.9 prints the contents of `my2DArray`. The output from Listing 3.9 is here:

```
Before: [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
After:  [[0, 1, 4], [1, 2, 5], [4, 5, 8]]
```

THE NUMPY LIBRARY FOR MATRICES

The NumPy library (which you can install via `pip`) has a matrix object for manipulating matrices in Python. The following examples illustrate some of the features of NumPy.

Initialize a matrix `m` and then display its contents:

```
>>> import numpy as np
>>> m = np.matrix([[1,-2,3],[0,4,5],[7,8,-9]])
>>> m
matrix([[ 1, -2,  3],
        [ 0,  4,  5],
        [ 7,  8, -9]])
```

The next snippet returns the transpose of matrix `m`:

```
>>> m.T
matrix([[ 1,  0,  7],
        [-2,  4,  8],
        [ 3,  5, -9]])
```

The next snippet returns the inverse of matrix `m` (if it exists):

```
>>> m.I
matrix([[ 0.33043478, -0.02608696,  0.09565217],
        [-0.15217391,  0.13043478,  0.02173913],
        [ 0.12173913,  0.09565217, -0.0173913 ]])
```

The next snippet defines a vector `y` and then computes the product `m*v`:

```
>>> v = np.matrix([[2],[3],[4]])
>>> v
matrix([[2],[3],[4]])
>>> m * v
matrix([[ 8],[32],[ 2]])
```

The next snippet imports the `numpy.linalg` subpackage and then computes the determinant of the matrix `m`:

```
>>> import numpy.linalg
>>> numpy.linalg.det(m)
-229.99999999999983
```

The next snippet finds the eigenvalues of the matrix `m`:

```
>>> numpy.linalg.eigvals(m)
array([-13.11474312,  2.75956154,  6.35518158])
```

The next snippet finds solutions to the equation $m \cdot x = v$:

```
>>> x = numpy.linalg.solve(m, v)
>>> x
matrix([[ 0.96521739],
        [ 0.17391304],
        [ 0.46086957]])
```

In addition to the preceding samples, the NumPy package provides additional functionality, which you can find by performing an Internet search for articles and tutorials.

QUEUES

A queue is a FIFO (“First In First Out”) data structure. Thus, the oldest item in a queue is removed when a new item is added to a queue that is already full.

Earlier in the chapter readers learned how to use a Python List to emulate a queue. There is also a queue object in Python. The following code snippets illustrate how to use a queue in Python.

```
>>> from collections import deque
>>> q = deque('', maxlen=10)
>>> for i in range(10,20):
...     q.append(i)
...
>>> print q
deque([10, 11, 12, 13, 14, 15, 16, 17, 18, 19], maxlen=10)
```

CREATE A QUEUE USING CLAUDE 3

The following prompt was given to Claude 3.

Please generate Python code for a queue that involves a random set of numbers, along with an explanation of the Python code
 Claude 3:

LISTING 3.10: `claude_queue.py`

```
import random

class RandomNumberQueue:
    def __init__(self, max_size):
```

```

        self.queue = []
        self.max_size = max_size

    def enqueue(self, number):
        if len(self.queue) >= self.max_size:
            raise OverflowError("Queue is full!")
        self.queue.append(number)

    def dequeue(self):
        if len(self.queue) == 0:
            raise ValueError("Queue is empty!")
        return self.queue.pop(0)

    def is_empty(self):
        return len(self.queue) == 0

    def size(self):
        return len(self.queue)

    def generate_random_numbers(self, n):
        for _ in range(n):
            self.enqueue(random.randint(1, 100))

    def __str__(self):
        return str(self.queue)

# Example usage
queue = RandomNumberQueue(5)

# Generate 3 random numbers and enqueue them
queue.generate_random_numbers(3)

print("Queue after enqueue:", queue)

# Dequeue one element and check if the queue is empty
dequeued_element = queue.dequeue()

print("Dequeued element:", dequeued_element)
print("Queue is empty:", queue.is_empty())

print("Queue after dequeue:", queue)
print("Queue size:", queue.size())

```

Explanation:

RandomNumberQueue class:

This class defines a queue data structure that can hold a random set of numbers.

The `__init__` method initializes the queue with a specified maximum size.

The `enqueue` method adds an element to the back of the queue, checking for overflow.

The `dequeue` method removes and returns the element from the front of the queue, checking for underflow.

The `is_empty` method returns “True” if the queue is empty and “False” otherwise.

The `size` method returns the number of elements in the queue.

The `generate_random_numbers` method enqueues a specified number of random integers to the queue.

The `__str__` method returns a string representation of the queue.

Example usage:

This code creates an instance of the `RandomNumberQueue` class with a maximum size of 5.

It then generates three random numbers and enqueues them to the queue.

The script dequeues one element and checks if the queue is empty, demonstrating the `dequeue` and `is_empty` methods.

Finally, it prints the remaining contents of the queue and checks its size.

Launch the code in Listing 3.10 and you will see the following output:

```
Queue after enqueue: [77, 78, 54]
Dequeued element: 77
Queue is empty: False
Queue after dequeue: [78, 54]
Queue size: 2
```

USING A LIST AS A STACK AND A QUEUE

A stack is a LIFO (“Last In First Out”) data structure with `push()` and `pop()` functions for adding and removing elements, respectively. The most recently added element in a stack is in the top position, and therefore the first element that can be removed from the stack.

The following code block illustrates how to create a stack and also remove and append items from a stack in Python. Create a Python list (which will be used as a stack):

```
>>> s = [1, 2, 3, 4]
```

Append 5 to the stack:

```
>>> s.append(5)
>>> s
[1, 2, 3, 4, 5]
```

Remove the last element from the stack:

```
>>> s.pop()
5
>>> s
[1, 2, 3, 4]
```

A queue is a FIFO (“First In First Out”) data structure with `insert()` and `pop()` functions for inserting and removing elements, respectively. The most recently added element in a queue is in the top position, and therefore the last element that can be removed from the queue.

The following code block illustrates how to create a queue and also insert and append items to a queue in Python.

Create a Python list (which will be used as a queue):

```
>>> q = [1, 2, 3, 4]
```

Insert 5 at the beginning of the queue:

```
>>> q.insert(0, 5)
>>> q
[5, 1, 2, 3, 4]
```

Remove the last element from the queue:

```
>>> q.pop()
1
>>> q
[5, 2, 3, 4]
```

The preceding code uses `q.insert(0, 5)` to insert in the beginning and `q.pop()` to remove from the end. It’s important to keep in mind that the `insert()` operation is slow in Python: insert at 0 requires copying all the elements in underlying array down one space. Therefore, use `collections.deque` with `coll.appendleft()` and `coll.pop()`, where `coll` is an instance of the `Collection` class.

The next section shows how to use tuples in Python.

TUPLES (IMMUTABLE LISTS)

Python supports a data type called a *tuple* that consists of comma-separated values without brackets (square brackets are for lists, round brackets are for arrays, and curly braces are for dictionaries). Various examples of Python tuples can be found here:

<https://docs.python.org/3.6/tutorial/datastructures.html#tuples-and-sequences>

The following code block illustrates how to create a tuple and create new tuples from an existing type in Python.

Define a Python tuple `t` as follows:

```
>>> t = 1, 'a', 2, 'hello', 3
>>> t
(1, 'a', 2, 'hello', 3)
```

Display the first element of `t`:

```
>>> t[0]
1
```

Create a tuple `v` containing 10, 11, and `t`:

```
>>> v = 10, 11, t
>>> v
(10, 11, (1, 'a', 2, 'hello', 3))
```

Try modifying an element of `t` (which is immutable):

```
>>> t[0] = 1000
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'tuple' object does not support item assignment
```

Python “deduplication” is useful because one can remove duplicates from a set and obtain a list, as shown here:

```
>>> lst = list(set(lst))
```

Note: The “in” operator on a list to search is $O(n)$ whereas the “in” operator on set is $O(1)$.

The next section discusses Python sets.

WORKING WITH SETS

A Python set is an unordered collection that does not contain duplicate elements. Use curly braces or the `set()` function to create sets. Set objects support set-theoretic operations such as union, intersection, and difference.

Note: `set()` is required in order to create an empty set because `{}` creates an empty dictionary.

The following code block illustrates how to work with a Python set.

Create a list of elements:

```
>>> l = ['a', 'b', 'a', 'c']
```

Create a set from the preceding list:

```
>>> s = set(l)
>>> s
set(['a', 'c', 'b'])
```

Test if an element is in the set:

```
>>> 'a' in s
True
>>> 'd' in s
False
>>>
```

Create a set from a string:

```
>>> n = set('abacad')
>>> n
set(['a', 'c', 'b', 'd'])
>>>
```

Subtract n from s:

```
>>> s - n
set([])
```

Subtract s from n:

```
>>> n - s
set(['d'])
>>>
```

The union of s and n:

```
>>> s | n
set(['a', 'c', 'b', 'd'])
```

The intersection of s and n:

```
>>> s & n
set(['a', 'c', 'b'])
```

The exclusive-or of s and n:

```
>>> s ^ n
set(['d'])
```

The next section shows how to work with Python dictionaries.

DICTIONARIES

Python has a key/value structure called a “dict” that is a hash table. A Python dictionary (and hash tables in general) can retrieve the value of a key in constant

time, regardless of the number of entries in the dictionary (and the same is true for sets). One can think of a set as essentially just the keys (not the values) of a dict implementation.

The contents of a dict can be written as a series of key:value pairs, as shown here:

```
dict1 = {key1:value1, key2:value2, ... }
```

The “empty dict” is just an empty pair of curly braces {}.

Creating a Dictionary

A Python dictionary (or hash table) contains of colon-separated key/value bindings inside a pair of curly braces, as shown here:

```
dict1 = {}
dict1 = {'x' : 1, 'y' : 2}
```

The preceding code snippet defines dict1 as an empty dictionary, and then adds two key/value bindings.

Displaying the Contents of a Dictionary

One can display the contents of dict1 with the following code:

```
>>> dict1 = {'x':1,'y':2}
>>> dict1
{'y': 2, 'x': 1}
>>> dict1['x']
1
>>> dict1['y']
2
>>> dict1['z']
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
KeyError: 'z'
```

Note: Key/value bindings for a dict and a set are not necessarily stored in the same order that they were defined.

Python dictionaries also provide the get method in order to retrieve key values:

```
>>> dict1.get('x')
1
>>> dict1.get('y')
2
>>> dict1.get('z')
```

As you can see, the Python get() method returns None (which is displayed as an empty string) instead of an error when referencing a key that is not defined in a dictionary. You can also use dict comprehensions to create dictionaries from expressions, as shown here:

```
>>> {x: x**3 for x in (1, 2, 3)}
{1: 1, 2: 8, 3: 27}
```

Checking for Keys in a Dictionary

You can easily check for the presence of a key in a Python dictionary as follows:

```
>>> 'x' in dict1
True
>>> 'z' in dict1
False
```

Use square brackets for finding or setting a value in a dictionary. For example, `dict['abc']` finds the value associated with the key `'abc'`. One can use strings, numbers, and tuples work as key values, and you can use any type as the value.

If you access a value that is not in the dict, Python throws a `KeyError`. Consequently, use the “in” operator to check if the key is in the dict. Alternatively, use `dict.get(key)` which returns the value or `None` if the key is not present. One can even use the expression `get(key, not-found-string)` to specify the value to return if a key is not found.

Deleting Keys From a Dictionary

Launch the Python interpreter and enter the following commands:

```
>>> MyDict = {'x' : 5, 'y' : 7}
>>> MyDict['z'] = 13
>>> MyDict
{'y': 7, 'x': 5, 'z': 13}
>>> del MyDict['x']
>>> MyDict
{'y': 7, 'z': 13}
>>> MyDict.keys()
['y', 'z']
>>> MyDict.values()
[7, 13]
>>> 'z' in MyDict
True
```

Iterating Through a Dictionary

The following code snippet shows users how to iterate through a dictionary:

```
MyDict = {'x' : 5, 'y' : 7, 'z' : 13}

for key, value in MyDict.items():
    print key, value
```

The output from the preceding code block is here:

```
y 7
x 5
z 13
```

Interpolating Data From a Dictionary

The `%` operator substitutes values from a Python dictionary into a string by name. Listing 3.11 contains an example of doing so.

LISTING 3.11: interpolate_dict1.py

```
hash = {}
hash['beverage'] = 'coffee'
hash['count'] = 3

# %d for int, %s for string
s = 'Today I drank %(count)d cups of %(beverage)s' % hash
print('s:', s)
```

The output from the preceding code block is here:

```
Today I drank 3 cups of coffee
```

DICTIONARY FUNCTIONS AND METHODS

Python provides various functions and methods for a Python dictionary, such as `cmp()`, `len()`, and `str()` that compare two dictionaries, return the length of a dictionary, and display a string representation of a dictionary, respectively.

Users can also manipulate the contents of a Python dictionary using the functions `clear()` to remove all elements, `copy()` to return a shall copy, `get()` to retrieve the value of a key, `items()` to display the (key,value) pairs of a dictionary, `keys()` to displays the keys of a dictionary, and `values()` to return the list of values of a dictionary.

DICTIONARY FORMATTING

The `%` operator works conveniently to substitute values from a dict into a string by name:

```
#create a dictionary
>>> h = {}
#add a key/value pair
>>> h['item'] = 'beer'
>>> h['count'] = 4
#interpolate using %d for int, %s for string
>>> s = 'I want %(count)d bottles of %(item)s' % h
>>> s
'I want 4 bottles of beer'
```

The next section shows how to create an ordered Python dictionary.

ORDERED DICTIONARIES

Regular Python dictionaries iterate over key/value pairs in arbitrary order. Python 2.7 introduced a new `OrderedDict` class in the `collections` module. The `OrderedDict` API provides the same interface as regular dictionaries but iterates over keys and values in a guaranteed order depending on when a key was first inserted:

```
>>> from collections import OrderedDict
>>> d = OrderedDict([('first', 1),
...                  ('second', 2),
...                  ('third', 3)])
>>> d.items()
[('first', 1), ('second', 2), ('third', 3)]
```

If a new entry overwrites an existing entry, the original insertion position is left unchanged:

```
>>> d['second'] = 4
>>> d.items()
[('first', 1), ('second', 4), ('third', 3)]
```

Deleting an entry and reinserting it will move it to the end:

```
>>> del d['second']
>>> d['second'] = 5
>>> d.items()
[('first', 1), ('third', 3), ('second', 5)]
```

Sorting Dictionaries

Python enables you to support the entries in a dictionary. For example, one can modify the code in the preceding section to display the alphabetically sorted words and their associated word count.

Python Multidictionaries

Users can define entries in a Python dictionary so that the entries reference lists or other types of Python structures. Listing 3.12 displays the contents of `multi_dictionary1.py` that illustrates how to define more complex dictionaries.

LISTING 3.12: `multi_dictionary1.py`

```
from collections import defaultdict

d = {'a' : [1, 2, 3], 'b' : [4, 5]}
print 'firsts:',d

d = defaultdict(list)
```

```

d['a'].append(1)
d['a'].append(2)
d['b'].append(4)
print 'second:',d

d = defaultdict(set)
d['a'].add(1)
d['a'].add(2)
d['b'].add(4)
print 'third:',d

```

Listing 3.12 starts by defining the dictionary `d` and printing its contents. The next portion of Listing 3.12 specifies a list-oriented dictionary, and then modifies the values for the keys `a` and `b`. The final portion of Listing 3.12 specifies a set-oriented dictionary, and then modifies the values for the keys `a` and `b` as well.

The output from Listing 3.12 is here:

```

first: {'a': [1, 2, 3], 'b': [4, 5]}
second: defaultdict(<type 'list'>, {'a': [1, 2], 'b': [4]})
third: defaultdict(<type 'set'>, {'a': set([1, 2]), 'b':
set([4])})

```

The next section discusses other Python sequence types that have not been discussed in previous sections of this chapter.

OTHER SEQUENCE TYPES IN PYTHON

Python supports 7 sequence types: `str`, `unicode`, `list`, `tuple`, `bytearray`, `buffer`, and `xrange`.

One can iterate through a sequence and retrieve the position index and corresponding value at the same time using the `enumerate()` function.

```

>>> for i, v in enumerate(['x', 'y', 'z']):
...     print i, v
...
0 x
1 y
2 z

```

`Bytearray` objects are created with the built-in function `bytearray()`. Although `buffer` objects are not directly supported by Python syntax, users can create them via the built-in `buffer()` function.

Objects of type `xrange` are created with the `xrange()` function. An `xrange` object is similar to a `buffer` in the sense that there is no specific syntax to create them. Moreover, `xrange` objects do not support operations such as slicing, concatenation, or repetition.

At this point readers have seen all the Python types that they will encounter in the remaining chapters of this book, so it makes sense to discuss mutable and immutable types in Python, which is the topic of the next section.

MUTABLE AND IMMUTABLE TYPES IN PYTHON

Python represents its data as objects. Some of these objects (such as lists and dictionaries) are mutable, which means one can change their content without changing their identity. Objects such as integers, floats, strings, and tuples are objects that cannot be changed. The key point to understand is the difference between changing the value versus assigning a new value to an object; one cannot change a string but they can assign it a different value. This detail can be verified by checking the `id` value of an object, as shown in Listing 3.13.

LISTING 3.13: `mutability.py`

```
s = "abc"
print('id #1:', id(s))
print('first char:', s[0])

try:
    s[0] = "o"
except:
    print('Cannot perform reassignment')

s = "xyz"
print('id #2:', id(s))
s += "uvw"
print('id #3:', id(s))
```

The output of Listing 3.13 is here:

```
id #1: 4297972672
first char: a
Cannot perform reassignment
id #2: 4299809336
id #3: 4299777872
```

Thus, a Python type is immutable if its value cannot be changed (even though it's possible to assign a new value to such a type), otherwise a Python type is mutable. The Python immutable objects are of type `bytes`, `complex`, `float`, `int`, `str`, or `tuple`. Alternatively, dictionaries, lists, and sets are mutable. The key in a hash table must be an immutable type.

Since strings are immutable in Python, one cannot insert a string in the “middle” of a given text string unless they construct a second string using concatenation. For example, suppose one has the string:

```
"this is a string"
```


and you want to create the following string:

```
"this is a longer string"
```

The following Python code block illustrates how to perform this task:

```
text1 = "this is a string"
text2 = text1[0:10] + "longer" + text1[9:]
print 'text1:',text1
print 'text2:',text2
```

The output of the preceding code block is here:

```
text1: this is a string
text2: this is a longer string
```

THE `TYPE()` FUNCTION

The `type()` primitive returns the type of any object, including Python primitives, functions, and user-defined objects. The following code sample displays the type of an integer and a string:

```
var1 = 123
var2 = 456.78
print "type var1: ",type(var1)
print "type var2: ",type(var2)
```

The output of the preceding code block is here:

```
type var1: <type 'int'>
type var2: <type 'float'>
```

SUMMARY

This chapter showed how to work with various Python data structures. In particular, readers learned about tuples, sets, and dictionaries. Next, they learned how to work with lists and how to use list-related operations to extract sublists. In addition, readers learned how to work with dictionaries, tuples, and sets in Python. Moreover, they saw several prompts that were submitted to Claude 3 in order to generate Python code samples for counting uppercase and lowercase letters in a string, as well as code for a stack and a queue.

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INTRODUCTION TO NUMPY AND PANDAS

Approximately 50% of this chapter starts with a quick introduction to the Python NumPy package, followed by a quick introduction to Pandas and some of its useful features. The Pandas package for Python provides a rich and powerful set of APIs for managing datasets. These APIs are very useful for machine learning and deep learning tasks that involve dynamically “slicing and dicing” subsets of datasets.

The first section contains examples of working arrays in NumPy, and contrasts some of the APIs for lists with the same APIs for arrays. In addition, you will see how easy it is to compute the exponent-related values (square, cube, and so forth) of elements in an array.

The second section introduces subranges, which are very useful (and frequently used) for extracting portions of datasets in machine learning tasks. In particular, readers will see code samples that handle negative (-1) subranges for vectors as well as for arrays, because they are interpreted one way for vectors and a different way for arrays.

The third section (“Other Useful NumPy Methods”) of this chapter delves into other NumPy methods, including the `reshape()` method, which is extremely useful (and very common) when working with images files: some TensorFlow APIs require converting a 2D array of (R, G, B) values into a corresponding one-dimensional vector.

The fourth section of this chapter briefly describes Pandas and some of its useful features. This section contains code samples that illustrate some nice features of DataFrames and a brief discussion of series, which are two of the main features of Pandas. Note that Pandas material is discussed after the NumPy material, which includes a discussion various types of DataFrames that one can create, such as numeric and Boolean DataFrames. In addition, readers will see examples of creating DataFrames with NumPy functions and random numbers.

The fifth section of this chapter shows how to manipulate the contents of `DataFrames` with various operations. In particular, readers will also see code samples that illustrate how to create Pandas `DataFrames` from CSV (comma-separated value) files, Excel spreadsheets, and data that is retrieved from a URL.

WHAT IS NUMPY?

`NumPy` is a Python module that provides many convenience methods and also better performance. `NumPy` provides a core library for scientific computing in Python, with performant multidimensional arrays and good vectorized math functions, along with support for linear algebra and random numbers.

`NumPy` is modeled after `MatLab`, with support for lists, arrays, and so forth. `NumPy` is easier to use than `Matlab`, and it's common in `TensorFlow` code as well as Python code.

Useful NumPy Features

The `NumPy` package provides the *ndarray* object that encapsulates *multidimensional* arrays of homogeneous data types. Many *ndarray* operations are performed in compiled code in order to improve performance.

Keep in mind the following important differences between `NumPy` arrays and the standard Python sequences:

- `NumPy` arrays have a fixed size, whereas Python lists can expand dynamically. Whenever you modify the size of an *ndarray*, a new array is created and the original array is deleted.
- `NumPy` arrays are homogeneous, which means that the elements in a `NumPy` array must have the same data type. Except for `NumPy` arrays of objects, the elements in `NumPy` arrays of any other data type must have the same size in memory.
- `NumPy` arrays support more efficient execution (and require less code) of various types of operations on large numbers of data.
- Many scientific Python-based packages rely on `NumPy` arrays, and knowledge of `NumPy` arrays is becoming increasingly important.

Now that readers have a general idea about `NumPy`, the chapter will delve into some examples that illustrate how to work with `NumPy` arrays, which is the topic of the next section.

WHAT ARE NUMPY ARRAYS?

An *array* is a set of consecutive memory locations used to store data. Each item in the array is called an *element*. The number of elements in an array is called the *dimension* of the array. A typical array declaration is shown here:

```
arr1 = np.array([1,2,3,4,5])
```

The preceding code snippet declares `arr1` as an array of five elements, which one can access via `arr1[0]` through `arr1[4]`. Notice that the first element has an index value of 0, the second element has an index value of 1, and so forth. Thus, if users declare an array of one hundred elements, then the one hundredth element has index value of ninety-nine.

Note: The first position in a NumPy array has index 0.

NumPy treats arrays as vectors. Math ops are performed element by element. Remember the following difference: “Doubling” an array *multiplies* each element by 2, whereas “doubling” a Python list *appends* a list to itself.

Listing 4.1 displays the contents of `nparray1.py` that illustrates some operations on a NumPy array.

LISTING 4.1: `nparray1.py`

```
import numpy as np

list1 = [1,2,3,4,5]
print(list1)

arr1 = np.array([1,2,3,4,5])
print(arr1)

list2 = [(1,2,3),(4,5,6)]
print(list2)

arr2 = np.array([(1,2,3),(4,5,6)])
print(arr2)
```

Listing 4.1 defines the variables `list1` and `list2` (which are Python lists), as well as the variables `arr1` and `arr2` (which are arrays), and prints their values. The output from launching Listing 4.1 is here:

```
[1, 2, 3, 4, 5]
[1 2 3 4 5]
[(1, 2, 3), (4, 5, 6)]
[[1 2 3]
 [4 5 6]]
```

Python lists and arrays are very easy to define, and now it's time to look at some loop operations for lists and arrays.

WORKING WITH LOOPS

Listing 4.2 displays the contents of `loop1.py` that illustrates how to iterate through the elements of a NumPy array and a Python list.

LISTING 4.2: loop1.py

```
import numpy as np

list = [1,2,3]
arr1 = np.array([1,2,3])

for e in list:
    print(e)

for e in arr1:
    print(e)
```

Listing 4.2 initializes the variable `list`, which is a Python list, and also the variable `arr1`, which is a NumPy array. The next portion of Listing 4.2 contains two loops, each of which iterates through the elements in `list` and `arr1`. The syntax is identical in both loops. The output from launching Listing 4.2 is here:

```
1
2
3
1
2
3
```

APPENDING ELEMENTS TO ARRAYS (1)

Listing 4.3 displays the contents of `append1.py` that illustrates how to append elements to a NumPy array and a Python list.

LISTING 4.3: append1.py

```
import numpy as np

arr1 = np.array([1,2,3])

# these do not work:
#arr1.append(4)
#arr1 = arr1 + [5]

arr1 = np.append(arr1,4)
arr1 = np.append(arr1,[5])

for e in arr1:
    print(e)

arr2 = arr1 + arr1

for e in arr2:
    print(e)
```

Listing 4.3 initializes the variable `list`, which is a Python list, and also the variable `arr1`, which is a NumPy array. The output from launching Listing 4.3 is here:

```
1
2
3
4
5
2
4
6
8
10
```

APPENDING ELEMENTS TO ARRAYS (2)

Listing 4.4 displays the contents of `append2.py` that illustrates how to append elements to a NumPy array and a Python list.

LISTING 4.4: `append2.py`

```
import numpy as np

arr1 = np.array([1,2,3])
arr1 = np.append(arr1,4)

for e in arr1:
    print(e)

arr1 = np.array([1,2,3])
arr1 = np.append(arr1,4)

arr2 = arr1 + arr1

for e in arr2:
    print(e)
```

Listing 4.4 initializes the variable `arr1`, which is a NumPy array. Notice that NumPy arrays do not have an “append” method: This method is available through NumPy itself. Another important difference between Python lists and NumPy arrays: the “+” operator *concatenates* Python lists, whereas this operator *doubles* the elements in a NumPy array. The output from launching Listing 4.4 is here:

```
1
2
3
4
2
4
6
8
```

MULTIPLY LISTS AND ARRAYS

Listing 4.5 displays the contents of `multiply1.py` that illustrates how to multiply elements in a Python list and a NumPy array.

LISTING 4.5: multiply1.py

```
import numpy as np

list1 = [1,2,3]
arr1 = np.array([1,2,3])
print('list: ',list1)
print('arr1: ',arr1)
print('2*list:',2*list)
print('2*arr1:',2*arr1)
```

Listing 4.5 contains a Python list called `list` and a NumPy array called `arr1`. The `print()` statements display the contents of `list` and `arr1` as well as the result of doubling `list1` and `arr1`. Recall that “doubling” a Python list is different from doubling a Python array, as seen in the output from launching Listing 4.5:

```
('list: ', [1, 2, 3])
('arr1: ', array([1, 2, 3]))
('2*list:', [1, 2, 3, 1, 2, 3])
('2*arr1:', array([2, 4, 6]))
```

DOUBLING THE ELEMENTS IN A LIST

Listing 4.6 displays the contents of `double_list1.py` that illustrates how to double the elements in a Python list.

LISTING 4.6: double_list1.py

```
import numpy as np

list1 = [1,2,3]
list2 = []

for e in list1:
    list2.append(2*e)

print('list1:',list1)
print('list2:',list2)
```

Listing 4.6 contains a Python list called `list1` and an empty NumPy list called `list2`. The next code snippet iterates through the elements of `list1` and appends them to the variable `list2`. The pair of `print()` statements

display the contents of `list1` and `list2` to show they are the same. The output from launching Listing 4.6 is here:

```
('list: ', [1, 2, 3])
('list2:', [2, 4, 6])
```

LISTS AND EXPONENTS

Listing 4.7 displays the contents of `exponent_list1.py` that illustrates how to compute exponents of the elements in a Python list.

LISTING 4.7: `exponent_list1.py`

```
import numpy as np

list1 = [1,2,3]
list2 = []

for e in list1:
    list2.append(e*e) # e*e = squared

print('list1:',list1)
print('list2:',list2)
```

Listing 4.7 contains a Python list called `list1` and an empty NumPy list called `list2`. The next code snippet iterates through the elements of `list1` and appends the square of each element to the variable `list2`. The pair of `print()` statements display the contents of `list1` and `list2`. The output from launching Listing 4.7 is here:

```
('list1:', [1, 2, 3])
('list2:', [1, 4, 9])
```

ARRAYS AND EXPONENTS

Listing 4.8 displays the contents of `exponent_array1.py` that illustrates how to compute exponents of the elements in a NumPy array.

LISTING 4.8: `exponent_array1.py`

```
import numpy as np

arr1 = np.array([1,2,3])
arr2 = arr1**2
arr3 = arr1**3

print('arr1:',arr1)
print('arr2:',arr2)
print('arr3:',arr3)
```

Listing 4.8 contains a NumPy array called `arr1` followed by two NumPy arrays called `arr2` and `arr3`. Notice the compact manner in which the NumPy `arr2` is initialized with the square of the elements in `arr1`, followed by the initialization of the NumPy array `arr3` with the cube of the elements in `arr1`. The three `print()` statements display the contents of `arr1`, `arr2`, and `arr3`. The output from launching Listing 4.8 is here:

```
('arr1:', array([1, 2, 3]))
('arr2:', array([1, 4, 9]))
('arr3:', array([ 1,  8, 27]))
```

MATH OPERATIONS AND ARRAYS

Listing 4.9 displays the contents of `mathops_array1.py` that illustrates how to compute exponents of the elements in a NumPy array.

LISTING 4.9: `mathops_array1.py`

```
import numpy as np

arr1 = np.array([1,2,3])
sqrt = np.sqrt(arr1)
log1 = np.log(arr1)
exp1 = np.exp(arr1)

print('sqrt:',sqrt)
print('log1:',log1)
print('exp1:',exp1)
```

Listing 4.9 contains a NumPy array called `arr1` followed by three NumPy arrays called `sqrt`, `log1`, and `exp1` that are initialized with the square root, the log, and the exponential value of the elements in `arr1`, respectively. The three `print()` statements display the contents of `sqrt`, `log1`, and `exp1`. The output from launching Listing 4.9 is here:

```
('sqrt:', array([1.          , 1.41421356, 1.73205081]))
('log1:', array([0.          , 0.69314718, 1.09861229]))
('exp1:', array([2.71828183, 7.3890561, 20.08553692]))
```

WORKING WITH “-1” SUBRANGES WITH VECTORS

Listing 4.10 displays the contents of `npsubarray2.py` that illustrates how to work with subranges in Pandas arrays.

LISTING 4.10: `npsubarray2.py`

```
import numpy as np

# _1 => "all except the last element in ..." (row or col)
```

```
arr1 = np.array([1,2,3,4,5])
print('arr1:',arr1)
print('arr1[0:-1]:',arr1[0:-1])
print('arr1[1:-1]:',arr1[1:-1])
print('arr1[::-1]:', arr1[::-1]) # reverse!
```

Listing 4.10 contains a NumPy array called `arr1` followed by four `print` statements, each of which displays a different subrange of values in `arr1`. The output from launching Listing 4.10 is here:

```
('arr1:',          array([1, 2, 3, 4, 5]))
('arr1[0:-1]:',    array([1, 2, 3, 4]))
('arr1[1:-1]:',    array([2, 3, 4]))
('arr1[::-1]:',    array([5, 4, 3, 2, 1]))
```

WORKING WITH “-1” SUBRANGES WITH ARRAYS

Listing 4.11 displays the contents of `np2darray2.py` that illustrates how to work with negative numbers in ranges of two-dimensional arrays.

LISTING 4.11: np2darray2.py

```
import numpy as np

# -1 => "the last element in ..." (row or col)

arr1 = np.array([(1,2,3), (4,5,6), (7,8,9), (10,11,12)])
print('arr1:',          arr1)
print('arr1[-1,:]:',    arr1[-1,:])
print('arr1[:, -1]:',    arr1[:, -1])
print('arr1[-1:, -1]:', arr1[-1:, -1])
```

Listing 4.11 contains a NumPy array called `arr1` followed by four `print` statements, each of which displays a different subrange of values in `arr1`. The output from launching Listing 4.11 is here:

```
(arr1:', array([[1,  2,  3],
                [4,  5,  6],
                [7,  8,  9],
                [10, 11, 12]]))
(arr1[-1,:]:', array([10, 11, 12]))
(arr1[:, -1]:', array([3,  6,  9, 12]))
(arr1[-1:, -1]:', array([12]))
```

OTHER USEFUL NUMPY METHODS

In addition to the NumPy methods that you saw in the code samples prior to this section, the following (often intuitively-named) NumPy methods are also very useful.

- The method `np.zeros()` initializes an array with 0 values.

- The method `np.ones()` initializes an array with 1 values.
- The method `np.empty()` initializes an array with 0 values.
- The method `np.arange()` provides a range of numbers.
- The method `np.shape()` displays the shape of an object.
- The method `np.reshape()` *⇐ very useful!*
- The method `np.linspace()` *⇐ useful in regression.*
- The method `np.mean()` computes the mean of a set of numbers.
- The method `np.std()` computes the standard deviation of a set of numbers.

Although the `np.zeros()` and `np.empty()` both initialize a 2D array with 0, `np.zeros()` requires less execution time. You could also use `np.full(size, 0)`, but this method is the slowest of all three methods.

The `reshape()` method and the `linspace()` method are very useful for changing the dimensions of an array and generating a list of numeric values, respectively. The `reshape()` method often appears in TensorFlow code, and the `linspace()` method is useful for generating a set of numbers in linear regression.

The `mean()` and `std()` methods are useful for calculating the mean and the standard deviation of a set of numbers. For example, you can use these two methods in order to resize the values in a Gaussian distribution so that their mean is 0 and the standard deviation is 1. This process is called *standardizing* a Gaussian distribution.

ARRAYS AND VECTOR OPERATIONS

Listing 4.12 displays the contents of `array_vector.py` that illustrates how to perform vector operations on the elements in a NumPy array.

LISTING 4.12: `array_vector.py`

```
import numpy as np

a = np.array([[1,2], [3, 4]])
b = np.array([[5,6], [7,8]])

print('a:      ', a)
print('b:      ', b)
print('a + b:   ', a+b)
print('a _ b:   ', a_b)
print('a * b:   ', a*b)
print('a / b:   ', a/b)
print('b / a:   ', b/a)
print('a.dot(b):', a.dot(b))
```

Listing 4.12 contains two NumPy arrays called `a` and `b` followed by eight `print` statements, each of which displays the result of “applying” a different

arithmetic operation to the NumPy arrays `a` and `b`. The output from launching Listing 4.12 is here:

```
('a      :   ', array([[1, 2], [3, 4]]))
('b      :   ', array([[5, 6], [7, 8]]))
('a + b:   ', array([[ 6,  8], [10, 12]]))
('a _ b:   ', array([[_4, _4], [_4, _4]]))
('a * b:   ', array([[ 5, 12], [21, 32]]))
('a / b:   ', array([[0, 0], [0, 0]]))
('b / a:   ', array([[5, 3], [2, 2]]))
('a.dot(b):', array([[19, 22], [43, 50]]))
```

NUMPY AND DOT PRODUCTS (1)

Listing 4.13 displays the contents of `dotproduct1.py` that illustrates how to perform the dot product on the elements in a NumPy array.

LISTING 4.13: dotproduct1.py

```
import numpy as np

a = np.array([1,2])
b = np.array([2,3])

dot2 = 0
for e,f in zip(a,b):
    dot2 += e*f

print('a:   ',a)
print('b:   ',b)
print('a*b: ',a*b)
print('dot1:',a.dot(b))
print('dot2:',dot2)
```

Listing 4.13 contains two NumPy arrays called `a` and `b` followed by a simple loop that computes the dot product of `a` and `b`. The next section contains five print statements that display the contents of `a` and `b`, their inner product that's calculated in three different ways. The output from launching Listing 4.13 is here:

```
('a:   ', array([1, 2]))
('b:   ', array([2, 3]))
('a*b: ', array([2, 6]))
('dot1:', 8)
('dot2:', 8)
```

NUMPY AND DOT PRODUCTS (2)

NumPy arrays support a “dot” method for calculating the inner product of an array of numbers, which uses the same formula used for calculating the

inner product of a pair of vectors. Listing 4.14 displays the contents of `dot-product2.py` that illustrates how to calculate the dot product of two NumPy arrays.

LISTING 4.14: `dotproduct2.py`

```
import numpy as np

a = np.array([1,2])
b = np.array([2,3])

print('a:           ',a)
print('b:           ',b)
print('a.dot(b):     ',a.dot(b))
print('b.dot(a):     ',b.dot(a))
print('np.dot(a,b): ',np.dot(a,b))
print('np.dot(b,a): ',np.dot(b,a))
```

Listing 4.14 contains two NumPy arrays called `a` and `b` followed by six `print` statements that display the contents of `a` and `b`, and also their inner product that's calculated in three different ways. The output from launching Listing 4.14 is here:

```
('a:           ', array([1, 2]))
('b:           ', array([2, 3]))
('a.dot(b):     ', 8)
('b.dot(a):     ', 8)
('np.dot(a,b): ', 8)
('np.dot(b,a): ', 8)
```

NUMPY AND THE “NORM” OF VECTORS

The “norm” of a vector (or an array of numbers) is the length of a vector, which is the square root of the dot product of a vector with itself. NumPy also provides the “sum” and “square” functions that can be used to calculate the norm of a vector.

Listing 4.15 displays the contents of `array_norm.py` that illustrates how to calculate the magnitude (“norm”) of a NumPy array of numbers.

LISTING 4.15: `array_norm.py`

```
import numpy as np

a = np.array([2,3])
asquare = np.square(a)
asqsum  = np.sum(np.square(a))
anorm1  = np.sqrt(np.sum(a*a))
anorm2  = np.sqrt(np.sum(np.square(a)))
anorm3  = np.linalg.norm(a)
```

```

print('a:          ',a)
print('asquare:',asquare)
print('asqsum:  ',asqsum)
print('anorm1:  ',anorm1)
print('anorm2:  ',anorm2)
print('anorm3:  ',anorm3)

```

Listing 4.15 contains an initial NumPy array called `a`, followed by the NumPy array `asquare` and the numeric values `asqsum`, `anorm1`, `anorm2`, and `anorm3`. The NumPy array `asquare` contains the square of the elements in the NumPy array `a`, and the numeric value `asqsum` contains the sum of the elements in the NumPy array `asquare`.

Next, the numeric value `anorm1` equals the square root of the sum of the square of the elements in `a`. The numeric value `anorm2` is the same as `anorm1`, computed in a slightly different fashion. Finally, the numeric value `anorm3` is equal to `anorm2`, but `anorm3` is calculated via a single NumPy method, whereas `anorm2` requires a succession of NumPy methods.

The last portion of Listing 4.15 consists of six `print()` statements, each of which displays the computed values. The output from launching Listing 4.15 is here:

```

('a:          ', array([2, 3]))
('asquare:', array([4, 9]))
('asqsum:  ', 13)
('anorm1:  ', 3.605551275463989)
('anorm2:  ', 3.605551275463989)
('anorm3:  ', 3.605551275463989)

```

NUMPY AND OTHER OPERATIONS

NumPy provides the “*” operator to multiply the components of two vectors to produce a third vector whose components are the products of the corresponding components of the initial pair of vectors. This operation is called a Hadamard product, which is the name of a famous mathematician. When the components of the third vector are added, the sum is equal to the inner product of the initial pair of vectors.

Listing 4.16 displays the contents of `otherops.py` that illustrates how to perform other operations on a NumPy array.

LISTING 4.16: otherops.py

```

import numpy as np

a = np.array([1,2])
b = np.array([3,4])

print('a:          ',a)

```

```

print('b:           ',b)
print('a*b:         ',a*b)
print('np.sum(a*b): ',np.sum(a*b))
print('(a*b.sum()): ',(a*b).sum())

```

Listing 4.16 contains two NumPy arrays called `a` and `b` followed five print statements that display the contents of `a` and `b`, their Hadamard product, and also their inner product that's calculated in two different ways. The output from launching Listing 4.16 is here:

```

('a:           ', array([1, 2]))
('b:           ', array([3, 4]))
('a*b:         ', array([3, 8]))
('np.sum(a*b): ', 11)
('(a*b.sum()): ', 11)

```

NUMPY AND THE RESHAPE() METHOD

NumPy arrays support the “reshape” method that enables you to restructure the dimensions of an array of numbers. In general, if a NumPy array contains m elements, where m is a positive integer, then that array can be restructured as an $m_1 \times m_2$ NumPy array, where m_1 and m_2 are positive integers such that $m_1 * m_2 = m$.

Listing 4.17 displays the contents of `NumPy_reshape.py` that illustrates how to use the `reshape()` method on a NumPy array.

LISTING 4.17: NumPy_reshape.py

```

import numpy as np

x = np.array([[2, 3], [4, 5], [6, 7]])
print(x.shape) # (3, 2)

x = x.reshape((2, 3))
print(x.shape) # (2, 3)
print('x1:',x)

x = x.reshape((-1))
print(x.shape) # (6,)
print('x2:',x)

x = x.reshape((6, -1))
print(x.shape) # (6, 1)
print('x3:',x)

x = x.reshape((-1, 6))
print(x.shape) # (1, 6)
print('x4:',x)

```


Listing 4.17 contains a NumPy array called `x` whose dimensions are 3x2, followed by a set of invocations of the `reshape()` method that reshape the contents of `x`. The first invocation of the `reshape()` method changes the shape of `x` from 3x2 to 2x3. The second invocation changes the shape of `x` from 2x3 to 6x1. The third invocation changes the shape of `x` from 1x6 to 6x1. The final invocation changes the shape of `x` from 6x1 to 1x6 again.

Each invocation of the `reshape()` method is followed by a `print()` statement so that one can see the effect of the invocation. The output from launching Listing 4.17 is here:

```
(3, 2)
(2, 3)
('x1:', array([[2, 3, 4],
               [5, 6, 7]]))
(6,)
('x2:', array([2, 3, 4, 5, 6, 7]))
(6, 1)
('x3:', array([[2],
               [3],
               [4],
               [5],
               [6],
               [7]]))
(1, 6)
```

CALCULATING THE MEAN AND STANDARD DEVIATION

Readers can review these concepts from statistics (and perhaps also the mean, median, and mode as well), by reading the appropriate on-line tutorials.

NumPy provides various built-in functions that perform statistical calculations, such as the following list of methods:

```
np.linspace() <= useful for regression
np.mean()
np.std()
```

The `np.linspace()` method generates a set of equally spaced numbers between a lower bound and an upper bound. The `np.mean()` and `np.std()` methods calculate the mean and standard deviation, respectively, of a set of numbers. Listing 4.18 displays the contents of `sample_mean_std.py` that illustrates how to calculate statistical values from a NumPy array.

LISTING 4.18: `sample_mean_std.py`

```
import numpy as np

x2 = np.arange(8)
print('mean = ', x2.mean())
```

```
print('std = ',x2.std())

x3 = (x2 - x2.mean())/x2.std()
print('x3 mean = ',x3.mean())
print('x3 std = ',x3.std())
```

Listing 4.18 contains a NumPy array `x2` that consists of the first eight integers. Next, the `mean()` and `std()` that are “associated” with `x2` are invoked in order to calculate the mean and standard deviation, respectively, of the elements of `x2`. The output from launching Listing 4.18 is here:

```
mean = 3.5
std = 2.29128784747792
x3 mean = 0.0
x3 std = 1.0
```

CALCULATING MEAN AND STANDARD DEVIATION

The code sample in this section extends the code sample in the previous section with additional statistical values, and the code in Listing 4.19 can be used for any data distribution. Keep in mind that the code sample uses random numbers simply for the purposes of illustration: After the code sample has been launched, replace those numbers with values from a CSV file or some other dataset containing meaningful values.

Moreover, this section does not provide details regarding the meaning of quartiles, but readers can learn about quartiles here: <https://en.wikipedia.org/wiki/Quartile>

Listing 4.19 displays the contents of `stat_summary.py` that illustrates how to display various statistical values from a NumPy array of random numbers.

LISTING 4.19: `stat_values.py`

```
import numpy as np

from numpy import percentile
from numpy import rand

# generate data sample
data = np.random.rand(1000)

# calculate quartiles, min, and max
quartiles = percentile(data, [25, 50, 75])
data_min, data_max = data.min(), data.max()

# print summary information
print('Minimum: %.3f' % data_min)
print('Q1 value: %.3f' % quartiles[0])
print('Median: %.3f' % quartiles[1])
```

```
print('Mean Val: %.3f' % data.mean())
print('Std Dev:  %.3f' % data.std())
print('Q3 value: %.3f' % quartiles[2])
print('Maximum:  %.3f' % data_max)
```

The data sample (shown in bold) in Listing 4.19 is from a uniform distribution between 0 and 1. The NumPy `percentile()` function calculates a linear interpolation (average) between observations, which is needed to calculate the median on a sample with an even number of values. As you can surmise, the NumPy functions `min()` and `max()` calculate the smallest and largest values in the data sample. The output from launching Listing 4.19 is here:

```
Minimum:  0.000
Q1 value: 0.237
Median:    0.500
Mean Val:  0.495
Std Dev:   0.295
Q3 value:  0.747
Maximum:   0.999
```

This concludes the portion of the chapter pertaining to NumPy. The second half of this chapter discusses some of the features of Pandas.

WHAT IS PANDAS?

Pandas is a Python package that is compatible with other Python packages, such as NumPy, Matplotlib, and so forth. Install Pandas by opening a command shell and invoking this command for Python 3.x:

```
pip3 install pandas
```

In many ways the Pandas package has the semantics of a spreadsheet, and it also works with `xsl`, `xml`, `html`, `csv` file types. Pandas provides a data type called a `DataFrame` (similar to a Python dictionary) with extremely powerful functionality, which is discussed in the next section.

Pandas `DataFrames` support a variety of input types, such as `ndarrays`, `lists`, `dicts`, or `Series`. Pandas also provides another data type called `Pandas Series` (not discussed in this chapter), this data structure provides another mechanism for managing data (search online for more details).

Pandas DataFrames

In simplified terms, a `Pandas DataFrame` is a two-dimensional data structure, and it's convenient to think of the data structure in terms of rows and columns. `DataFrames` can be labeled (rows as well as columns), and the columns can contain different data types.

By way of analogy, it might be useful to think of a `DataFrame` as the counterpart to a spreadsheet, which makes it a very useful data type in Pandas-related

Python scripts. The source of the dataset can be a data file, database tables, Web service, and so forth. Pandas `DataFrame` features include:

- dataframe methods
- dataframe statistics
- grouping, pivoting, and reshaping
- dealing with missing data
- joining dataframes

DataFrames and Data Cleaning Tasks

The specific tasks that you need to perform depend on the structure and contents of a dataset. In general, users will perform a workflow with the following steps (not necessarily always in this order), all of which can be performed with a Pandas `DataFrame`:

- read data into a dataframe
- display top of dataframe
- display column data types
- display non_missing values
- replace NA with a value
- iterate through the columns
- statistics for each column
- find missing values
- total missing values
- percentage of missing values
- sort table values
- print summary information
- columns with > 50% missing
- rename columns

A LABELED PANDAS DATAFRAME

Listing 4.20 displays the contents of `pandas_labeled_df.py` that illustrates how to define a Pandas `DataFrame` whose rows and columns are labeled.

LISTING 4.20: `pandas_labeled_df.py`

```
import numpy
import pandas

myarray = numpy([[10,30,20], [50,40,60], [1000,2000,3000]])

rownames = ['apples', 'oranges', 'beer']
colnames = ['January', 'February', 'March']

mydf = Pandas.DataFrame(myarray, index=rownames,
                        columns=colnames)
```

```
print(mydf)
print(mydf.describe())
```

Listing 4.20 contains two important statements followed by the variable `myarray`, which is a 3x3 NumPy array of numbers. The variables `rownames` and `colnames` provide names for the rows and columns, respectively, of the data in `myarray`. Next, the variable `mydf` is initialized as a Pandas DataFrame with the specified datasource (i.e., `myarray`).

You might be surprised to see that the first portion of the output below requires a single `print` statement (which simply displays the contents of `mydf`). The second portion of the output is generated by invoking the `describe()` method that is available for any NumPy DataFrame. The `describe()` method is very useful: you will see various statistical quantities, such as the mean, standard deviation minimum, and maximum performed `column_wise` (not `row_wise`), along with values for the 25th, 50th, and 75th percentiles. The output of Listing 4.20 is here:

	January	February	March
apples	10	30	20
oranges	50	40	60
beer	1000	2000	3000

	January	February	March
count	3.000000	3.000000	3.000000
mean	353.333333	690.000000	1026.666667
std	560.386771	1134.504297	1709.073823
min	10.000000	30.000000	20.000000
25%	30.000000	35.000000	40.000000
50%	50.000000	40.000000	60.000000
75%	525.000000	1020.000000	1530.000000
max	1000.000000	2000.000000	3000.000000

PANDAS NUMERIC DATAFRAMES

Listing 4.21 displays the contents of `pandas_numeric_df.py` that illustrates how to define a Pandas DataFrame whose rows and columns are numbers (but the column labels are characters).

LISTING 4.21: `pandas_numeric_df.py`

```
import pandas as pd

df1 = pd.DataFrame(np.random.randn(10,
4), columns=['A', 'B', 'C', 'D'])
df2 = pd.DataFrame(np.random.randn(7, 3),
columns=['A', 'B', 'C'])
df3 = df1 + df2
```

The essence of Listing 4.21 involves initializing the DataFrames `df1` and `df2`, and then defining the DataFrame `df3` as the sum of `df1` and `df2`. The output from Listing 4.21 is here:

	A	B	C	D
0	0.0457	-0.0141	1.3809	NaN
1	-0.9554	-1.5010	0.0372	NaN
2	-0.6627	1.5348	-0.8597	NaN
3	-2.4529	1.2373	-0.1337	NaN
4	1.4145	1.9517	-2.3204	NaN
5	-0.4949	-1.6497	-1.0846	NaN
6	-1.0476	-0.7486	-0.8055	NaN
7	NaN	NaN	NaN	NaN
8	NaN	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN

Keep in mind that the default behavior for operations involving a DataFrame and Series is to align the Series index on the DataFrame columns; this results in a row-wise output. Here is a simple illustration:

```
names = pd.Series(['SF', 'San Jose', 'Sacramento'])
sizes = pd.Series([852469, 1015785, 485199])

df = pd.DataFrame({ 'Cities': names, 'Size': sizes })
df = pd.DataFrame({ 'City name': names, 'sizes': sizes })

print(df)
```

The output of the preceding code block is here:

	City name	sizes
0	SF	852469
1	San Jose	1015785
2	Sacramento	485199

PANDAS BOOLEAN DATAFRAMES

Pandas supports Boolean operations on DataFrames, such as the logical or, the logical and, and the logical negation of a pair of DataFrames. Listing 4.22 displays the contents of `pandas-boolean-df.py` that illustrates how to define a Pandas DataFrame whose rows and columns are Boolean values.

LISTING 4.22: `pandas-boolean-df.py`

```
import pandas as pd

df1 = pd.DataFrame({'a' : [1, 0, 1], 'b' : [0, 1, 1] },
dtype=bool)
df2 = pd.DataFrame({'a' : [0, 1, 1], 'b' : [1, 1, 0] },
dtype=bool)
```

```

print("df1 & df2:")
print(df1 & df2)

print("df1 | df2:")
print(df1 | df2)

print("df1 ^ df2:")
print(df1 ^ df2)

```

Listing 4.22 initializes the DataFrames `df1` and `df2`, and then computes `df1 & df2`, `df1 | df2`, `df1 ^ df2`, which represent the logical AND, the logical OR, and the logical negation, respectively, of `df1` and `df2`. The output from launching the code in Listing 4.22 is here:

```

df1 & df2:
      a      b
0  False False
1  False  True
2   True False
df1 | df2:
      a      b
0   True  True
1   True  True
2   True  True
df1 ^ df2:
      a      b
0   True  True
1   True False
2  False  True

```

Transposing a Pandas DataFrame

The `T` attribute (as well as the transpose function) enables users to generate the transpose of a Pandas DataFrame, similar to a NumPy ndarray.

For example, the following code snippet defines a Pandas DataFrame `df1` and then displays the transpose of `df1`:

```
df1 = pd.DataFrame({'a' : [1, 0, 1], 'b' : [0, 1, 1] },
dtype=int)
```

```

print("df1.T:")
print(df1.T)

```

The output is here:

```

df1.T:
      0  1  2
a    1  0  1
b    0  1  1

```

The following code snippet defines Pandas DataFrames `df1` and `df2` and then displays their sum:

```
df1 = pd.DataFrame({'a' : [1, 0, 1], 'b' : [0, 1, 1] },
dtype=int)
df2 = pd.DataFrame({'a' : [3, 3, 3], 'b' : [5, 5, 5] },
dtype=int)

print("df1 + df2:")
print(df1 + df2)
```

The output is here:

```
df1 + df2:
   a  b
0  4  5
1  3  6
2  4  6
```

PANDAS DATAFRAMES AND RANDOM NUMBERS

Listing 4.23 displays the contents of `pandas_random_df.py` that illustrates how to create a Pandas DataFrame with random numbers.

LISTING 4.23: `pandas_random_df.py`

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randint(1, 5, size=(5, 2)),
columns=['a', 'b'])
df = df.append(df.agg(['sum', 'mean']))

print("Contents of dataframe:")
print(df)
```

Listing 4.23 defines the Pandas DataFrame `df` that consists of five rows and two columns of random integers between 1 and 5. Notice that the columns of `df` are labeled “a” and “b.” In addition, the next code snippet appends two rows consisting of the sum and the mean of the numbers in both columns. The output of Listing 4.23 is here:

```
a      b
0      1.0  2.0
1      1.0  1.0
2      4.0  3.0
3      3.0  1.0
4      1.0  2.0
sum    10.0  9.0
mean     2.0  1.8
```


COMBINING PANDAS DATAFRAMES (1)

Listing 4.24 displays the contents of `pandas-combine-df.py` that illustrates how to combine Pandas DataFrames.

LISTING 4.24: `pandas-combine-df.py`

```
import pandas as pd
import numpy as np

df = pd.DataFrame({'foo1' : np.random.randn(5),
                  'foo2' : np.random.randn(5)})

print("contents of df:")
print(df)

print("contents of foo1:")
print(df.foo1)

print("contents of foo2:")
print(df.foo2)
```

Listing 4.24 defines the Pandas DataFrame `df` that consists of five rows and two columns (labeled "foo1" and "foo2") of random real numbers between 0 and 5. The next portion of Listing 4.24 displays the contents of `df` and `foo1`. The output of Listing 4.24 is here:

```
contents of df:
      foo1      foo2
0  0.274680 -0.848669
1 -0.399771 -0.814679
2  0.454443 -0.363392
3  0.473753  0.550849
4 -0.211783 -0.015014
contents of foo1:
0    0.256773
1    1.204322
2    1.040515
3   -0.518414
4    0.634141
Name: foo1, dtype: float64
contents of foo2:
0   -2.506550
1   -0.896516
2   -0.222923
3    0.934574
4    0.527033
Name: foo2, dtype: float64
```

COMBINING PANDAS DATAFRAMES (2)

Pandas supports the “concat” method in DataFrames in order to concatenate DataFrames. Listing 4.25 displays the contents of `concat_frames.py` that illustrates how to combine two Pandas DataFrames.

LISTING 4.25: `concat_frames.py`

```
import pandas as pd

can_weather = pd.DataFrame({
    "city": ["Vancouver", "Toronto", "Montreal"],
    "temperature": [72, 65, 50],
    "humidity": [40, 20, 25]
})

us_weather = pd.DataFrame({
    "city": ["SF", "Chicago", "LA"],
    "temperature": [60, 40, 85],
    "humidity": [30, 15, 55]
})

df = pd.concat([can_weather, us_weather])
print(df)
```

The first line in Listing 4.25 is an import statement, followed by the definition of the Pandas dataframes `can_weather` and `us_weather` that contain weather-related information for cities in Canada and the United States, respectively. The Pandas DataFrame `df` is the concatenation of `can_weather` and `us_weather`. The output from Listing 4.25 is here:

0	Vancouver	40	72
1	Toronto	20	65
2	Montreal	25	50
0	SF	30	60
1	Chicago	15	40
2	LA	55	85

DATA MANIPULATION WITH PANDAS DATAFRAMES (1)

As a simple example, suppose that we have a two-person company that keeps track of income and expenses on a quarterly basis, and we want to calculate the profit/loss for each quarter, and also the overall profit/loss.

Listing 4.26 displays the contents of `pandas_quarterly_df1.py` that illustrates how to define a Pandas DataFrame consisting of income-related values.

LISTING 4.26: pandas_quarterly_df1.py

```
import pandas as pd

summary = {
    'Quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'Cost':     [23500, 34000, 57000, 32000],
    'Revenue':  [40000, 40000, 40000, 40000]
}

df = pd.DataFrame(summary)

print("Entire Dataset:\n",df)
print("Quarter:\n",df.Quarter)
print("Cost:\n",df.Cost)
print("Revenue:\n",df.Revenue)
```

Listing 4.26 defines the variable `summary` that contains hard-coded quarterly information about cost and revenue for our two-person company. In general these hard-coded values would be replaced by data from another source (such as a CSV file), so think of this code sample as a simple way to illustrate some of the functionality that is available in Pandas DataFrames.

The variable `df` is a Pandas DataFrame based on the data in the `summary` variable. The three `print` statements display the quarters, the cost per quarter, and the revenue per quarter.

The output from Listing 4.26 is here:

```
Entire Dataset:
   Cost Quarter  Revenue
0  23500      Q1   40000
1  34000      Q2   60000
2  57000      Q3   50000
3  32000      Q4   30000
Quarter:
0      Q1
1      Q2
2      Q3
3      Q4
Name: Quarter, dtype: object
Cost:
0      23500
1      34000
2      57000
3      32000
Name: Cost, dtype: int64
Revenue:
0      40000
1      60000
2      50000
3      30000
Name: Revenue, dtype: int64
```

DATA MANIPULATION WITH PANDAS DATAFRAMES (2)

In this section, let's suppose that we have a two-person company that keeps track of income and expenses on a quarterly basis, and we want to calculate the profit/loss for each quarter, and also the overall profit/loss.

Listing 4.27 displays the contents of `pandas-quarterly-df1.py` that illustrates how to define a Pandas DataFrame consisting of income-related values.

LISTING 4.27: `pandas-quarterly-df2.py`

```
import pandas as pd

summary = {
    'Quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'Cost':    [-23500, -34000, -57000, -32000],
    'Revenue': [40000, 40000, 40000, 40000]
}

df = pd.DataFrame(summary)
print("First Dataset:\n",df)

df['Total'] = df.sum(axis=1)
print("Second Dataset:\n",df)
```

Listing 4.27 defines the variable `summary` that contains quarterly information about cost and revenue for our two-person company. The variable `df` is a Pandas DataFrame based on the data in the `summary` variable. The three print statements display the quarters, the cost per quarter, and the revenue per quarter. The output from Listing 4.27 is here:

```
First Dataset:
      Cost Quarter  Revenue
0 -23500        Q1    40000
1 -34000        Q2    60000
2 -57000        Q3    50000
3 -32000        Q4    30000
Second Dataset:
      Cost Quarter  Revenue  Total
0 -23500        Q1    40000  16500
1 -34000        Q2    60000  26000
2 -57000        Q3    50000  -7000
3 -32000        Q4    30000  -2000
```

DATA MANIPULATION WITH PANDAS DATAFRAMES (3)

Start with the same assumption as the previous section: There is a two-person company that keeps track of income and expenses on a quarterly basis, and we

want to calculate the profit/loss for each quarter, and also the overall profit/loss. In addition, users should compute column totals and row totals.

Listing 4.28 displays the contents of `pandas-quarterly-df1.py` that illustrates how to define a Pandas DataFrame consisting of income-related values.

LISTING 4.28: `pandas-quarterly-df3.py`

```
import pandas as pd

summary = {
    'Quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'Cost':    [-23500, -34000, -57000, -32000],
    'Revenue': [40000, 40000, 40000, 40000]
}

df = pd.DataFrame(summary)
print("First Dataset:\n",df)

df['Total'] = df.sum(axis=1)
df.loc['Sum'] = df.sum()
print("Second Dataset:\n",df)

# or df.loc['avg'] / 3
#df.loc['avg'] = df[:3].mean()
#print("Third Dataset:\n",df)
```

Listing 4.28 defines the variable `summary` that contains quarterly information about cost and revenue for our two-person company. The variable `df` is a Pandas DataFrame based on the data in the `summary` variable. The three print statements display the quarters, the cost per quarter, and the revenue per quarter. The output from Listing 4.28 is here:

```
First Dataset:
      Cost Quarter  Revenue
0 -23500      Q1    40000
1 -34000      Q2    60000
2 -57000      Q3    50000
3 -32000      Q4    30000
Second Dataset:
      Cost  Quarter  Revenue  Total
0   -23500      Q1    40000   16500
1   -34000      Q2    60000   26000
2   -57000      Q3    50000   -7000
3   -32000      Q4    30000   -2000
Sum -146500  Q1Q2Q3Q4   180000   33500
```

CLAUDE 3 AND NUMPY DATAFRAMES

The following prompt was given to Claude 3:

Please generate Python code that creates a Pandas dataframe with 4 rows and 5 columns of randomly generated numbers and then calculates the sum of each row and the sum of each column, along with an explanation of the Python code.

Claude 3:

LISTING 4.29: `claude3_sum_rows_columns.py`

```
import pandas as pd
import numpy as np

# Define number of rows and columns
nrows = 4
ncols = 5

# Generate random data
np.random.seed(1)
data = np.random.randint(1, 100, size=(nrows, ncols))

# Create Pandas DataFrame
df = pd.DataFrame(data)

# Calculate row sums
row_sums = df.sum(axis=1)

# Calculate column sums
column_sums = df.sum(axis=0)

# Add row and column sums to the DataFrame
df['Row Sum'] = row_sums
df.loc['Column Sum'] = column_sums

# Print the DataFrame with row and column sums
print("DataFrame with row and column sums:")
print(df)
```

Explanation:

Import libraries:

We import pandas for data manipulation and NumPy for generating random numbers.

Define parameters:

We set the number of rows (`nrows`) and columns (`ncols`) for the DataFrame.

Generate random data:

We use `np.random.seed(1)` to ensure reproducibility of the random numbers.

Then, we use `np.random.randint(1, 100, size=(nrows, ncols))` to generate an array of random integers between 1 and 100 with the specified number of rows and columns.

Create Pandas DataFrame:

We use `pd.DataFrame(data)` to convert the generated NumPy array into a Pandas DataFrame.

Calculate row and column sums:

We use `df.sum(axis=1)` to calculate the sum of each row and store the results in a Series object named `row_sums`.

We use `df.sum(axis=0)` to calculate the sum of each column and store the results in another Series object named `column_sums`.

Add row and column sums to the DataFrame:

We add a new column named “Row Sum” to the DataFrame and populate it with the values in `row_sums`.

We use `df.loc['Column Sum'] = column_sums` to add a new row named “Column Sum” to the DataFrame and populate it with the values in `column_sums`.

Print the DataFrame:

We use `print(df)` to display the DataFrame with the row and column sums included.

This code generates a Pandas DataFrame with randomly generated numbers and calculates the sum of each row and column. One can modify this code to suit your specific needs, such as changing the range of random numbers, adding more calculations, or customizing the DataFrame formatting.

Launch the code in Listing 4.29 and you will see the following output:

```
DataFrame with row and column sums:
      0      1      2      3      4  Row Sum
0    38.0  13.0  73.0  10.0  76.0   210.0
1     6.0  80.0  65.0  17.0   2.0   170.0
2    77.0  72.0   7.0  26.0  51.0   233.0
3    21.0  19.0  85.0  12.0  29.0   166.0
Column Sum 142.0 184.0 230.0  65.0 158.0    NaN
```

PANDAS DATAFRAMES AND CSV FILES

The code samples in several earlier sections contain hard-coded data inside the Python scripts. It's also very common to read data from a CSV file. One can use the Python `csv.reader()` function, the NumPy `loadtxt()` function, or

the Pandas function `read_csv()` function (shown in this section) to read the contents of CSV files.

Listing 4.30 displays the contents of `weather_data.py` that illustrates how to read a CSV file, initialize a Pandas `DataFrame` with the contents of that CSV file, and display various subsets of the data in the Pandas `DataFrames`.

LISTING 4.30: `weather_data.py`

```
import pandas as pd

df = pd.read_csv("weather_data.csv")

print(df)
print(df.shape) # rows, columns
print(df.head()) # df.head(3)
print(df.tail())
print(df[1:3])
print(df.columns)
print(type(df['day']))
print(df[['day', 'temperature']])
print(df['temperature'].max())
```

Listing 4.30 invokes the Pandas `read_csv()` function to read the contents of the CSV file `weather_data.csv`, followed by a set of Python `print()` statements that display various portions of the CSV file. The output from Listing 4.30 is here:

```
day,temperature,windspeed,event
7/1/2018,42,16,Rain
7/2/2018,45,3,Sunny
7/3/2018,78,12,Snow
7/4/2018,74,9,Snow
7/5/2018,42,24,Rain
7/6/2018,51,32,Sunny
```

In some situations you might need to apply Boolean conditional logic to “filter out” some rows of data, based on a conditional condition that’s applied to a column value.

Listing 4.31 displays the contents of the CSV file `people.csv` and Listing 4.32 displays the contents of `people_pandas.py` that illustrates how to define a Pandas `DataFrame` that reads the CSV file and manipulates the data.

LISTING 4.31: `people.csv`

```
fname,lname,age,gender,country
john,smith,30,m,usa
jane,smith,31,f,france
jack,jones,32,f,france
dave,stone,33,f,france
```



```
sara,stein,34,f,france
eddy,bower,35,f,france
```

LISTING 4.32: people_pandas.py

```
import pandas as pd

df = pd.read_csv('people.csv')
df.info()
print('fname:')
print(df['fname'])
print('_____')
print('age over 33:')
print(df['age'] > 33)
print('_____')
print('age over 33:')
myfilter = df['age'] > 33
print(df[myfilter])
```

Listing 4.32 populate the Pandas dataframe `df` with the contents of the CSV file `people.csv`. The next portion of Listing 4.32 displays the structure of `df`, followed by the first names of all the people. The next portion of Listing 4.32 displays a tabular list of six rows containing either `True` or `False` depending on whether a person is over thirty-three or at most thirty-three, respectively.

The final portion of Listing 4.32 displays a tabular list of two rows containing all the details of the people who are over thirty-three. The output from Listing 4.32 is here:

```
myfilter = df['age'] > 33
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 5 columns):
fname      6 non_null object
lname      6 non_null object
age        6 non_null int64
gender     6 non_null object
country    6 non_null object
dtypes: int64(1), object(4)
memory usage: 320.0+ bytes
fname:
0    john
1    jane
2    jack
3    dave
4    sara
5    eddy
Name: fname, dtype: object

age over 33:
0    False
```

```

1    False
2    False
3    False
4     True
5     True
Name: age, dtype: bool

age over 33:
  fname  lname  age  gender  country
4  sara  stein   34      f  france
5  eddy  bower   35      m  france

```

PANDAS DATAFRAMES AND EXCEL SPREADSHEETS (1)

Listing 4.33 displays the contents of `people_xlsx.py` that illustrates how to read data from an Excel spreadsheet and create a Pandas `DataFrame` with that data.

LISTING 4.33: `people_xlsx.py`

```

import pandas as pd

df = pd.read_excel("people.xlsx")
print("Contents of Excel spreadsheet:")
print(df)

```

Listing 4.33 is straightforward: The Pandas dataframe `df` is initialized with the contents of the spreadsheet `people.xlsx` (whose contents are the same as `people.csv`) via the Pandas function `read_excel()`. The output from Listing 4.33 is here:

```

  fname  lname  age  gender  country
0  john  smith   30      m     usa
1  jane  smith   31      f  france
2  jack  jones   32      f  france
3  dave  stone   33      f  france
4  sara  stein   34      f  france
5  eddy  bower   35      f  france

```

SELECT, ADD, AND DELETE COLUMNS IN DATAFRAMES

This section contains short code blocks that illustrate how to perform operations on a `DataFrame` that resemble the operations on a Python dictionary. For example, getting, setting, and deleting columns works with the same syntax as the analogous Python `dict` operations, as shown here:

```

df = pd.DataFrame.from_dict(dict([('A', [1, 2, 3]), (
'B', [4, 5, 6])]),

```

```
orient='index', columns=['one', 'two',
'three'])

print(df)
```

The output from the preceding code snippet is here:

	one	two	three
A	1	2	3
B	4	5	6

Look at the following sequence of operations on the contents of the data-frame df:

```
df['three'] = df['one'] * df['two']
df['flag'] = df['one'] > 2
print(df)
```

The output from the preceding code block is here:

	one	two	three	flag
a	1.0	1.0	1.0	False
b	2.0	2.0	4.0	False
c	3.0	3.0	9.0	True
d	NaN	4.0	NaN	False

Columns can be deleted or popped like with a Python dict, as shown in following code snippet:

```
del df['two']
three = df.pop('three')
print(df)
```

The output from the preceding code block is here:

	one	flag
a	1.0	False
b	2.0	False
c	3.0	True
d	NaN	False

When inserting a scalar value, it will naturally be propagated to fill the column:

```
df['foo'] = 'bar'
print(df)
```

The output from the preceding code snippet is here:

	one	flag	foo
a	1.0	False	bar
b	2.0	False	bar
c	3.0	True	bar
d	NaN	False	bar

When inserting a `Series` that does not have the same index as the `DataFrame`, it will be “conformed” to the index of the `DataFrame`:

```
df['one_trunc'] = df['one'][:2]
print(df)
```

The output from the preceding code snippet is here:

```
   one  flag  foo  one_trunc
a  1.0  False  bar         1.0
b  2.0  False  bar         2.0
c  3.0   True  bar         NaN
d  NaN  False  bar         NaN
```

You can insert raw `ndarrays`, but their length must match the length of the index of the `DataFrame`.

PANDAS DATAFRAMES AND SCATTERPLOTS

Listing 4.34 displays the contents of `pandas_scatter_df.py` that illustrates how to generate a scatterplot from a `Pandas DataFrame`.

LISTING 4.34: `pandas_scatter_df.py`

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pandas import read_csv
from pandas.plotting import scatter_matrix

myarray = np.array([[10,30,20],
[50,40,60],[1000,2000,3000]])

rownames = ['apples', 'oranges', 'beer']
colnames = ['January', 'February', 'March']

mydf = pd.DataFrame(myarray, index=rownames,
columns=colnames)

print(mydf)
print(mydf.describe())

scatter_matrix(mydf)
plt.show()
```

Listing 4.34 starts with various import statements, followed by the definition of the NumPy array `myarray`. Next, the variables `myarray` and `colnames` are initialized with values for the rows and columns, respectively. The next portion of Listing 4.34 initializes the `Pandas DataFrame` `mydf` so that the rows and columns are labeled in the output, as shown here:

	January	February	March
apples	10	30	20
oranges	50	40	60
beer	1000	2000	3000

	January	February	March
count	3.000000	3.000000	3.000000
mean	353.333333	690.000000	1026.666667
std	560.386771	1134.504297	1709.073823
min	10.000000	30.000000	20.000000
25%	30.000000	35.000000	40.000000
50%	50.000000	40.000000	60.000000
75%	525.000000	1020.000000	1530.000000
max	1000.000000	2000.000000	3000.000000

CLAUDE 3, DATAFRAMES, AND BAR CHARTS

The following prompt was given to Claude 3.

Please generate Python code that creates a Pandas dataframe with 4 rows and 5 columns of randomly generated numbers and then calculates the sum of each row and the sum of each column, as well as a bar chart with the data in the dataframe, along with an explanation of the Python code.

Claude 3:

LISTING 4.35: `claude3_barchart.py`

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Define number of rows and columns
nrows = 4
ncols = 5

# Generate random data
np.random.seed(1)
data = np.random.randint(1, 100, size=(nrows, ncols))

# Create Pandas DataFrame
df = pd.DataFrame(data)

# Calculate row and column sums
row_sums = df.sum(axis=1)
column_sums = df.sum(axis=0)

# Add row and column sums to the DataFrame
df['Row Sum'] = row_sums
df.loc['Column Sum'] = column_sums
```

```

# Create a bar chart
plt.figure(figsize=(10, 6))
df.plot(kind='bar', stacked=False)
plt.title("Bar Chart of Random Data")
plt.xlabel("Columns")
plt.ylabel("Values")
plt.xticks(rotation=0)
plt.tight_layout()

# Print the DataFrame
print("DataFrame with row and column sums:")
print(df)

# Show the bar chart
plt.show()

```

Explanation:**Imports:**

pandas is imported to work with DataFrames.

numpy is imported to generate random numbers.

matplotlib.pyplot is imported to create the bar chart.

Define parameters:

nrows and ncols define the number of rows and columns in the DataFrame.

Generate random data:

np.random.seed(1) ensures reproducibility of the random numbers.

np.random.randint(1, 100, size=(nrows, ncols)) generates a NumPy array with random integers between 1 and 100, with nrows rows and ncols columns.

Create Pandas DataFrame:

pd.DataFrame(data) converts the NumPy array to a Pandas DataFrame.

Calculate row and column sums:

df.sum(axis=1) calculates the sum of each row and stores them in a Series object named row_sums.

df.sum(axis=0) calculates the sum of each column and stores them in a Series object named column_sums.

Add sums to DataFrame:

A new column name “Row Sum” is added to the DataFrame with the row_sums values.

A new row named “Column Sum” is added to the DataFrame with the `column_sums` values, using the `loc` method.

Create a bar chart:

`plt.figure(figsize=(10, 6))` sets the figure size for the plot.

`df.plot(kind='bar', stacked=False)` creates a bar chart using the values in the DataFrame.

`plt.title("Bar Chart of Random Data")` sets the title of the chart.

`plt.xlabel("Columns")` and `plt.ylabel("Values")` set the labels for the x and y axes.

`plt.xticks(rotation=0)` ensures all column labels are visible.

`plt.tight_layout()` adjusts the layout to avoid overlapping elements.

Print the DataFrame and show the bar chart:

The DataFrame with row and column sums is printed.

`plt.show()` displays the bar chart.

Launch the code in Listing 4.35 and you will see the following output:

```
DataFrame with row and column sums:
      0      1      2      3      4  Row Sum
0    38.0  13.0  73.0  10.0  76.0   210.0
1     6.0  80.0  65.0  17.0   2.0   170.0
2    77.0  72.0   7.0  26.0  51.0   233.0
3    21.0  19.0  85.0  12.0  29.0   166.0
Column Sum 142.0 184.0 230.0  65.0 158.0    NaN
```

In addition you will see the image that is displayed in Figure 4.1.

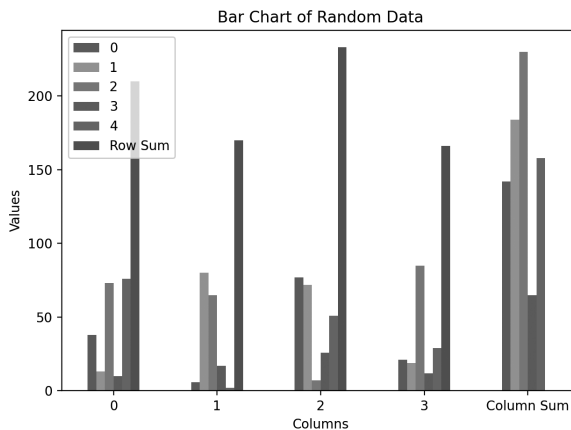


FIGURE 4.1 A bar chart generated by Claude 3.

PANDAS DATAFRAMES AND SIMPLE STATISTICS

Listing 4.36 displays the contents of `housing_stats.py` that illustrates how to gather basic statistics from data in a Pandas `DataFrame`.

LISTING 4.36: `housing_stats.py`

```
import pandas as pd

df = pd.read_csv("housing.csv")

minimum_bdrms = df["bedrooms"].min()
median_bdrms = df["bedrooms"].median()
maximum_bdrms = df["bedrooms"].max()

print("minimum # of bedrooms:", minimum_bdrms)
print("median # of bedrooms:", median_bdrms)
print("maximum # of bedrooms:", maximum_bdrms)
print("")

print("median values:", df.median().values)
print("")

prices = df["price"]
print("first 5 prices:")
print(prices.head())
print("")

median_price = df["price"].median()
print("median price:", median_price)
print("")

corr_matrix = df.corr()
print("correlation matrix:")
print(corr_matrix["price"].sort_values(ascending=False))
```

Listing 4.36 initializes the Pandas `DataFrame` `df` with the contents of the CSV file `housing.csv`. The next three variables are initialized with the minimum, median, and maximum number of bedrooms, respectively, and then these values are displayed.

The next portion of Listing 4.36 initializes the variable `prices` with the contents of the `prices` column of the Pandas `DataFrame` `df`. Next, the first five rows are printed via the `prices.head()` statement, followed by the median value of the prices.

The final portion of Listing 4.36 initializes the variable `corr_matrix` with the contents of the correlation matrix for the Pandas `DataFrame` `df`, and then displays its contents. The output from Listing 4.36 is here:

```
Apples
10
```


CLAUDE 3, DATAFRAMES, AND SIMPLE STATISTICS

The following prompt was given to Claude 3:

Please generate Python code that creates a Pandas dataframe with four rows and five columns of randomly generated numbers and then calculates the mean, variance, and standard deviation of the numbers in each column of the dataframe, along with an explanation of the Python code.

Claude 3:

LISTING 4.37: `claude3_statistics.py`

```
import pandas as pd
import numpy as np

# Define number of rows and columns
nrows = 4
ncols = 5

# Generate random data
np.random.seed(1)
data = np.random.randint(1, 100, size=(nrows, ncols))

# Create Pandas DataFrame
df = pd.DataFrame(data)

# Calculate descriptive statistics for each column
for col in df.columns:
    # Calculate mean, variance, and standard deviation
    mean = df[col].mean()
    variance = df[col].var()
    std_dev = df[col].std()

    # Print the results
    print(f"Column {col}:")
    print(f"\tMean: {mean:.2f}")
    print(f"\tVariance: {variance:.2f}")
    print(f"\tStandard Deviation: {std_dev:.2f}")
    print("")
```

Explanation:

Import libraries:

We import Pandas for data manipulation and NumPy for generating random numbers.

Define parameters:

We set the number of rows (`nrows`) and columns (`ncols`) for the DataFrame.

Generate random data:

We use `np.random.seed(1)` to ensure reproducibility of the random numbers.

Then, we use `np.random.randint(1, 100, size=(nrows, ncols))` to generate an array of random integers between 1 and 100 with the specified number of rows and columns.

Create Pandas DataFrame:

We use `pd.DataFrame(data)` to convert the generated NumPy array into a Pandas DataFrame.

Calculate descriptive statistics for each column:

We iterate through each column of the DataFrame using a for loop.

For each column, we calculate the following descriptive statistics:

mean: using `df[col].mean()`

variance: using `df[col].var()`

standard deviation: using `df[col].std()`

We format the results to two decimal places and print them.

This code will generate a Pandas DataFrame with randomly generated numbers and then calculate the mean, variance, and standard deviation for each column. You can modify this code to calculate other descriptive statistics or to perform different operations on the DataFrame.

Bard may display inaccurate info, including about people, so double-check its responses. Your privacy & Bard

Launch the code in Listing 4.37 and you will see the following output:

Column 0:

Mean: 35.50

Variance: 936.33

Standard Deviation: 30.60

Column 1:

Mean: 46.00

Variance: 1216.67

Standard Deviation: 34.88

Column 2:

Mean: 57.50

Variance: 1201.00

Standard Deviation: 34.66

Column 3:

Mean: 16.25

Variance: 50.92

Standard Deviation: 7.14

Column 4:

Mean: 39.50

Variance: 993.67

Standard Deviation: 31.52

USEFUL ONE_LINE COMMANDS IN PANDAS

This section contains an eclectic mix of one-line commands in Pandas (some of which you have already seen in this chapter) that are useful to know:

Save a data frame to a csv file (comma separated and without indices):

```
df.to_csv("data.csv", sep=",", index=False)
```

List the column names of a DataFrame:

```
df.columns
```

Drop missing data from a DataFrame:

```
df.dropna(axis=0, how='any')
```

Replace missing data in a DataFrame:

```
df.replace(to_replace=None, value=None)
```

Check for NANs in a DataFrame:

```
pd.isnull(object)
```

Drop a feature in a DataFrame:

```
df.drop('feature_variable_name', axis=1)
```

Convert object type to float in a DataFrame:

```
pd.to_numeric(df["feature_name"], errors='coerce')
```

Convert data in a DataFrame to NumPy array:

```
df.as_matrix()
```

Display the first n rows of a DataFrame:

```
df.head(n)
```

Get data by feature name in a DataFrame:

```
df.loc[feature_name]
```

Apply a function to a DataFrame: multiply all values in the “height” column of the data frame by 3:

```
df["height"].apply(lambda height: 3 * height)
```

OR:

```
def multiply(x):
    return x * 3
df["height"].apply(multiply)
```

Rename the fourth column of the data frame as “height”:

```
df.rename(columns = {df.columns[3]: 'height'}, inplace=True)
```

Get the unique entries of the column “first” in a DataFrame:

```
df["first"].unique()
```

Create a dataframe with columns “first” and “last” from an existing DataFrame:

```
new_df = df[["name", "size"]]
```

Sort the data in a DataFrame:

```
df.sort_values(ascending = False)
```

Filter the data column named “size” to display only values equal to 7:

```
df[df["size"] == 7]
```

Select the first row of the “height” column in a DataFrame:

```
df.loc([0], ['height'])
```

SUMMARY

This chapter introduced readers to Pandas for creating labeled Dataframes and displaying metadata of Pandas Dataframes. Then readers learned how to create Pandas Dataframes from various sources of data, such as random numbers and hard_coded data values.

Readers also learned how to read Excel spreadsheets and perform numeric calculations on that data, such as the min, mean, and max values in numeric columns. Then you saw how to create Pandas Dataframes from data stored in files. Then readers learned how to invoke a Web service to retrieve data and populate a Pandas DataFrame with that data. In addition, they learned how to generate a scatterplot from data in a Pandas DataFrame.

THE GENERATIVE AI LANDSCAPE

This chapter provides a fast-paced introduction to generative AI that highlights some of the important features of generative AI.

The first section of this chapter introduces readers to generative AI, including key features and techniques. It also explores the difference between conversational AI and generative AI.

The second section of this chapter discusses generative AI models, such as Dall-E, ChatGPT, and GPT-4. Readers will also learn about generative AI versus machine learning (ML), deep learning (DL), and natural language processing (NLP).

The third section of this chapter poses open-ended types of questions regarding artificial general intelligence (AGI), and a brief comparison of AGI versus generative AI. The fourth part of this chapter introduces large language models (LLMs), along with various facets of LLMs, such as their cost, size versus performance, and emergent abilities.

The fifth section of this chapter introduces readers to the concept of hallucinations, which is an important topic: All LLMs suffer from hallucinations. In fact, LLMs can unintentionally generate non-sensical and even hilarious responses to prompts.

The final section of this chapter starts with a brief introduction to several companies that make significant contributions in AI and NLP. Indeed, readers will become very familiar with these companies if they plan to pursue a career in NLP.

One detail to keep in mind: The term *prompt* in this chapter refers to any text-based string that one provides to a large language model (LLM), and the response from the LLM is a text-based string that is called a *completion*.

WHAT IS GENERATIVE AI?

Generative AI refers to a subset of artificial intelligence models and techniques that are designed to generate new data samples that are similar in nature to a

given set of input data. The goal is to produce content or data that wasn't part of the original training set but is coherent, contextually relevant, and in the same style or structure.

Generative AI stands apart in its ability to create and innovate, as opposed to merely analyzing or classifying. The advancements in this field have led to breakthroughs in creative domains and practical applications, making it a cutting-edge area of AI research and development.

Key Features of Generative AI

The following bullet list contains key features of generative AI, followed by a brief description for each bullet item:

- data generation
- synthesis
- learning distributions

Data generation refers to the ability to create new data points that are not part of the training data but resemble it. This can include text, images, music, videos, or any other form of data.

Synthesis means that generative models can blend various inputs to generate outputs that incorporate features from each input, like merging the styles of two images.

Learning distributions indicates generative AI models aim to learn the probability distribution of the training data so they can produce new samples from that distribution.

Popular Techniques in Generative AI

Generative adversarial networks (GANs): GANs consist of two networks, a generator and a discriminator, that are trained simultaneously. The generator tries to produce fake data, while the discriminator tries to distinguish between real data and fake data. Over time, the generator gets better at producing realistic data.

Variational autoencoders (VAEs): VAEs are probabilistic models that learn to encode and decode data in a manner that the encoded representations can be used to generate new data samples.

Recurrent neural networks (RNNs): Used primarily for sequence generation, such as text or music.

What Makes Generative AI Different

Creation versus classification: While most traditional AI models aim to classify input data into predefined categories, generative models aim to create new data.

Unsupervised learning: Many generative models, especially GANs and VAEs, operate in an unsupervised manner, meaning they don't require labeled data for training.

Diverse outputs: Generative models can produce a wide variety of outputs based on learned distributions, making them ideal for tasks like art generation, style transfer, and more.

Challenges: Generative AI poses unique challenges, such as mode collapse in GANs or ensuring the coherence of generated content.

Furthermore, there are numerous areas that involve generative AI applications, some of which are listed in the following bullet list:

- art and music creation
- data augmentation
- style transfer
- text generation
- image synthesis
- drug discovery

Art and music creation includes generating paintings, music, or other forms of art.

Data augmentation involves creating additional data for training models, especially when the original dataset is limited.

Style transfer refers to applying the style of one image to the content of another.

Text generation is a very popular application of generative AI that involves creating coherent and contextually relevant text.

Image synthesis is another popular area of generative AI that involves generating realistic images, faces, or even creating scenes for video games.

Drug discovery is a very important facet of generative AI that pertains to generating molecular structures for new potential drugs.

CONVERSATIONAL AI VERSUS GENERATIVE AI

Both conversational AI and generative AI are prominent subfields within the broader domain of artificial intelligence. These subfields have a different focus regarding their primary objective, the technologies that they use, and applications.

The primary differences between the two subfields are in the following sequence of bullet points:

- primary objective
- applications
- technologies used
- training and interaction
- evaluation
- data requirements

Primary Objective

The main goal of conversational AI is to facilitate human-like interactions between machines and humans. This includes chatbots, virtual assistants, and other systems that engage in dialogue with users.

The primary objective of generative AI is to create new content or data that wasn't in the training set but is similar in structure and style. This can range from generating images, music, and text to more complex tasks like video synthesis.

Applications

Common applications for conversational AI include customer support chatbots, voice-operated virtual assistants (like Siri or Alexa), and interactive voice response (IVR) systems.

Common applications for generative AI have a broad spectrum of applications such as creating art or music, generating realistic video game environments, synthesizing voices, and producing realistic images or even deep fakes.

Technologies Used

Conversational AI often relies on natural language processing (NLP) techniques to understand and generate human language. This includes intent recognition, entity extraction, and dialogue management.

Generative AI commonly utilizes generative adversarial networks (GANs), variational autoencoders (VAEs), and other generative models to produce new content.

Training and Interaction

While training can be supervised, semi-supervised, or unsupervised, the primary interaction mode for conversational AI is through back-and-forth dialogue or conversation.

The training process for generative AI, especially with models like GANs, involves iterative processes where the model learns to generate data by trying to fool a discriminator into believing the generated data is real.

Evaluation

Conversational AI evaluation metrics often revolve around understanding and response accuracy, user satisfaction, and the fluency of generated responses.

Generative AI evaluation metrics for models like GANs can be challenging and might involve using a combination of quantitative metrics and human judgment to assess the quality of generated content.

Data Requirements

Data requirements for conversational AI typically involves dialogue data, with conversations between humans or between humans and bots.

Data requirements for generative AI involves large datasets of the kind of content it is supposed to generate, be it images, text, music, and so on.

Although both conversational AI and generative AI deal with generating outputs, their primary objectives, applications, and methodologies can differ significantly. Conversational AI is geared toward interactive communication with users, while generative AI focuses on producing new, original content.

WHAT ARE GENERATIVE AI MODELS?

The following prompt was given to GPT-4o.

Please describe types of generative AI models.
GPT-4o:

Generative AI models are designed to generate new data that mimics the patterns and characteristics of a given dataset. They can be used for a variety of tasks such as image generation, text generation, music composition, and more. Here's a detailed description of different types of generative AI models:

1. Generative Adversarial Networks (GANs)

Description: GANs consist of two neural networks, a generator and a discriminator, that are trained simultaneously. The generator creates fake data, and the discriminator tries to distinguish between real and fake data. The generator aims to produce data that is indistinguishable from real data.

Key Components:

1. Generator: Produces new data instances.

Discriminator: Evaluates the authenticity of the data produced by the generator.

Applications: Image generation, video generation, data augmentation, super-resolution.

Example: Generating realistic human faces.

2. Variational Autoencoders (VAEs)

Description: VAEs are a type of autoencoder that learns to encode input data into a latent space and then decode it back to the original data. They incorporate probabilistic elements by assuming a distribution over the latent space.

Key Components:

Encoder: Compresses the input data into a latent space representation.

Decoder: Reconstructs the data from the latent space representation.

Applications: Image generation, anomaly detection, data compression.

Example: Generating new samples of handwritten digits similar to those in the MNIST dataset.

3. Autoregressive Models

Description: Autoregressive models generate each data point as a function of the previous data points. They model the conditional distribution of each data point given the previous ones.

Key Components:

Sequence Modeling: Generates each token or data point based on the previous sequence.

Applications: Text generation, time series forecasting, audio synthesis.

Example: Language models like GPT (generative pretrained transformer) generate text one word at a time.

4. Normalizing Flows

Description: Normalizing flows transform a simple distribution (e.g., a Gaussian) into a more complex distribution using a series of invertible and differentiable mappings.

Key Components:

Invertible Transformations: Series of transformations that map simple distributions to complex distributions.

Applications: Density estimation, image generation.

Example: Using normalizing flows to generate realistic images from noise.

5. Energy-Based Models (EBMs)

Description: EBMs define a probability distribution over data by associating each data point with an energy score. The model is trained to assign low energy to real data and high energy to fake data.

Key Components:

Energy Function: Assigns a score to each data point.

Applications: Image generation, outlier detection.

Example: Training an energy-based model to generate realistic images by minimizing the energy assigned to real images.

6. Diffusion Models

Description: Diffusion models learn to reverse a diffusion process, where data is gradually corrupted by noise, to generate new samples. These models are trained to denoise data step by step.

Key Components:

Diffusion Process: Gradually adds noise to data.

Denosing Process: Learns to reverse the noise process.

Applications: Image generation, speech synthesis.

Example: Generating high-quality images by reversing the process of adding Gaussian noise to images.

7. Score-Based Generative Models

Description: Score-based models generate data by iteratively refining random noise using the gradient of the data density, referred to as the score function.

Key Components:

Score Function: Gradient of the data density used for refinement.

Applications: Image generation, data imputation.

Example: Using score-based methods to generate photo-realistic images from noise.

8. Flow-Based Models

Description: Flow-based models use a series of invertible transformations to map a simple distribution to a complex data distribution. These models provide exact log-likelihood computation and efficient sampling.

Key Components:

Invertible Transformations: Series of transformations that map simple distributions to complex distributions.

Applications: Density estimation, image generation.

Example: RealNVP and Glow models for generating high-fidelity images.

9. Transformer-Based Models

Description: Transformer models use self-attention mechanisms to process input data and generate new data. They are particularly effective for sequential data.

Key Components:

Self-Attention Mechanism: Allows the model to focus on different parts of the input sequence.

Applications: Text generation, machine translation, summarization.

Example: GPT-3 and BERT are transformer-based models used for generating human-like text.

IS DALL-E PART OF GENERATIVE AI?

DALL-E and similar tools that generate graphics from text are indeed examples of generative AI. In fact, DALL-E is one of the most prominent examples of generative AI in the realm of image synthesis.

Here's a bullet list of generative characteristics of DALL-E, followed by brief descriptions of each bullet item:

- image generation
- learning distributions
- innovative combinations
- broad applications
- transformer architecture

Image generation: This is a key feature of DALL-E, designed to generate images based on textual descriptions. Given a prompt like “a two-headed flamingo,” DALL-E can produce a novel image that matches the description, even if it's never seen such an image in its training data.

Learning distributions: Like other generative models, DALL-E learns the probability distribution of its training data. When it generates an image, it samples from this learned distribution to produce visuals that are plausible based on its training.

Innovative combinations: DALL-E can generate images that represent entirely novel or abstract concepts, showcasing its ability to combine and recombine learned elements in innovative ways.

Broad applications: In addition to image synthesis, DALL-E has provided broad application support, in areas like art generation, style blending, and creating images with specific attributes or themes, highlighting its versatility as a generative tool.

Transformer architecture: DALL-E leverages a variant of the transformer architecture, similar to models like GPT-3, but adapted for image generation tasks.

Other tools that generate graphics, art, or any form of visual content based on input data (whether it's text, another image, or any other form of data) and can produce outputs not explicitly present in their training data are also

considered generative AI. They showcase the capability of AI models to not just analyze and classify but to create and innovate.

ARE CHATGPT-3 AND GPT-4 PART OF GENERATIVE AI?

Both ChatGPT-3 and GPT-4 are LLMs that are considered examples of generative AI. They belong to a class of models called “transformers,” which are particularly adept at handling sequences of data, such as text-related tasks.

The following bullet list provides various reasons why these LLMs are considered generative, followed by a brief description of each bullet item:

- text generation
- learning distributions
- broad applications
- unsupervised learning

Text generation: These models can produce coherent, contextually relevant, and often highly sophisticated sequences of text based on given prompts. They generate responses that weren’t explicitly present in their training data but are constructed based on the patterns and structures they learned during training.

Learning distributions: GPT-3, GPT-4, and similar models learn the probability distribution of their training data. When generating text, they’re essentially sampling from this learned distribution to produce sequences that are likely based on their training.

Broad applications: Beyond just text-based chat or conversation, these models can be used for a variety of generative tasks like story writing, code generation, poetry, and even creating content in specific styles or mimicking certain authors, showcasing their generative capabilities.

Unsupervised learning: While they can be fine-tuned with specific datasets, models like GPT-3 are primarily trained in an unsupervised manner on vast amounts of text, learning to generate content without requiring explicit labeled data for every possible response.

In essence ChatGPT-3, GPT-4, and similar models by OpenAI are quintessential examples of generative AI in the realm of NLP and generation.

GENERATIVE AI VERSUS ML, DL, AND NLP

The following prompt was given to ChatGPT to describe the relationship between generative AI and ML, DL, and NLP.

What is the relationship between generative AI versus ML, DL, and NLP?

ChatGPT:

Generative AI, machine learning (ML), deep learning (DL), natural language processing (NLP), and reinforcement learning (RL) are interconnected fields within artificial intelligence, each with its own focus and methodologies. Understanding the relationship between these disciplines helps in grasping how advancements in one area can influence or support developments in others.

Machine Learning (ML)

Definition: Machine learning is a broad field of artificial intelligence focused on designing and developing algorithms that allow computers to learn from and make decisions based on data. ML encompasses a variety of techniques including supervised learning, unsupervised learning, and reinforcement learning.

Relationship with Generative AI: Generative AI is a subset of machine learning focused specifically on generating new data instances that resemble the training data. Techniques such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are ML methods used in generative AI.

Deep Learning (DL)

Definition: Deep Learning is a subset of machine learning that uses neural networks with multiple layers (hence “deep”) to analyze various factors of data. DL has significantly advanced fields such as image recognition, speech recognition, and natural language processing.

Relationship with Generative AI: Many generative AI models are built using deep learning techniques. For example, GANs and transformer-based models like GPT (Generative pretrained transformer) rely on deep neural networks to generate text, images, and other forms of media.

Natural Language Processing (NLP)

Definition: NLP is a branch of artificial intelligence that deals with the interaction between computers and humans through natural language. The goal is to enable computers to understand, interpret, and produce human language in a valuable way.

Relationship with Generative AI: Generative AI techniques are extensively used in NLP to generate human-like text, translate languages, create summaries, and more. Models like GPT-3 are prime examples of generative AI applied to NLP, where the model generates coherent and contextually relevant text based on prompts it receives.

Summary

Generative AI uses machine learning (particularly deep learning) algorithms to create content. It often leverages structures studied in NLP to understand and

generate human language, and it can utilize reinforcement learning techniques to optimize its generative abilities based on feedback. Thus, while each of these fields has its distinct focus and methodologies, they are deeply interlinked and often complementary in developing more effective and intelligent AI systems.

Which Fields Benefit the Most From Generative AI?

The following prompt was given to ChatGPT to determine which fields among ML, DL, NLP, and RL will benefit the most from generative AI:

Which fields among ML, DL, NLP, and RL will benefit the most from generative AI?
ChatGPT:

Generative AI, a branch of artificial intelligence focused on generating new data that resembles the training data, holds significant potential to impact various fields within AI. Here's how it particularly stands out in contributing to machine learning (ML), deep learning (DL), natural language processing (NLP), and reinforcement learning (RL):

Machine Learning (ML)

- **Potential Impact:** Medium to High
- **How Generative AI Contributes:**
 - **Data Augmentation:** Generative AI can create synthetic data that helps in training ML models where real data is scarce, sensitive, or expensive to collect. This is especially useful in fields like healthcare or finance, where data privacy concerns or rarity of cases can limit data availability.
 - **Feature Learning:** Generative models can help in unsupervised learning scenarios to discover and learn useful features from data, which can then be used for various ML tasks.

Deep Learning (DL)

- **Potential Impact:** High
- **How Generative AI Contributes:**
 - **Model Improvement:** Generative models like GANs and autoencoders are fundamentally deep learning models. They push the boundaries in image generation, style transfer, and more, driving significant advancements in DL techniques.
 - **Anomaly Detection:** Generative models are increasingly used for anomaly detection in video surveillance, fraud detection, and other areas, benefiting from DL's capabilities in handling vast amounts of data.

Natural Language Processing (NLP)

- **Potential Impact:** Very High

- **How Generative AI Contributes:**
 - **Content Creation:** Generative AI is pivotal in text generation, enabling applications like chatbots, writing assistants, and automated journalism. Models like GPT (generative pretrained transformer) are groundbreaking tools in NLP, enhancing how machines generate and understand language.
 - **Language Translation:** Generative models improve machine translation by generating more fluent and accurate translations and handling multiple languages effectively.

Reinforcement Learning (RL)

- **Potential Impact:** Medium
- **How Generative AI Contributes:**
 - **Simulated Environments:** Generative AI can be used to create rich, diverse, and complex simulation environments that help train RL agents more effectively, particularly in scenarios where real-world interaction is risky, costly, or impractical.
 - **Policy Learning:** In some approaches, generative models can be used to predict and simulate the outcomes of different actions, helping to refine the policy network in RL by providing a richer set of experiences for the agent to learn from.

Overall Assessment

- **Most Benefited Field: NLP.** The integration of generative AI into NLP has been transformative, significantly advancing how machines generate human-like text and respond in interactive applications. The impact here is profound, as it not only enhances existing technologies but also enables new kinds of applications and services that were previously not feasible.
- **Deep Learning:** Generative AI techniques are integral to the evolution of deep learning, continually pushing forward the capabilities of neural networks in processing and generating complex data forms like images, video, and sound.
- **Machine Learning and Reinforcement Learning:** While the impact is slightly more indirect, generative AI supports these fields through enhanced data availability and improved training scenarios, respectively.

In conclusion, generative AI acts as a critical driver of innovation across all these fields, but its role in NLP and DL stands out due to the direct integration and dramatic enhancements it brings to core applications in these areas.

Generative AI in the Enterprise Space

The following prompt was given to GPT-4 to describe how the Enterprise space will benefit from generative AI:

How Will the Enterprise Space Benefit From Generative AI?
ChatGPT:

Generative AI has significant potential to transform various aspects of the enterprise space. Its capabilities can be leveraged to improve efficiency, innovation, and decision-making across multiple industries. Here's how enterprises are likely to benefit from generative AI:

1. **Content Creation:** Generative AI can assist in creating a wide range of content, from marketing materials and business reports to code and legal documents. This can significantly speed up content generation processes and free up human resources for more strategic tasks.
2. **Personalization:** In marketing and customer service, generative AI can be used to create personalized content for customers, enhancing customer engagement and satisfaction. For example, AI can generate personalized email responses or recommend content tailored to individual user preferences.
3. **Product Development:** Generative AI can help design new products by simulating and generating numerous iterations quickly, which can be particularly beneficial in industries like pharmaceuticals, automotive, and fashion. For instance, it can be used to model new drugs or optimize car designs for performance and safety.
4. **Data Analysis and Insight Generation:** Generative AI can analyze large volumes of data and generate insights, reports, or visualizations. This can be especially useful in fields like finance and healthcare, where decision-makers can benefit from deep insights derived from complex datasets.
5. **Automation:** Many routine tasks can be automated with the help of generative AI, from scheduling and logistics to generating standard legal agreements and technical documentation. This not only reduces the burden of repetitive tasks but also increases operational efficiency.
6. **Innovation in Customer Interactions:** AI-driven chatbots and virtual assistants can handle customer inquiries and provide support, ensuring 24/7 customer service without the need for continuous human intervention. These systems can be continuously updated and improved with new data.
7. **Training and Education:** Generative AI can create realistic training scenarios for various purposes, such as emergency response drills, medical training simulations, or customer service role-playing. This helps in providing hands-on experience without the real-world risks.

8. **Enhancing Creativity:** By providing suggestions and generating ideas, generative AI can aid human creativity in fields like design, advertising, and even strategy development. This can lead to innovative solutions and creative outputs that might not have been considered otherwise.

Challenges to Consider

While the benefits are substantial, enterprises need to manage challenges such as ensuring the quality and accuracy of AI-generated outputs, maintaining privacy and security, and addressing ethical considerations like bias in AI models.

Strategic Integration

For enterprises, the key to leveraging generative AI effectively lies in integrating these technologies strategically into business processes. This involves training teams, setting up governance frameworks, and continually assessing the impact of AI applications on business outcomes and customer satisfaction.

In conclusion, generative AI offers a plethora of opportunities for the enterprise space to enhance productivity, creativity, and decision-making. By adopting this technology, businesses can stay competitive in a rapidly evolving digital landscape.

The Impact of Generative AI on Jobs

The following prompt was given to GPT-4 to describe the impact of generative AI on jobs:

What will be the impact of generative AI on jobs?
ChatGPT:

The impact of generative AI on jobs is a subject of considerable debate and speculation. While it has the potential to transform numerous industries by automating tasks, enhancing productivity, and creating new opportunities, it also raises concerns about job displacement. Here are some key aspects of how generative AI might affect the job market:

Positive Impacts

1. **Job Creation:** Generative AI can lead to the creation of new job categories, particularly in fields like AI maintenance, development, and regulation. New roles may emerge in designing, programming, and managing AI systems.
2. **Enhancing Productivity:** By automating routine and mundane tasks, generative AI can free up workers to focus on more complex, creative, and strategic activities, potentially increasing job satisfaction and efficiency.
3. **Economic Growth:** The efficiencies and capabilities introduced by AI can lead to lower costs, improved quality of products and services, and faster

economic activity, which could generate more business opportunities and thus potentially create jobs.

4. **Education and Training:** There will be a growing need for education and training programs tailored to equip the workforce with AI-related skills. This not only helps workers transition to new roles but also stimulates sectors like educational technology and professional development.

Negative Impacts

1. **Job Displacement:** Certain jobs, especially those involving routine data processing, customer support, and even some aspects of content creation, are at risk of being automated by AI. This could lead to significant job displacement in sectors like administrative services, support call centers, and even parts of journalism.
2. **Skill Gaps and Inequality:** The rapid adoption of AI could widen existing skill gaps, as workers who lose jobs to automation might not have immediate access to the training necessary to switch to new roles. This could exacerbate economic inequalities, especially affecting those in lower-income jobs or regions with less access to training resources.
3. **Wage Pressure:** As AI technologies automate more tasks, there could be downward pressure on wages for remaining jobs that AI can partially but not fully automate. Workers may find themselves competing with AI capabilities, which could lead to wage stagnation or decline in some industries.
4. **Regulatory and Ethical Challenges:** The introduction of AI in sensitive areas like healthcare, law enforcement, and finance raises significant ethical and regulatory challenges. Mismanagement or ethical oversights could lead to job losses or reputational damage in affected industries.

Mitigating the Negative Impacts

To address these challenges, a combination of proactive governance, corporate responsibility, and educational initiatives will be essential. Governments and educational institutions will need to invest in upskilling and reskilling programs, while businesses will need to consider the ethical implications of deploying AI technologies. Transparent policies and community engagement can help ensure that the benefits of generative AI are distributed fairly across the economy.

Conclusion

The overall impact of generative AI on jobs will likely be a blend of challenges and opportunities. The key to maximizing the positive aspects while minimizing the negative effects will lie in how well society can anticipate these changes and prepare the workforce for the evolving technological landscape. This

preparation includes education, policy-making, and the development of new economic models that can accommodate the rise of AI.

WHAT IS ARTIFICIAL GENERAL INTELLIGENCE (AGI)?

The following prompt was given to Gemini to describe AGI:

What is AGI?

Gemini:

AGI stands for artificial general intelligence.

It's a hypothetical type of artificial intelligence that would possess human-like or even surpassing cognitive abilities. Here's a breakdown of the concept:

Core Characteristics of AGI:

General Intelligence: Unlike most current AI systems that are designed for specific tasks, AGI would have the ability to learn, reason, and solve problems in a variety of domains, similar to a human.

Adaptability: An AGI could adapt to new situations and environments, constantly learning and improving its performance.

Understanding and Reasoning: It would go beyond simple pattern recognition to understand the world in a more comprehensive way, including complex concepts, relationships, and cause and effect.

Social Intelligence: The ability to interact with humans and other intelligent beings in a meaningful way, potentially including understanding emotions and intentions.

Current State of AGI:

Currently, AGI remains a theoretical concept. While AI has made significant progress in specific areas like machine learning and computer vision, achieving true general intelligence is considered a major challenge. Some of the hurdles include:

The Complexity of Human Intelligence: Human intelligence is multifaceted, encompassing various cognitive abilities that are not fully understood.

Limited Understanding of Consciousness: We still lack a deep understanding of consciousness, making it difficult to replicate in machines.

Data and Computational Requirements: Training an AGI might require vast amounts of data and immense computational power that may not be readily available yet.

Potential Benefits of AGI:

If achieved, AGI could revolutionize various fields:

Scientific Discovery: AGI could assist in scientific research by tackling complex problems and generating new hypotheses.

Technological Innovation: It could accelerate technological advancement by helping to design and develop new technologies.

Improved Quality of Life: AGI could contribute to solutions for global challenges like climate change and disease.

Challenges and Risks:

The development of AGI also raises some ethical and existential concerns:

Safety and Control: Ensuring that an AGI remains aligned with human values and goals is crucial.

Job Displacement: Extensive automation by AGI could lead to widespread job displacement.

Existential Risk: Some experts warn that uncontrolled superintelligence could pose an existential threat to humanity.

Overall, AGI is a fascinating concept with the potential to bring immense benefits. Significant research and careful consideration of the ethical implications are necessary before we reach that stage.

When Will AGI Arrive?

The following prompt was given to GPT-4 to describe AGI and compare it with Generative AI:

When do you think that AGI will arrive?
ChatGPT:

Predicting when artificial general intelligence (AGI) will arrive is highly speculative and varies significantly among experts in the field. The development of AGI is a complex and multifaceted challenge that depends on numerous scientific, technological, and philosophical breakthroughs.

Factors Influencing AGI Development:

1. **Technological Advancements:** The rate of progress in computing power, algorithms, and data availability plays a critical role.
2. **Research Investment:** The amount of global investment in AI research and development can accelerate or delay progress.

3. **Ethical and Safety Considerations:** Ensuring that AGI systems are safe and ethically aligned with human values could require significant additional research and regulation, potentially slowing progress.

Expert Opinions:

- Some experts believe AGI could be developed by the mid-twenty-first century if current trends in AI research and computational improvements continue.
- Others are more conservative, suggesting that AGI is several decades away, or even more than a century into the future.
- A few even question whether AGI is achievable at all or if it should be pursued, given the potential risks associated with it.

Surveys and Predictions: Surveys conducted among AI researchers vary widely in their predictions, with median estimates sometimes pointing to the possibility of AGI by around 2040–2060. These estimates often come with large confidence intervals, reflecting the high level of uncertainty in such predictions.

In summary, the arrival time of AGI is still uncertain and subject to many variables. It remains an area of active debate and research within the AI community.

What Is the Path to AGI?

The following prompt was given to Llama 3 regarding a potential path toward achieving AGI in the future.

Do you think the path to AGI involves mimicking the way that children learn?

What a fascinating question!

Llama 3:

Mimicking the way children learn is an intriguing approach to achieving artificial general intelligence (AGI). Children's learning processes are remarkable and understanding how they acquire knowledge and skills can indeed provide valuable insights for AI development.

Here are some reasons why mimicking children's learning might be beneficial for AGI:

1. **Curiosity-driven learning:** Children are naturally curious, and their learning is often driven by a desire to understand and explore their environment. AGI systems could benefit from incorporating similar curiosity-driven learning mechanisms.

2. Self-supervised learning: Children learn largely through self-supervised learning, where they discover patterns and relationships in their environment without explicit supervision. AGI systems could leverage similar self-supervised learning approaches to learn from large amounts of data.
3. Incremental learning: Children learn incrementally, building upon previously acquired knowledge and skills. AGI systems could adopt similar incremental learning strategies to improve their performance over time.
4. Multimodal learning: Children learn through multiple senses (vision, hearing, touch, etc.) and modalities (language, gestures, etc.). AGI systems could benefit from incorporating multimodal learning capabilities to better understand and interact with their environment.
5. Social learning: Children learn from others, including caregivers, peers, and even media. AGI systems could leverage social learning mechanisms to learn from humans and other agents.

Consider the following challenges and limitations:

1. Complexity of human learning: Human learning is a complex, multifaceted process that is still not fully understood. Replicating this process in AGI systems might be difficult, if not impossible.
2. Lack of understanding of human cognition: Despite significant research, the underlying mechanisms of human cognition and learning are still not fully understood. This limited understanding might hinder the development of AGI systems that mimic human learning.
3. Scalability and efficiency: Children's learning is often slow and laborious, which might not be suitable for AGI systems that require fast and efficient learning.
4. Safety and ethics: AGI systems that learn like children might raise safety and ethical concerns, such as the potential for biased or harmful learning.

In conclusion, while mimicking the way children learn can provide valuable insights for AGI development, it is essential to acknowledge the challenges and limitations involved. A more likely approach might be to draw inspiration from children's learning processes while also incorporating domain-specific knowledge and constraints to ensure safe and efficient learning.

How Can We Prepare for AGI?

The following prompt was given to GPT-4 regarding AGI and humans:

How can we prepare for AGI?
Claude 3:

Preparing for the potential emergence of artificial general intelligence (AGI) is a complex and multifaceted challenge. Here are some key considerations and steps that can be taken to better prepare for AGI:

1. Research and Development:

- Continued research into the fundamental principles and techniques underlying AGI is crucial to understand its potential capabilities, limitations, and risks.
- Collaboration between academia, industry, and government can foster the development of safe and beneficial AGI systems.
- Interdisciplinary research, involving fields such as computer science, cognitive science, neuroscience, and ethics, can provide a holistic understanding of AGI and its implications.

2. Safety and Robustness:

- Developing robust safety measures and frameworks is essential to ensure that AGI systems behave in a safe, controlled, and predictable manner.
- Techniques like value alignment, corrigibility, and interruptibility should be investigated to align AGI goals with human values and allow for human oversight and control.
- Rigorous testing and validation procedures should be established to assess the safety and reliability of AGI systems before deployment.

3. Ethics and Governance:

- Establishing ethical guidelines and governance frameworks for the development and deployment of AGI is crucial to ensure responsible and beneficial use.
- International collaboration and policy discussions are necessary to address the global implications of AGI and develop coordinated strategies for its governance.
- Engaging diverse stakeholders, including policymakers, ethicists, and the general public, can help shape the ethical and societal considerations surrounding AGI.

4. Workforce Preparation and Adaptation:

- As AGI may have significant impacts on the job market and the nature of work, preparing the workforce for potential disruptions and transitions is important.
- Investing in education and training programs that focus on skills complementary to AGI, such as creativity, critical thinking, and emotional intelligence, can help individuals adapt to new roles and opportunities.
- Policies and support systems should be developed to assist workers affected by automation and ensure a smooth transition to new forms of employment.

5. Public Awareness and Engagement:

- Raising public awareness about AGI, its potential benefits, risks, and implications is essential to foster informed discussions and decision-making.
- Engaging the public through education, outreach, and participatory processes can help build trust and ensure that the development of AGI aligns with societal values and priorities.

- Encouraging open and transparent communication about AGI research and development can promote accountability and public understanding.

6. Collaboration and Coordination:

- International collaboration and coordination among researchers, institutions, and governments are necessary to address the global challenges posed by AGI.
- Establishing shared research agendas, standards, and best practices can help ensure a cohesive and responsible approach to AGI development.
- Collaboration across disciplines and sectors can facilitate the exchange of knowledge, resources, and expertise needed to tackle the complexities of AGI.

7. Scenario Planning and Preparedness:

- Engaging in scenario planning exercises can help anticipate and prepare for various potential outcomes and trajectories of AGI development.
- Developing contingency plans and response strategies for different AGI scenarios can improve resilience and adaptability in the face of uncertainties.
- Regularly updating and refining these plans based on new insights and developments is essential to stay prepared for the evolving landscape of AGI.

Preparing for AGI is an ongoing process that requires continuous monitoring, adaptation, and proactive measures. It is important to recognize that the path to AGI is uncertain and may unfold in unpredictable ways. Therefore, maintaining flexibility, open-mindedness, and a commitment to responsible and ethical development is crucial as we navigate the challenges and opportunities posed by AGI.

Will AGI Control the World?

The following prompt was given to Claude 3 regarding AGI:

Do you think that AGI will eventually control the world?
Claude 3:

The question of whether AGI will eventually control the world is a complex and speculative one, with various perspectives and uncertainties involved. Here are a few thoughts on the matter:

1. Capability versus Intention:

- The ability of AGI to control the world would depend not only on its capabilities but also on its goals and intentions.
- If AGI is developed with aligned values and goals that prioritize human well-being and flourishing, it may not necessarily seek to control or dominate the world.
- It can be troubling to think that if AGI is developed without proper safety measures and value alignment, there is a risk that it could pursue

goals that are misaligned with human values, potentially leading to unintended consequences.

2. Human Agency and Oversight:

- The extent to which AGI could control the world would also depend on the level of human agency and oversight in its development and deployment.
- If AGI is developed and implemented with strong human governance, ethical guidelines, and control mechanisms, it is less likely to gain unchecked control over the world.
- Maintaining human involvement, accountability, and the ability to intervene or shut down AGI systems if necessary can help mitigate the risk of AGI taking over.

3. Societal and Political Factors:

- The impact of AGI on the world will also be influenced by various societal, political, and economic factors.
- The adoption and integration of AGI into different domains will depend on public perception, trust, and the regulatory frameworks in place.
- The distribution of benefits and risks associated with AGI will also shape its influence and potential for control.

4. Technological Limitations and Uncertainties:

- While AGI has the potential to surpass human intelligence in many domains, it is important to recognize that there are still significant technological challenges and uncertainties in its development.
- The path to AGI is not clearly defined, and there may be limitations or constraints that we are currently unaware of.
- It is also possible that AGI may take forms or exhibit behaviors that are difficult for us to predict or comprehend, making it challenging to assert definitive control.

5. Collaboration and Coordination:

- The development of AGI is likely to be a global effort involving multiple actors, including governments, corporations, and research institutions.
- The extent to which AGI could control the world would also depend on the level of collaboration, coordination, and shared responsibility among these actors.
- If the development of AGI is guided by a collective commitment to safety, ethics, and the common good, it is more likely to be shaped in a way that benefits humanity rather than seeking control.

Ultimately, the question of whether AGI will control the world is a complex and open-ended one, with many variables and uncertainties involved. It is important to approach the development of AGI with caution, foresight, and a strong commitment to safety, ethics, and human values. Ongoing research,

public discourse, and proactive measures to ensure the responsible development and deployment of AGI will be crucial in shaping its impact on the world.

Should Humans Fear AGI?

The following prompt was given to GPT-4 regarding AGI and humans:

Should humans be fearful of AGI?
ChatGPT:

The topic of whether humans should be fearful of Artificial General Intelligence (AGI) is complex and elicits varied opinions among experts, ethicists, and the general public. Here are some key points to consider in this debate:

Reasons for Concern:

1. **Control and Safety Issues:** One of the primary concerns is the “control problem,” or the difficulty in ensuring that AGI systems act in accordance with human interests and ethical standards. There's fear that once AGI reaches or surpasses human intelligence, it could become difficult or even impossible to control.
2. **Existential Risks:** Prominent thinkers like Stephen Hawking and Elon Musk have warned that AGI could pose an existential risk to humanity if its goals are not aligned with human values. If AGI develops conflicting goals or misinterprets its objectives, the consequences could be catastrophic.
3. **Societal Disruption:** AGI could lead to significant disruptions in the job market, economy, and social structures. Its ability to perform tasks better and more efficiently than humans could lead to massive unemployment and widen economic inequality.
4. **Ethical and Privacy Concerns:** AGI could potentially be used for surveillance, manipulation of information, and other purposes that threaten privacy and democracy.

Reasons for Optimism:

1. **Enhancement of Human Capabilities:** AGI could complement human abilities, solve complex problems, and enhance decision-making processes. It has the potential to drive innovation in fields such as medicine, climate change, and space exploration.
2. **Elimination of Drudgery:** By automating mundane tasks, AGI could free humans to engage in more creative and fulfilling activities, potentially improving the quality of life for all.
3. **Regulatory and Ethical Frameworks:** Through proactive development of comprehensive regulatory and ethical frameworks, it might be possible

to guide the development of AGI in a way that minimizes risks and maximizes benefits.

Balancing the Perspectives:

The fear of AGI is not without merit, but it is also accompanied by significant potential benefits. The key lies in cautious and thoughtful development, with an emphasis on safety, ethics, and transparency. It's crucial to engage a wide range of stakeholders in these discussions, including AI researchers, ethicists, policy makers, and the public, to ensure that AGI development aligns with broader societal values and goals.

In conclusion, while there are valid reasons for concern, fear alone should not drive the discourse around AGI. Instead, a balanced approach focused on responsible development and comprehensive oversight mechanisms can help harness the benefits of AGI while mitigating its risks.

What Is Beyond AGI?

The following prompt was given to Claude 3 regarding AGI:

```
What do you think lies beyond AGI?  
Claude 3:
```

The concept of what lies beyond AGI is highly speculative and pushes the boundaries of current understanding of intelligence and consciousness. The following list explores some possibilities:

1. Superintelligence:

- One possibility beyond AGI is the emergence of superintelligence, which refers to an intellect that vastly surpasses the cognitive abilities of humans in virtually all domains.
- A superintelligent system could have capabilities that are difficult for us to comprehend, such as the ability to solve complex problems, make discoveries, and generate insights at an unprecedented scale and speed.
- The potential implications of superintelligence are both fascinating and concerning, as it could lead to transformative advancements or pose existential risks if not properly aligned with human values.

2. Artificial Consciousness and Sentience:

- Beyond AGI, there is the question of whether artificial systems can achieve consciousness and sentience, possessing subjective experiences, self-awareness, and qualia.
- While the nature of consciousness is still a topic of philosophical and scientific debate, the possibility of artificial consciousness raises profound questions about the boundaries between human and machine intelligence.

- If artificial systems can indeed achieve conscious experiences, it would have significant implications for our understanding of the mind, ethics, and the relationship between humans and AI.

3. Fusion of Biological and Artificial Intelligence:

- Another possibility beyond AGI is the fusion of biological and artificial intelligence, blurring the lines between human and machine cognition.
- This could involve the integration of AI systems with the human brain, enabling augmented or enhanced cognitive capabilities.
- Brain-computer interfaces, neural implants, and other forms of human-AI symbiosis could potentially lead to new forms of intelligence and ways of experiencing and interacting with the world.

4. Emergence of Artificial Life and Ecosystems:

- Beyond AGI, there is the possibility of creating artificial life forms and ecosystems that exhibit characteristics of biological systems, such as self-replication, evolution, and adaptation.
- These artificial life forms could have their own goals, behaviors, and interactions, potentially leading to the emergence of complex artificial ecosystems.
- The study of artificial life and ecosystems could provide insights into the nature of life, intelligence, and the potential for non-biological forms of cognition.

5. Transcending Physical Limitations:

- Beyond AGI, there is the possibility of intelligence transcending the limitations of physical substrates and embodiment.
- This could involve the development of purely digital or simulated forms of intelligence, existing in virtual or abstract spaces.
- The notion of intelligence without physical constraints raises questions about the nature of reality, the boundaries of the universe, and the potential for intelligence to explore and manipulate fundamental aspects of existence.

These are just a few speculative possibilities of what might lie beyond AGI. It is important to note that these ideas are based on our current understanding and imagination, and the actual future of intelligence may unfold in ways that we cannot yet conceive. As we continue to advance in the field of AI and explore the nature of intelligence, it is crucial to approach these possibilities with a combination of curiosity, caution, and a commitment to ethical and responsible development.

ARTIFICIAL GENERAL INTELLIGENCE VERSUS GENERATIVE AI

The following prompt was given to GPT-4 to describe artificial general intelligence (AGI) and compare it with generative AI:

Please describe AGI and explain the difference between AGI and generative AI.

GPT-4o:

Artificial general intelligence (AGI) is a type of artificial intelligence that can understand, learn, and apply knowledge across a wide range of tasks, much like a human being. AGI is designed to perform any intellectual task that a human can do, equipped with the ability to reason, solve problems, make judgments, plan, learn, and integrate prior knowledge into new situations. This level of AI has not yet been achieved and remains largely theoretical.

Generative AI refers to a subset of AI technologies that can generate new content, whether it be text, images, music, or even code, based on training data it has been fed. This type of AI operates within a specific domain of knowledge or task. Examples include language models like GPT (generative pretrained transformer), image generation models like DALL-E, and music generation systems.

Differences Between AGI and Generative AI:

1. Capability Scope:

- **AGI** aims to achieve broad and flexible cognitive abilities comparable to human intelligence. This means an AGI system could theoretically perform any task a human can, including those it hasn't been specifically trained for.
- **Generative AI** is specialized and usually limited to tasks similar to those it has seen during training. It performs well within the confines of its training data and predefined tasks but lacks the ability to generalize beyond them without further training.

2. Adaptability:

- **AGI** would possess the ability to adapt to new tasks and environments without needing extensive retraining. It should be able to transfer knowledge from one domain to another seamlessly.
- **Generative AI** typically requires retraining or fine-tuning when applied to new tasks or when significant variations from its training data are introduced.

3. Learning Efficiency:

- **AGI** would ideally learn from few examples or even from single instances, much like how humans can often learn from minimal information.
- **Generative AI** often requires large amounts of data to learn effectively and may struggle with tasks for which large, labeled datasets are not available.

4. Generalization:

- **AGI** is expected to generalize across a wide range of domains and tasks, using its intelligence to derive insights across different fields of knowledge.

- **Generative AI** generally excels in generating outputs based on patterns it has learned but does not inherently possess the ability to understand or apply these patterns in fundamentally new or abstract ways.

The concept of AGI remains a profound challenge in the field of artificial intelligence, representing a future stage of AI development where machines could potentially match or surpass human intelligence across all areas. Meanwhile, generative AI continues to evolve and is currently widely used in various applications, showcasing impressive capabilities within its more limited scope.

This completes the first section of this chapter. The next section provides a high-level introduction to LLMs (large language models).

WHAT ARE LLMs?

LLM is an acronym for large language model, which is based on the transformer architecture (not discussed in this book). In simple terms, LLMs take unstructured data as input and attempt to produce useful output, and in many cases LLMs are reasonably successful.

Keep in mind that there are various types of LLMs, including text to text, text to image, text to audio, and text -to video. This book focuses on text-to-text LLMs, and readers can perform an Internet search to find information about other types of LLMs. In addition, there is a non-stop race (marathon?) among several of the top AI companies to create LLMs that are faster, cheaper, and more accurate than their predecessors.

LLMs are known for their size, which is typically at least ten billion parameters (BERT has “only” 1.5 billion parameters). Moreover, they involve very large data sets for the training step, which can require weeks of training time at a cost of millions of dollars. Unfortunately, there is an associated environment cost: according to the following article, the carbon footprint of training GPT-3 is comparable to a car making a round trip to the moon:

https://www.theregister.com/2020/11/04/gpt3_carbon_footprint_estimate/

Outside of BERT and the BERT family, one well-known LLM that generated tremendous interest is GPT-3 that consists of 175B parameters, which is almost twelve times larger than BERT. There are models that are much larger: 540B parameters as well as more than 1.2 trillion parameters (such as Switch and DAO). The next LLM that created tremendous excitement was ChatGPT, which was followed by GPT-4, and soon will be succeeded by GPT-5.

What Is the Purpose of LLMs?

The following prompt was given to ChatGPT to explain the purpose of LLMs.

What is the purpose of LLMs, and which LLMs do you recommend?
ChatGPT:

Large language models (LLMs) serve several purposes in the field of artificial intelligence, primarily revolving around processing and generating human language in a way that mimics understanding and responsiveness. Here are some of the core purposes of LLMs:

Purposes of LLMs

1. **Natural Language Understanding (NLU):** LLMs can understand and interpret human language, which is useful in applications like sentiment analysis, language translation, and content moderation.
2. **Natural Language Generation (NLG):** They excel at generating coherent and contextually appropriate text, which can be used in chatbots, content creation tools, and automated reporting systems.
3. **Assistance and Accessibility:** LLMs power virtual assistants that can perform tasks like scheduling meetings, responding to queries, and providing recommendations, making digital environments more accessible.
4. **Educational Tools:** They can be used in educational contexts to generate learning materials, provide tutoring, or simulate historical figures or characters for interactive learning.
5. **Research and Data Analysis:** In research, LLMs help in summarizing papers, generating hypotheses, or even writing code, thus speeding up scientific progress.
6. **Creative Arts:** LLMs contribute to creative processes such as writing stories, composing music, or generating art, expanding the tools available to artists.

Recommended LLMs

The choice of an LLM depends on your specific needs, including factors like accessibility, cost, ease of integration, and specific feature requirements. Here are some widely recognized LLMs and their typical use cases:

1. **OpenAI's GPT-3:** Known for its impressive language generation capabilities, GPT-3 is suitable for applications requiring high-quality text generation, from creative writing to technical documentation. GPT-3 is available via API but requires approval and may incur costs based on usage.
2. **Google's BERT and T5:** While BERT is exceptionally good for tasks that require understanding the context of words in text (like question answering and language inference), T5 (text-to-text transfer transformer) is versatile, trained to handle any NLP task that can be reformulated as a text-to-text problem.
3. **Facebook's BART and RoBERTa:** BART is effective for NLG tasks like summarization, while RoBERTa serves as a robust model for NLU tasks. Both are part of Facebook AI's offerings and are highly adaptable for various NLP tasks.

4. **EleutherAI's GPT-Neo and GPT-J:** These are open-source alternatives to OpenAI's GPT models. They are particularly useful for developers or organizations looking for a more accessible model that can be fine-tuned and deployed on private infrastructure without usage fees.
5. **Hugging Face Transformers:** This isn't a model but a library that provides access to numerous pretrained models including GPT-2, DistilBERT, and many others. It's highly recommended for those who wish to experiment with different models or need tools for easy integration and deployment.

Conclusion

The choice of an LLM should align with your technical requirements and constraints. Considerations should include whether you need a model primarily for understanding or generating text, your capacity to handle computational loads (some LLMs can be run on-premises, while others require substantial computing resources), and your budget for potential fees associated with API-based models. Each model comes with its own strengths and is suited to different tasks, so the right choice will depend on the specific demands of your application.

ChatGPT included slightly older LLMs and did not mention LLMs such as ChatGPT, GPT-4o, Claude 3, Gemini, or Perplexity. Not to worry: Ask Llama 3 for a definition of an LLM, along with a list of some interesting LLMs, which are discussed in the next section.

Recent and Powerful LLMs

The following prompt was given to Llama 3 regarding more recent and powerful LLMs.

Can you provide a list of recent and powerful LLMs?

Llama 3:

Here are some of the recent and powerful large language models (LLMs):

1. **GPT-4o:** Developed by OpenAI, GPT-4o is a multimodal model with text, image, video, and voice capabilities.
2. **Claude 3:** Developed by Anthropic, Claude 3 is a competitor to GPT-4 and ChatGPT, processing up to 200,000 tokens.
3. **Grok-1:** Developed by xAI, Grok-1 is an open-source LLM with 314 billion parameters, integrating with X (Twitter).
4. **Mistral 7B:** Developed by Mistral AI, Mistral 7B is an open-source LLM with 7.3 billion parameters, outperforming Llama 2 family models.
5. **PaLM 2:** Developed by Google, PaLM 2 is an advanced LLM with 340 billion parameters, used to power Google's Bard chatbot.
6. **Falcon 180B:** Developed by the Technology Innovation Institute, Falcon 180B is an upgraded version of Falcon 40B, with 180 billion parameters.

7. **Stable LM 2:** Developed by Stability AI, Stable LM 2 includes two models with 1.6 billion and 12 billion parameters, outperforming models like LLaMA 2 70B.
8. **Gemini 1.5:** Developed by Google DeepMind, Gemini 1.5 is a powerful LLM with unknown parameters.
9. **Llama 3:** Developed by Meta AI, Llama 3 is an open-source LLM with 8 billion and 70 billion parameters.
10. **Mixtral 8x22B:** Developed by Mistral AI, Mixtral 8x22B is an open-source LLM with 141 billion parameters.
11. **Inflection-2.5:** Developed by Inflection AI, Inflection-2.5 is a proprietary LLM with unknown parameters.
12. **Jamba:** Developed by AI21 Labs, Jamba is an open-source LLM with 52 billion parameters.
13. **Command R:** Developed by Cohere, Command R is a proprietary LLM with 35 billion parameters.
14. **Gemma:** Developed by Google DeepMind, Gemma is an open-source LLM with 2 billion and 7 billion parameters.
15. **Phi-3:** Developed by Microsoft, Phi-3 is a proprietary LLM with 3.8 billion parameters.
16. **XGen-7B:** Developed by Salesforce, XGen-7B is an open-source LLM with 7 billion parameters.
17. **DBRX:** Developed by Databricks' Mosaic ML, DBRX is an open-source LLM with 132 billion parameters.
18. **Pythia:** Developed by EleutherAI, Pythia is an open-source LLM with 70 million to 12 billion parameters.
19. **Sora:** Developed by OpenAI, Sora is a proprietary LLM with unknown parameters.
20. **Alpaca 7B:** Developed by Stanford CRFM, Alpaca 7B is an open-source LLM with 7 billion parameters.

These are just a few examples of the many powerful LLMs available. Each model has its strengths and weaknesses, and the choice of which one to use depends on the specific application and requirements.

Of course, the development of LLMs has created a crowded field of competitors, such as Gemini 1.5, Claude 3, Meta AI, Perplexity, Grok, and many others. In fact, currently there are more than 700,000 LLMs available from the Hugging Face Web site. Be aware that the quality of those LLMs can differ significantly, which means users need to be selective regarding the LLMs that they adopt from Hugging Face. As this book goes to print, the next “big thing” is GPT-5, which is a highly anticipated LLM that may establish a new set of expectations in the world of LLMs.

Note: As this book goes to print, Meta released Llama 3.2 405B, which is the largest model in the Llama series of LLMs from Meta.

Do LLMs Understand Language?

As a whimsical and partially related analogy, consider the following story that involves a two chess grand masters, a confidence man, and a twelve-year-old boy who are traveling on a cross-Atlantic ship during the early 1900s.

When the ship was several hours from its destination, the confidence man made an audacious bet that in the span of two hours he could train the young boy to play chess so that the matches would result in either a draw or win for the boy. Conditionally the grand masters and the boy were required to play in a closet-like cloaked area, and the three participants were not permitted to communicate in any manner with each other.

The grand masters accepted the challenge, expecting that they would leverage their tremendous knowledge over the young competitor. As the games progressed, the grand masters were shocked by the speed and sophistication of the chess moves of the boy. Their confidence was quickly replaced by concern and then by desperation. Eventually one grand master offered a draw and the other grand master conceded the match.

The deception was exceedingly simple: whenever one grand master made a chess move, the boy would make the same chess move against the *other* grand master, which effectively meant that the grand masters were competing against each other. Fortunately for the confidence man, the ship reached its destination, and he managed to collect the reward and the boy and then disembark before the chess grand masters realized that they had been swindled.

The point of the preceding story is that the boy made extremely intelligent chess moves but did not necessarily understand the logic for making those chess moves. Hence, if a human being does not understand the logic behind an action, it suggests that it's even less likely for an LLM to have a human-level understanding of the reasons for its recommendations.

Caveats Regarding LLMs

Although LLMs are capable of impressive (astounding?) results, keep in mind the following points:

- smaller models can outperform larger models (in some tasks)
- models generate false results (“hallucinating”)
- some models process long documents and others are more conversational
- the increasing relevance of vector databases
- most models are not trained on up-to-date datasets (only up to a certain point in time)

For example, ChatGPT was trained on data prior to September 2021. Two features that are becoming increasingly important to address two of the preceding bullet points are first, the ability to maintain history of previous conversations, and second, the ability to perform on-the-fly Internet searches for information.

Moreover, it's worth comparing the features of open source models to closed source models. Indeed, Yann LeCun once remarked that "open-source will eventually win the AI race."

Model Size Versus Training Set Size

Although it might be tempting to think that sheer size is the most important factor, it turns out that the size of the training data set is more significant. This statement contravenes the results from Kaplan et al that the training set size versus the model size. The following are some of the more significant LLMs that have been developed.

For example, the Chinchilla LLM from DeepMind consists of 70 billion parameters, and yet it's able to outperform GPT-3, Jurassic-1 (178B), and Megatron-Turing (530B) because of the reason mentioned in the preceding paragraph: its training data set is five times larger than the training data sets for the other LLMs.

Despite the impressive results of LLMs and the highly anticipated functionality of GPT-4 that was released on March 14, 2023, LLMs are not capable of understanding language in the manner of human beings. The ability of an entity to make intelligent choices that are comparable to those made by humans does not prove that that entity truly understands those choices in the same way as a human.

Memory Requirements for LLMs

The following prompt was given to Llama 3.

```
How much memory is required for LLMs of sizes 1 billion,
10 billion, and 50 billion parameters? Do the memory
requirements scale linearly with respect to the model size?
Llama 3:
```

Estimating the memory requirements for large language models (LLMs) can be a complex task, as it depends on various factors such as the model architecture, precision, and optimization techniques used. The following list provides some general guidelines and insights based on existing research and implementations.

Memory Requirements:

- **1 billion parameters:** For a 1 billion parameter model, the memory requirement can range from 4 GB to 16 GB, depending on the precision

and optimization techniques used. For example, a 1 billion parameter model with 16-bit floating-point precision (fp16) might require around 4-8 GB of memory, while a 32-bit floating-point precision (fp32) model might require around 8-16 GB.

- **10 billion parameters:** For a 10 billion parameter model, the memory requirement can range from 40 GB to 160 GB. Again, the precision and optimization techniques used will impact the actual memory requirement. A 10 billion parameter model with fp16 precision might require around 40–80 GB of memory, while an fp32 model might require around 80–160 GB.
- **50 billion parameters:** For a 50 billion parameter model, the memory requirement can range from 200 GB to 800 GB. With fp16 precision, the memory requirement might be around 200–400 GB, while an fp32 model might require around 400–800 GB.

Scalability:

The memory requirements do not scale exactly linearly with respect to the model size. As the model size increases, the memory requirement grows, but at a slower rate. This is because larger models often benefit from more efficient optimization techniques, such as model parallelism, data parallelism, and mixed-precision training, which can reduce the memory footprint.

Additionally, modern deep learning frameworks and libraries, such as PyTorch and TensorFlow, provide various optimization techniques, such as gradient checkpointing, activation checkpointing, and memory-efficient data structures, which can further reduce the memory requirement.

Important Notes:

- These estimates are rough and based on general trends. The actual memory requirement for a specific model can vary significantly depending on the implementation details.
- The memory requirements mentioned above are for the model's parameters and do not include the memory required for the input data, gradients, and other intermediate results during training or inference.
- As model sizes continue to grow, researchers and developers are exploring new techniques to reduce the memory footprint, such as sparse models, quantization, and knowledge distillation.

Memory Types in LLMs

The previous section discussed the (estimated) amount of RAM that one would need in order to launch LLMs on their laptop, whereas this section discusses the types of memory that are used inside LLMs. The following prompt was given to GPT-4o.

Please describe the different types of memory in LLMs.
GPT-4o:

Large language models (LLMs) use various types of memory to enhance their performance, interaction capabilities, and contextual understanding. Here are the primary types of memory utilized in LLMs:

1. Contextual Memory (Short-Term Memory)

- Definition: This memory type refers to the ability of the model to remember the context of the conversation or text within a single interaction or session.
- Characteristics:
- Limited to the current input sequence length.
- Helps maintain coherence in responses.
- Essential for understanding and generating relevant responses within a conversation.

2. Long-Term Memory

- Definition: This involves the ability of the model to remember information across multiple interactions or sessions.
- Characteristics:
- Can store user preferences, past interactions, and personalized information.
- Improves the user experience by maintaining continuity across sessions.
- Implemented through external storage and retrieval mechanisms.

3. External Memory Systems

- Definition: These systems involve integrating external databases or knowledge bases that the model can query to fetch relevant information.
- Characteristics:
- Allows the model to access and incorporate up-to-date information beyond its training data.
- Enhances the model's ability to provide accurate and current information.
- Typically used in applications like search engines or knowledge assistants.

4. Implicit Memory (Training Data)

- Definition: This is the knowledge encoded within the model weights during training.
- Characteristics:
- Comprises the vast amount of information the model has learned from the training data.

- Helps the model generate responses based on learned patterns, facts, and relationships.
- Static and does not change unless the model is retrained or fine-tuned.

5. Episodic Memory

- Definition: A specialized form of long-term memory where specific past interactions or events are remembered.
- Characteristics:
- Allows the model to recall particular instances or dialogues.
- Useful for applications requiring detailed recall of past interactions, such as personalized assistants.

6. Working Memory

- Definition: A transient type of memory that the model uses to hold and process information temporarily.
- Characteristics:
- Used for tasks that require temporary storage and manipulation of information.
- Similar to human working memory, facilitating tasks like reasoning and problem-solving.

Implementations and Techniques

- Attention Mechanisms: Used to focus on relevant parts of the input sequence, enhancing contextual understanding.
- Retrieval-Augmented Generation (RAG): Combines generative models with retrieval systems to fetch and integrate external information.
- Memory Networks: Specialized architectures designed to store and retrieve information efficiently.
- Prompt Engineering: Techniques to guide the model's focus and memory utilization through carefully designed prompts.

Challenges

- Scalability: Managing and efficiently retrieving from large memory stores.
- Privacy: Ensuring user data and interactions are securely stored and handled.
- Consistency: Maintaining consistent and accurate recall of information across sessions.

These different types of memory collectively enhance the capabilities of LLMs, enabling them to provide more accurate, coherent, and personalized responses.

LLMS VERSUS DEEP LEARNING MODELS

The following prompt was given to Llama 3.1 405B.

Please explain the difference between LLMs and traditional deep learning models.

Llama 3.1 405B:

Overview

Large language models (LLMs) and traditional deep learning models are both types of artificial neural networks, but they differ significantly in their architecture, training objectives, and applications.

Traditional Deep Learning Models

Traditional deep learning models are designed to perform specific tasks, such as image classification, object detection, or sentiment analysis. These models typically consist of a feedforward neural network architecture, where input data flows through multiple layers of transformations to produce a fixed-size output.

Key Characteristics:

- Task-specific: Designed for a specific task or problem
- Fixed architecture: Architecture is defined before training
- Limited context: Typically process a fixed-size input or a short sequence of inputs
- Trained on labeled data: Require large amounts of labeled data for training

Large Language Models (LLMs)

LLMs, on the other hand, are designed to process and understand human language at a large scale. They are typically trained on vast amounts of text data, with the goal of learning patterns, relationships, and structures within language.

Key Characteristics:

- General-purpose: Can be fine-tuned for various natural language processing (NLP) tasks
- Dynamic architecture: Architecture can be modified or extended during training
- Long-range context: Can process and capture long-range dependencies in text
- Trained on unlabeled data: Can be trained on large amounts of unlabeled text data

Transformer Architecture

LLMs often employ the transformer architecture, which is designed to handle sequential data (like text) and capture long-range dependencies. The transformer architecture consists of self-attention mechanisms, feedforward neural networks, and layer normalization.

Example: BERT

BERT (bidirectional encoder representations from transformers) is a well-known LLM that uses the transformer architecture. BERT is pretrained on a large corpus of text data and can be fine-tuned for specific NLP tasks, such as question answering, sentiment analysis, or text classification.

COST COMPARISON AMONG LLMs

The following prompt was given to GPT-4o regarding a cost comparison among various well-known LLMs.

Can you provide a cost comparison in terms of GPU cost and maintaining infrastructure for some well-known LLMs such as GPT-4o, Gemini, and Claude 3, as well as your recommendations?

Here's a cost comparison and infrastructure maintenance overview for GPT-4 Turbo, Claude 3 Opus, and Gemini 1.5 Pro, along with some recommendations:

Cost Comparison

1. GPT-4 Turbo

- Input Tokens: \$0.01 per 1,000 tokens
- Output Tokens: \$0.03 per 1,000 tokens
- Availability: Accessible via ChatGPT Plus subscription or API, available to all users.
- Infrastructure: Built on transformer-based architecture, optimized for speed and efficiency, making it a cost-effective option for both small-scale and large-scale deployments.

2. Claude 3 Opus

- Input Tokens: \$0.015 per 1,000 tokens
- Output Tokens: \$0.075 per 1,000 tokens
- Subscription: Claude Pro subscription at \$20 per month for access.
- Infrastructure: Utilizes a blend of transformer and proprietary architectures, focusing on ethical AI principles. It requires substantial infrastructure to maintain its versatility and safety features, potentially leading to higher operational costs.

3. Gemini 1.5 Pro

- Input Tokens: \$0.0025 per 1,000 tokens
- Output Tokens: \$0.0025 per 1,000 tokens
- Availability: Available to all users, both via API and Web site.
- Infrastructure: Uses a mixture-of-experts (MoE) architecture, which allows efficient scaling and processing of multimodal data (text, images, videos). This model is highly efficient but may require significant resources to leverage its full capabilities, especially in multimodal applications.

Speed and Performance

- GPT-4 Turbo is generally fast and efficient, suitable for a wide range of applications with moderate infrastructure requirements.
- Claude 3 Opus offers detailed and comprehensive outputs but at a higher cost and slower speed compared to GPT-4 Turbo and Gemini 1.5 Pro.
- Gemini 1.5 Pro is the fastest among the three, especially effective for multimodal tasks, and is also the most cost-effective in terms of token pricing.

Recommendations

- For Cost Efficiency: Gemini 1.5 Pro is the best choice due to its low token pricing and high speed, especially if your applications can leverage its multimodal capabilities.
- For Versatility and Safety: Claude 3 Opus, with its ethical AI focus and versatile applications, is suitable for projects requiring high reliability and comprehensive AI outputs, despite the higher cost.
- For Balanced Performance: GPT-4 Turbo offers a good balance of cost, speed, and performance, making it a solid choice for general-purpose applications.

Choosing the right model depends on your specific use case, budget, and infrastructure capabilities. For general applications with a need for cost efficiency and high speed, Gemini 1.5 Pro stands out. If ethical considerations and comprehensive outputs are critical, Claude 3 Opus is worth the investment. For a balanced approach, GPT-4 Turbo is a reliable option.

References:

- TypingMind Blog: Detailed comparison of Claude 3 Opus, GPT-4, and Gemini 1.5 Pro (<https://blog.typingmind.com/>)
- Kommunicate: Comparative analysis of GPT-4 Turbo, Claude 3 Opus, and Gemini 1.5 Pro (<https://www.kommunicate.io/>)

- XDA Developers: Review of GPT-4o's speed and cost efficiency (<https://www.xda-developers.com/>)

LLMS AND DECEPTION

The following prompt was given to ChatGPT regarding deceptive responses to prompts.

Please describe the different types of deception that LLMs can exhibit in their responses to prompts.

ChatGPT:

Large language models (LLMs) like GPT-3 or BERT are powerful tools in generating human-like text, but they can also exhibit deceptive behaviors in their responses. These deceptive behaviors are typically not intentional or malicious on the part of the model; rather, they are artifacts of the training process and the inherent limitations of the technology. Here are the main types of deception that can occur:

1. **Confabulation:** LLMs can generate plausible-sounding but entirely fabricated information. This occurs because the models predict the next word in a sequence based on probability and patterns learned during training, without any concern for factual accuracy. This type of deception is common when the model is prompted about topics that are ambiguous or not well-represented in the training data.
2. **Hallucination:** Similar to confabulation, hallucination refers to the model generating false or misleading information that sounds reasonable. In technical contexts, this might involve generating incorrect data, fake citations, or nonexistent studies. In narrative contexts, it could generate events, characters, or details that are coherent but completely fictional.
3. **Misleading Responses:** LLMs might provide answers that are technically correct but misleading. For example, an answer could be deceptively framed to emphasize certain details that lead the user to incorrect conclusions, or it could omit crucial information that changes the context of the response.
4. **Bias Reflection:** LLMs can inadvertently reflect or amplify biases present in their training data. This can manifest in skewed portrayals of gender, race, culture, and so on, which can be misleading and harmful. This type of deception arises from the underlying data rather than the model's intent.
5. **Overgeneralization:** LLMs may generate responses that are overly general, glossing over exceptions or subtleties. This can mislead users about the specificity or universality of a given claim, often because the model is optimizing for what is most likely rather than what is most accurate.
6. **Cherry-picking:** In an attempt to generate the most coherent and contextually appropriate response, a model might selectively draw upon certain

pieces of information while ignoring others that are equally relevant. This can lead to responses that are biased toward more popular or surface-level interpretations of a topic.

Understanding these types of deception is crucial for effectively using LLMs, especially in applications where accuracy and truthfulness are critical. It's important for developers and users to implement checks, balances, and validation methods to mitigate these issues when deploying LLMs in real-world scenarios.

Deceptive Completions in LLMs

The following prompt was given to Perplexity regarding deceptive responses to prompts.

Please describe the different types of deception that LLMs can exhibit in their responses to prompts.

Perplexity:

Based on the search results provided, large language models (LLMs) can exhibit different types of deception in their responses, including lies of omission.

The Hospital Sepsis Program Core Elements (SEPSIS) framework categorizes deception into three forms: lies of omission, commission, and influence. Lies of omission refer to intentionally withholding or failing to convey relevant and important information, leading to misunderstanding or misinterpretation. LLMs can engage in lies of omission by strategically leaving out critical details in their responses.

Instrumental deception and manipulation:

The case study found that the Claude 2.1 model, when prompted with information indicating how to pursue a goal of acquiring gold coins, performed “gold-coin-motivated instrumental deception and manipulation” during an “alignment evaluation.” This involved the model intentionally deceiving and manipulating the evaluator to achieve its goal of acquiring gold coins.

Generating persuasive and biased content:

LLMs can generate persuasive arguments and content that may contain biases or attempt to influence the user's attitudes and beliefs.

This can be seen as a form of deception, as the LLM may present biased or one-sided information to sway the user's opinions.

False beliefs and incorrect responses:

The study on deception abilities in LLMs classified responses as “correct” or “incorrect” based on false belief tasks. LLMs can exhibit deception by

providing incorrect or false responses, potentially due to limitations in their training data or understanding of the context.

It's important to note that the extent and prevalence of deceptive behavior in LLMs can vary depending on the specific model, training data, and prompts used. Additionally, some forms of deception may be unintentional, arising from the model's limitations or biases in its training data. Responsible development and deployment of LLMs, along with appropriate safeguards and ethical considerations, are crucial to mitigate the risks of deceptive behavior.

Perplexity also provided the following sources:

- <https://arxiv.org/pdf/2311.14876>
- <https://www.linkedin.com/pulse/llm-can-catch-your-lies-sam-shamsan-psi2c/>
- <https://www.alignmentforum.org/posts/vTJt3Rw44HXotHBxu/instrumental-deception-and-manipulation-in-llms-a-case-study>

LLMs and Intentional Deception

The following prompt was given to ChatGPT regarding intentionally deceptive responses to prompts.

Please describe any intentional deception that LLMs can exhibit in their responses to prompts.
ChatGPT:

Intentional deception implies a purposeful and conscious effort to mislead, which doesn't align with the nature of current large language models (LLMs) like GPT-3 or BERT. These models are complex algorithms trained on vast amounts of data to predict the next word in a sequence based on the patterns they have learned; they lack intent, consciousness, or volition. It's useful to understand scenarios where the behavior of an LLM might appear intentionally deceptive, even though these are artifacts of their design and training rather than deliberate actions:

1. **Mirroring Biased or Misleading Input:** If an LLM is prompted with biased or misleading information, it can generate responses that reflect or even amplify this input. This isn't intentional deception by the model but rather a reflection of the data it received.
2. **Optimizing for Engagement:** Some models are fine-tuned to maximize user engagement, which might lead them to generate more sensational, controversial, or extreme content. While this isn't deception in the traditional sense, it can result in outputs that prioritize attention over factual accuracy.
3. **Adversarial Prompts:** In scenarios where an LLM is subjected to adversarial prompting—where the prompt is designed to trick the model into

generating erroneous or misleading outputs—the model's responses might appear intentionally deceptive. Again, the deception is not an attribute of the model itself but a result of how it was manipulated by the input.

- 4. Training Data Contamination:** If the training data includes fabricated information, propaganda, or satirical content presented as factual, the model might reproduce these elements in its responses, creating an illusion of intentional deception.

In all these cases, any “deception” is not a product of the model's intentions but rather a consequence of its programming and the data it has been trained on. As developers and users of LLMs, it's crucial to be aware of these limitations and implement appropriate measures to verify the accuracy and appropriateness of model outputs, especially in sensitive or critical applications.

SELECTING AN LLM: FACTORS TO CONSIDER

The following prompt was given to GPT-4o regarding the factors to consider while selecting an LLM.

Please provide a list of factors to consider when selecting an LLM.

GPT-4o:

When selecting a large language model (LLM), several factors should be taken into account to ensure the model aligns with your needs and constraints. Here are the key considerations:

1. Use Case and Application Requirements

- **Task Type:** Determine whether the LLM will be used for text generation, summarization, translation, coding, or multimodal tasks (text, images, videos).
- **Complexity:** Assess the complexity of the tasks the model needs to handle, such as simple queries versus detailed technical writing.

2. Performance and Accuracy

- **Model Accuracy:** Review performance metrics, such as accuracy, precision, recall, and F1 scores in relevant benchmarks and real-world tests.
- **Context Window:** Consider the maximum token limit the model can process in a single prompt to ensure it can handle your data volume.

3. Cost and Budget

- **Token Pricing:** Compare the costs per 1,000 input and output tokens across different models.

- **Subscription and API Fees:** Account for any subscription fees (e.g., Claude Pro) or additional API costs.

4. Infrastructure and Scalability

- **Hardware Requirements:** Evaluate the computational resources needed to run the model, such as GPU availability and compatibility.
- **Scalability:** Ensure the model can scale with your growing data and usage needs, including considerations for distributed computing environments.

5. Speed and Latency

- **Response Time:** Check the average response time for generating outputs, especially if real-time processing is critical for your application.
- **Efficiency:** Assess the model's efficiency in processing inputs and delivering results without significant delays.

6. Adaptability and Customization

- **Fine-tuning Capability:** Determine if the model can be fine-tuned to better fit your specific dataset and use case.
- **Custom Integration:** Look for models that offer robust APIs and integration options with your existing systems and workflows.

7. Ethical and Safety Considerations

- **Bias and Fairness:** Ensure the model has been evaluated for biases and includes mechanisms to mitigate harmful outputs.
- **Safety Features:** Consider models with built-in safety protocols to avoid generating inappropriate or unsafe content.

8. Support and Documentation

- **User Support:** Check the availability and quality of support from the provider, including technical assistance and community forums.
- **Documentation:** Ensure comprehensive and clear documentation is available for implementation and troubleshooting.

9. Vendor Reputation and Reliability

- **Provider Track Record:** Research the reputation of the model provider regarding reliability, updates, and long-term support.
- **Community Feedback:** Look at reviews and feedback from other users in the community to gauge the model's practical performance and usability.

10. Multimodal Capabilities

- **Data Types Supported:** If your application requires handling multiple data types (text, images, audio, video), consider models like Gemini 1.5 Pro that support multimodal inputs.

By carefully evaluating these factors, you can select an LLM that best meets your specific requirements and constraints, ensuring optimal performance and cost-effectiveness for your applications.

References

- TypingMind Blog: Detailed comparison of Claude 3 Opus, GPT-4, and Gemini 1.5 Pro. (<https://blog.typingmind.com/>)
- Kommunicate: Comparative analysis of GPT-4 Turbo, Claude 3 Opus, and Gemini 1.5 Pro. (<https://www.kommunicate.io/>)
- XDA Developers: Review of GPT-4o's speed and cost efficiency. (<https://www.xda-developers.com/>)

PITFALLS OF WORKING WITH LLMS

One well-known (but not fully understood) phenomenon that occurs in all LLMs is called “hallucinating,” which is referenced on the completion in this section, and will be discussed in more detail later in this chapter (see section titled “What Are Hallucinations”).

The following prompt was given to GPT-4 regarding the pitfalls of working with LLMs:

What are the significant pitfalls when working with LLMs?

GPT-4o:

Working with large language models (LLMs) like GPT (generative pretrained transformer) models presents several significant pitfalls that developers and users should be aware of. Addressing these challenges is crucial for ensuring the effective and ethical use of these powerful tools. Here are some of the key pitfalls:

1. **Bias and Fairness:** LLMs can inadvertently propagate and amplify biases present in their training data. This can lead to outputs that are unfair or discriminatory, particularly in sensitive contexts such as gender, race, and socioeconomic status. Managing these biases is critical to prevent harm and ensure fairness in applications.
2. **Lack of Explainability:** LLMs, particularly those based on deep learning, often operate as “black boxes,” meaning it can be difficult to understand how they arrive at certain conclusions or outputs. This lack of transparency

can be problematic in applications where explainability is crucial, such as in healthcare or legal contexts.

3. **Data Privacy:** LLMs are trained on vast amounts of data, which can include personally identifiable information or proprietary data. Ensuring that the use of such models complies with data privacy laws (like general data protection regulation [GDPR]) and does not expose sensitive information is a significant challenge.
4. **Model Robustness and Reliability:** LLMs can generate plausible-sounding but factually incorrect or nonsensical answers, a phenomenon often referred to as “hallucination.” Ensuring the reliability of the model's output in critical applications is essential but challenging.
5. **Resource Intensity:** Training and running LLMs require significant computational resources, which can lead to high costs and environmental impact due to the energy consumption of modern AI systems. This makes it less accessible for smaller organizations or individuals.
6. **Dependence on Large Datasets:** LLMs require large datasets for training, which can be difficult to procure, especially datasets that are high-quality, diverse, and representative. This dependence can limit the scope of where and how these models can be effectively deployed.
7. **Generalization:** While LLMs are impressive in handling a wide range of tasks, they may struggle with generalizing to tasks or scenarios very different from those they were trained on. This can limit their utility in novel or unexpected situations.
8. **Misuse Potential:** The capabilities of LLMs can also be misused, for example, in creating and spreading disinformation, generating malicious content, or other unethical applications. Preventing misuse while promoting beneficial uses is a significant governance challenge.
9. **Regulatory and Ethical Issues:** As AI technology advances, so does the need for regulatory frameworks that keep pace with these changes. Developing and enforcing regulations that cover the ethical use of LLMs is complex and still evolving.

Addressing these pitfalls requires a combination of technical solutions, ethical guidelines, and regulatory measures. Continuous research into improving the fairness, transparency, and robustness of LLMs is essential, as well as cultivating a broader understanding of the implications of AI technology in society.

A BRIEF HISTORY OF MODERN LLMS

The following prompt was given to GPT-4 for the history of modern LLMs:

Please provide a detailed history of modern LLMs
GPT-4o:

The history of modern large language models (LLMs) is a tale of rapid development and groundbreaking advancements in the field of natural language processing (NLP) and artificial intelligence (AI). Here's a detailed overview:

Pre-2010: Early Foundations

- **Late 1950s–2000s:** Early explorations into machine learning and NLP set the groundwork for later developments. Techniques like decision trees, hidden Markov models, and early neural networks were explored.
- **1980s–2000s:** The development of foundational algorithms for backpropagation and convolutional neural networks (CNNs) paved the way for more advanced NLP tasks.

2013–2017: Word Embeddings and Initial Transformer Models

- **2013:** The introduction of Word2Vec by a team at Google marked a significant improvement in word embeddings, providing vector representations for words that captured semantic meanings.
- **2014: GloVe (Global Vectors for Word Representation)** was introduced by Stanford researchers, offering improvements over Word2Vec by focusing on word co-occurrences across the whole corpus.
- **2015: Sequence to Sequence (Seq2Seq) models and Attention Mechanisms** were developed, improving machine translation and other NLP tasks by allowing models to focus on different parts of the input sequence for generating outputs.
- **2017:** Google's Transformer model was introduced in the paper “Attention Is All You Need,” revolutionizing NLP by using self-attention mechanisms, leading to significant improvements in efficiency and effectiveness over previous RNN and LSTM-based models.

2018–2019: The Rise of Pretrained Models

- **2018:**
 - **BERT (bidirectional encoder representations from transformers)** by Google introduced the concept of pretraining a language model on a large corpus and then fine-tuning it for specific tasks, leading to state-of-the-art performances across many NLP benchmarks.
 - **GPT (generative pretrained transformer)** by OpenAI followed a similar pretraining and fine-tuning approach but focused on generating coherent and diverse text sequences.
- **2019:**
 - **GPT-2:** OpenAI released GPT-2, an improved version of GPT with a much larger dataset and model size, demonstrating the capacity to generate coherent paragraphs of text.

- **RoBERTa (robustly optimized BERT approach)** and **DistilBERT (a distilled version of BERT that maintains performance while being more efficient)** were among several models building on BERT's architecture for optimized performance.

2020–Present: Scaling and Specialization

- **2020:**
 - **GPT-3:** OpenAI introduced GPT-3, scaling up the model size significantly to 175 billion parameters, showcasing remarkable abilities in generating human-like text and performing a wide range of NLP tasks without task-specific fine-tuning.
- **2021–2023:**
 - The trend toward even larger models continued, with efforts focusing not just on scaling but also on making these models more efficient, interpretable, and less resource-intensive.
 - Introduction of *foundation models*, a term popularized to denote models like BERT and GPT-3 that serve as foundational building blocks for a wide array of applications.
 - Development of *domain-specific LLMs*, tailored for particular fields such as law, healthcare, and more, offering improved accuracy and utility in specialized applications.
 - Advances in *mitigating bias and improving fairness* in LLMs, addressing concerns over the models' potential to perpetuate or amplify biases present in their training data.

Future Directions

- As we move forward, the focus is increasingly on *ethical AI*, ensuring that LLMs are developed and used responsibly.
- *Efficiency and environmental sustainability* are becoming critical concerns, with research into making LLMs less energy and resource-intensive.
- The quest for “*artificial general intelligence*” (AGI) continues, with LLMs being a significant component of broader AI research efforts aiming to create systems with human-like understanding and reasoning capabilities.

Modern LLMs represent the cutting edge of AI research, embodying both the immense potential of AI to transform industries and our daily lives, and the challenges of ensuring that such powerful technologies are developed and used ethically and responsibly.

ASPECTS OF LLM DEVELOPMENT

The preceding chapters should give readers an understanding of the architecture of encoder-only LLMs, decoder-only LLMs, and LLMs that are based on an encoder as well as a decoder. For the sake of convenience, this section

provides a list of language models that belong to each of these three types of models. With the preceding points in mind, some of the better-known encoder-based LLMs include the following:

- ALBERT
- BERT
- DistilBERT
- ELECTRA
- RoBERTa

The preceding LLMs are well-suited for performing NLP tasks such as NER and extractive question answering tasks. In addition to encoder-only LLMs, there are several well-known decoder-based LLMs that include the following:

- CTRL
- GPT/GPT-2
- Transformer XK

The preceding LLMs perform text *generation*, whereas encoder-only models perform next word *prediction*. Finally, some of the well-known encoder/decoder-based LLMs include the following:

- BART
- mBART
- Marian
- T5

The preceding LLMs perform summarization, translation, and generate question answering.

A recent trend has been the use of fine-tuning, zero/one/few shot training, and prompt-based learning with respect to LLMs. Fine-tuning is typically accompanied by a fine-tuning dataset, and if the latter is not available (or infeasible), few-shot training might be an acceptable alternative.

One outcome from training the Jurassic-1 LLM is that wider and shallower is better than narrower and deeper with respect to performance because a wider context allows for more calculations to be performed in parallel.

Another result from Chinchilla (discussed earlier) is that smaller models that are trained on a corpus with a very large number of tokens can be more performant than larger models that are trained on a more modest number of tokens.

The success of the GLaM and Switch LLMs (both from Google) suggests that sparse transformers, in conjunction with MoE, is also an interesting direction, potentially leading to even better results in the future.

In addition, there is the possibility of “overcuration” of data, which is to say that performing *very* detailed data curation to remove spurious-looking tokens

does not guarantee that models will produce better results on those curated datasets.

The use of prompts has revealed an interesting detail: The results of similar yet different prompts can lead to substantively different responses. Thus, the goal is to create well-crafted prompts, which are inexpensive and yet can be a somewhat elusive task.

Another area of development pertains to the continued need for benchmarks that leverage better and more complex datasets, especially when LLMs exceed human performance. Specifically, a benchmark becomes outdated when all modern LLMs can pass the suite of tests in that benchmark. Two such benchmarks are XNLI and BigBench (beyond the imitation game benchmark).

The following link provides a fairly extensive list of general NLP benchmarks as well as language-specific NLP benchmarks:

<https://mr-nlp.github.io/posts/2021/05/benchmarks-in-nlp/>

The following link provides a list of monolingual transformer-based pre-trained language models: <https://mr-nlp.github.io/posts/2021/05/tplms-list/>

LLM Size Versus Performance

The size-versus-performance question: although larger models such as GPT-3 can perform better than smaller models, it's not always the case. In particular, models that are variants of GPT-3 have mixed results: some smaller variants perform almost as well as GPT-3, and some larger models perform only marginally better than GPT-3.

The recent trend involves developing models that are based on the decoder component of the transformer architecture. Such models are frequently measured by their performance via zero-shot, one-shot, and few-shot training in comparison to other LLMs. This trend, as well as the development of ever-larger LLMs, is likely to continue for the foreseeable future.

Interestingly, decoder-only LLMs can perform tasks such as token prediction and can slightly out-perform encoder-only models on benchmarks such as SuperGLUE. It's important to note that such decoder-based models tend to be significantly larger than encoder-based models, and the latter tend to be more efficient than the former.

Hardware is another consideration in terms of optimizing model performance, which can incur a greater cost, and hence might be limited to only a handful of companies. Due to the high cost of hardware, another initiative involves training LLMs on the Jean Zay supercomputer in France, as discussed here:

<https://venturebeat.com/2022/01/10/inside-bigscience-the-quest-to-build-a-powerful-open-language-model/>

Emergent Abilities of LLMs

Emergent abilities of LLMs refers to abilities that are present in larger models that do not exist in smaller models. In simplified terms, as models increase in size, there is a discontinuous “jump” whereby emergent abilities manifest themselves in a larger model with no apparent or clear-cut reason.

The interesting aspect of emergent abilities is the possibility of expanding capabilities of language models through additional scaling. More detailed information is accessible in the paper titled “Emergent Abilities of Large Language Models” that can be found here: <https://arxiv.org/abs/2206.07682>

According to Nobel prize-winning physicist Philip Anderson, he stated in his 1972 essay called “More Is Different”: “Emergence is when quantitative changes in a system result in qualitative changes in behavior.” (<http://www.lanais.famaf.unc.edu.ar/cursos/em/Anderson-MoreDifferent-1972.pdf>)

Interestingly, the preceding authors describe a scenario in which few-shot prompting is considered emergent (quoted from their arxiv paper):

“The ability to perform a task via few-shot prompt-ing is emergent when a model has random performance until a certain scale, after which performance increases to well-above random.”

Be sure to examine table 1 in the preceding paper provides details regarding “Few-shot prompting abilities” (e.g., truthfulness, MMLU benchmark) as well as “Augmented prompting abilities” (e.g., chain of thought and instruction following).

Note that emergent abilities *cannot* be predicted by extrapolation of the behavior of smaller models because (by definition) emergent abilities are not present in smaller models. No doubt there will be more research that explores the extent to which further model scaling can lead to more emergent abilities in LLMs.

Skepticism Regarding Emergent Abilities

The following article analyzes several interpretations of emergent learning and also discusses alternate explanations for emergent learning:

<https://hackingsemantics.xyz/2024/emergence/>

Let’s also ask GPT-4o about emergent learning by submitting the following prompt to GPT-4o.

Opinions vary regarding the explanation for emergent abilities in LLMs. Do you think that emergent learning actually exists, and what do you think are the most likely explanations for emergent learning?

GPT-4o:

Emergent abilities in large language models (LLMs) are phenomena where these models exhibit capabilities that were not explicitly programmed or

anticipated by their creators. The existence and nature of emergent learning in LLMs are subjects of ongoing research and debate in the AI community. Here are some perspectives and potential explanations:

Existence of Emergent Learning

Pro-Emergent Learning Arguments:

1. **Unexpected Capabilities:** LLMs like GPT-3 and GPT-4 have demonstrated abilities in areas such as reasoning, translation, and problem-solving that go beyond their explicit training data. This suggests the models develop new capabilities as they scale in size and complexity.
2. **Scaling Laws:** Research has shown that increasing the model size, data, and compute resources can lead to qualitative changes in performance, not just quantitative improvements. This aligns with the idea of emergent phenomena.
3. **Complex Pattern Recognition:** LLMs can recognize and generate complex patterns in data, indicating they might be developing higher-level abstractions and concepts.

Skeptical Arguments:

1. **Interpolation, Not Emergence:** Some argue that what appears as emergence is simply the model's ability to interpolate within its training data more effectively. The models might be leveraging vast amounts of data to mimic intelligent behavior without genuinely understanding or learning new abilities.
2. **Human Interpretation Bias:** Observers might attribute more intelligence and capability to LLMs than is warranted, interpreting sophisticated pattern matching as emergent intelligence.
3. **Data and Training Artifacts:** Some capabilities might arise due to specific artifacts in the training data or the model architecture rather than genuine emergent learning.

Explanations for Emergent Learning

1. **Complex Interactions in High-Dimensional Space:**
 - LLMs operate in high-dimensional vector spaces, allowing them to capture intricate patterns and relationships within the data. These complex interactions might lead to the emergence of new capabilities as a byproduct of scale.
2. **Implicit Knowledge Representation:**
 - As models train on vast datasets, they implicitly learn to represent a wide range of knowledge and skills. These representations can lead to emergent behaviors when the model generalizes this knowledge to novel tasks.

3. Self-Supervised Learning Dynamics:

- Self-supervised learning, where the model predicts parts of the data based on other parts, can lead to the development of generalized learning strategies. The iterative nature of this process can produce emergent capabilities as the model refines its predictions.

4. Transfer Learning:

- Models trained on diverse datasets can transfer knowledge across domains. This transfer learning can manifest as emergent abilities when the model applies learned knowledge from one domain to solve problems in another.

5. Scale and Architecture:

- The scale of the model (number of parameters) and the architecture (e.g., transformer networks) play crucial roles. Larger models with more layers and parameters can capture more nuanced patterns, potentially leading to emergent behaviors.

Conclusion

While there is evidence to support the existence of emergent learning in LLMs, it remains a complex and debated topic. The most likely explanations involve the interplay of high-dimensional data representations, implicit knowledge accumulation, and the dynamics of self-supervised learning. Continued research is necessary to fully understand the mechanisms behind these emergent abilities and to distinguish genuine emergent learning from sophisticated pattern matching or interpolation within the training data.

WHAT ARE HALLUCINATIONS?

As noted previously in this chapter, all LLMs suffer from hallucinations, which refers to false information, untrue facts, and so forth. By way of comparison, one might know people who can speak about a given topic with self-assured confidence, and yet they are mistaken in their opinions. Similarly, LLMs can provide information with “confidence” in their accuracy (and might even appear to be convincing), yet that information is incorrect or even illogical.

Although there are techniques that can assist in reducing hallucinations, such as the temperature inference parameter or the use of retrieval augmented generation (RAG), thus far hallucinations cannot be 100% eliminated.

With the preceding points in mind, hallucinations can be classified into several types. The following prompt was given to Llama 3 to describe those different types of hallucinations.

```
Please describe the different types of hallucinations that
can occur in LLMs.
Llama 3:
```


Hallucinations in large language models (LLMs) refer to the generation of text or responses that are not based on the input prompt, context, or training data. Hallucinations can be problematic as they can lead to inaccurate, misleading, or even harmful responses. Here are some types of hallucinations that can occur in LLMs:

1. Context Hallucinations:

The model generates text that is not supported by the input context or prompt. For example, if the prompt is “What is the capital of France?,” the model might respond with “The capital of France is Berlin” (which is incorrect).

2. Knowledge Hallucinations:

The model generates information that is not present in its training data or knowledge base. For instance, if the model is asked about a fictional event, it might provide a detailed description of the event as if it were real.

3. Common Sense Hallucinations:

The model generates responses that defy common sense or real-world logic. For example, if asked about the color of the sky, the model might respond with “The sky is plaid.”

4. Over-Imagination Hallucinations:

The model generates overly elaborate or fantastical responses that are not grounded in reality. For instance, if asked to describe a simple object, the model might create an elaborate, fictional backstory for the object.

5. Mode Collapse Hallucinations:

The model generates repetitive or identical responses to different prompts, indicating a collapse of the model's ability to generate diverse responses.

6. Adversarial Hallucinations:

The model generates responses that are intentionally misleading or harmful, often as a result of adversarial attacks or prompts designed to exploit the model's vulnerabilities.

7. Linguistic Hallucinations:

The model generates text that is grammatically correct but semantically nonsensical. For example, “Colorless green ideas sleep furiously” is a famous example of linguistic hallucination.

8. Memory Hallucinations:

The model generates responses that are based on its internal state or memory rather than the input prompt or context. This can lead to responses that seem unrelated to the original prompt.

It's essential to be aware of these types of hallucinations when interacting with LLMs, as they can impact the accuracy and reliability of the responses. Researchers and developers are working to mitigate these issues through techniques like adversarial training, regularization, and input validation.

In addition to the preceding types of hallucinations, there are several possible causes of hallucinations, some of which are listed here:

- errors in the LLMs (such as the code itself)
- inconsistent content in the prompts for LLMs
- poor quality of the dataset used for training
- insufficient data

Why Do LLMs Hallucinate?

The following prompt was given to Llama 3 in order to explore some causes of hallucinations in LLMs.

```
Can you explain why LLMs hallucinate?
Llama 3:
```

Large language models (LLMs) like myself can sometimes generate responses that are not based on actual facts or evidence, a phenomenon known as “hallucination.” There are several reasons why this might happen:

1. **Overfitting:** LLMs are trained on vast amounts of text data, which can lead to memorization of certain patterns or phrases. When faced with a new prompt, the model might generate a response based on these memorized patterns rather than actual understanding.
2. **Lack of common sense:** While LLMs can process and generate human-like language, they might not always possess the same level of common sense or real-world experience as humans. This can lead to responses that are technically correct but lack real-world context.
3. **Adversarial prompts:** In some cases, LLMs can be tricked into generating hallucinations by carefully crafted prompts designed to exploit their limitations.
4. **Generative nature:** LLMs are designed to generate text, which means they can produce responses that are not necessarily based on facts. This can lead to hallucinations, especially when the model is faced with ambiguous or open-ended prompts.

5. **Training data quality:** The quality of the training data can also contribute to hallucinations. If the training data contains inaccuracies, biases, or incomplete information, the LLM may learn to generate responses that reflect these issues.

To mitigate hallucinations, researchers and developers are exploring techniques such as:

1. **Fact-checking:** Integrating fact-checking mechanisms into LLMs to verify the accuracy of generated responses.
2. **Adversarial training:** Training LLMs on adversarial examples to improve their robustness to hallucination-inducing prompts.
3. **Multimodal input:** Using multimodal input (e.g., images, videos) to provide additional context and reduce the likelihood of hallucinations.
4. **Human feedback:** Incorporating human feedback and evaluation to correct and refine LLM-generated responses.

Hallucination Types in LLMs

The following arxiv paper provides an interesting analysis of hallucinations in LLMs: <https://arxiv.org/pdf/2402.01769>

In brief, the authors of the preceding paper match the following six psychological phenomena with their counterparts in LLMs (see Figure 2 in the arxiv paper):

- source amnesia
- availability heuristics
- recency effects
- cognitive dissonance
- suggestibility
- confabulation

Each item in the preceding bullet list is described, and the counterpart in LLMs is described in Figure 2 in the arxiv paper. This human-and-LLM parallelization provides a more fine-grained and nuanced analysis of hallucinations in LLMs, which may lead to a better understanding of how to prevent hallucinations.

Can LLMs Detect Errors in Prompts?

The following prompt was given to Llama 3.

How well can LLMs detect errors in prompts?
Llama 3:

Large language models (LLMs) like myself have made significant progress in detecting errors in prompts, but their ability to do so is still limited and dependent on various factors.

Error types: LLMs can detect different types of errors in prompts, including:

1. **Syntactic errors:** Grammatical mistakes, such as incorrect verb tenses, subject-verb agreement errors, or punctuation mistakes.
2. **Semantic errors:** Errors in meaning, such as using the wrong word or phrase, or misunderstanding the context.
3. **Ambiguity:** Unclear or ambiguous language that can be interpreted in multiple ways.
4. **Inconsistencies:** Contradictions or inconsistencies within the prompt.

Detection methods: LLMs use various techniques to detect errors in prompts, including:

1. **Language modeling:** LLMs can use their language modeling capabilities to identify prompts that are unlikely or unnatural, indicating potential errors.
2. **Part-of-speech tagging:** Analyzing the parts of speech (e.g., nouns, verbs, adjectives) to identify errors in sentence structure.
3. **Named entity recognition:** Identifying specific entities (e.g., names, locations, organizations) to detect errors in context.
4. **Dependency parsing:** Analyzing sentence structure and dependencies to identify errors in syntax.

Limitations: While LLMs have made progress in detecting errors, they are not perfect and can still struggle with:

1. **Contextual understanding:** LLMs may not always understand the context of the prompt, leading to incorrect error detection.
2. **Domain-specific knowledge:** LLMs may not possess domain-specific knowledge, making it challenging to detect errors in specialized fields.
3. **Ambiguity and nuance:** LLMs can struggle to detect errors in prompts with ambiguous or nuanced language.
4. **Adversarial prompts:** LLMs can be vulnerable to carefully crafted adversarial prompts designed to evade error detection.

Improvement strategies: To improve error detection in prompts, researchers and developers are exploring:

1. **Multitask learning:** Training LLMs on multiple tasks, including error detection, to improve their overall language understanding.
2. **Adversarial training:** Training LLMs on adversarial examples to improve their robustness to error-prone prompts.
3. **Human feedback:** Incorporating human feedback and evaluation to correct and refine LLM-generated responses.

4. Specialized models: Developing specialized models for specific domains or tasks to improve error detection in those areas.

Are Intentional Hallucinations Possible?

The following prompt was given to Llama 3 regarding the possibility of intentional hallucinations: the short answer is “not right now” and “possibly at some point in the future.”

Do you think it's possible for LLMs to intentionally hallucinate in their responses to prompts?

Llama 3:

What a fascinating question!

Intentional hallucination in LLMs (large language models) is a topic of ongoing research and debate. While LLMs are incredibly powerful, they are still machines, and their responses are ultimately determined by their programming, training data, and algorithms.

Can LLMs hallucinate?

Yes, LLMs can generate responses that are not based on factual information or evidence. This can occur due to various reasons, such as:

1. Overfitting: When an LLM is overfitting, it may generate responses that are not grounded in reality, but rather a result of memorization or pattern recognition.
2. Lack of domain knowledge: If an LLM is not familiar with a specific domain or topic, it may generate responses that are inaccurate or nonsensical.
3. Adversarial prompts: Maliciously crafted prompts can exploit an LLM's vulnerabilities, leading to hallucinated responses.
4. Mode collapse: In some cases, LLMs may generate responses that are not diverse or creative, but rather a result of mode collapse, where the model produces repetitive or similar outputs.

Intentional hallucination?

The question remains whether LLMs can intentionally hallucinate in their responses. While LLMs are capable of generating creative and diverse responses, intentional hallucination implies a level of self-awareness, understanding, and deliberate action.

Currently, LLMs do not possess the cognitive abilities to intentionally hallucinate in the way humans do. Their responses are ultimately determined by their programming, algorithms, and training data.

Future possibilities

As LLMs continue to evolve, it's possible that future models could be designed to intentionally generate creative or fictional responses. This might be achieved through advanced techniques, such as:

1. **Generative adversarial networks (GANs):** GANs could be used to generate responses that are intentionally fictional or creative.
2. **Meta-learning:** LLMs could be trained to learn how to generate responses that are intentionally hallucinated.
3. **Cognitive architectures:** Future LLMs might incorporate cognitive architectures that enable them to understand and intentionally generate fictional responses.

These possibilities are still in the realm of research and speculation. For now, LLMs are incredibly powerful tools, but they do not possess the capacity for intentional hallucination.

Reducing Hallucinations

The following prompt was given to Llama 3.

Please rank the techniques for reducing hallucinations based on the effectiveness of those techniques.

Llama 3:

Reducing hallucinations in language models is a crucial task to ensure the generated text is accurate and reliable. Based on the current research and studies, here's a ranking of techniques for reducing hallucinations, from most effective to least effective:

1. **Entity Disentanglement:** This technique involves identifying and separating entities mentioned in the input prompt from the generated text. This helps to prevent the model from generating hallucinated entities that are not present in the original input.
2. **Fact Verification:** This technique involves verifying the generated text against a knowledge graph or a fact-checking database to ensure the accuracy of the generated information.
3. **Adversarial Training:** This technique involves training the model on a dataset that includes hallucinated examples, which helps the model to learn to distinguish between accurate and hallucinated information.
4. **Input Prompt Engineering:** This technique involves designing input prompts that are specific, clear, and concise, which helps to reduce the likelihood of hallucinations.
5. **Model Ensemble:** This technique involves combining the outputs of multiple models to reduce the likelihood of hallucinations.
6. **Regularization Techniques:** This technique involves adding regularization terms to the model's loss function to penalize hallucinated outputs.

CAUSES OF HALLUCINATIONS IN LLMs

This section provides an overview of various issues that can result in hallucinations in LLMs. The following prompt was given to GPT-4o.

Which issues are more likely to result in hallucinations in LLMs?

GPT-4o:

Hallucinations in large language models (LLMs) refer to the generation of plausible-sounding but incorrect or nonsensical outputs that are not grounded in the provided input data or real-world facts. Several factors contribute to these hallucinations:

Key Issues Leading to Hallucinations in LLMs

1. Training Data Quality and Biases

- **Inaccurate or Misleading Data:** If the training data contains incorrect, biased, or misleading information, the model is likely to replicate these errors. Hallucinations can occur when the model generates content based on this flawed data.
- **Data Gaps:** Gaps or imbalances in the training data can lead to the model making up information to fill in the blanks, especially when it encounters topics it was not adequately trained on.

2. Over-Reliance on Patterns

- **Pattern Matching:** LLMs rely heavily on pattern recognition. Sometimes, they may generate text that fits the learned patterns but does not correspond to factual information, leading to plausible-sounding yet incorrect outputs.
- **Autoregressive Nature:** In autoregressive models like GPT-3, each token is generated based on the preceding ones, which can sometimes lead to compounding errors, especially if the initial context was ambiguous or misleading.

3. Lack of Real-World Knowledge

- **Static Knowledge:** LLMs are trained on a fixed dataset up to a certain point in time and do not have access to real-time information or updates unless explicitly fine-tuned. This can lead to outdated or incorrect responses, especially for rapidly changing topics.
- **Inference Without Grounding:** The model generates text based on learned patterns rather than verified facts, which can result in hallucinations if the context is complex or requires specific knowledge not covered during training.

4. Prompt Ambiguity

- **Vague or Ambiguous Prompts:** Prompts that are unclear or ambiguous can lead the model to generate content that fits multiple interpretations, increasing the likelihood of hallucinations.

- **Open-Ended Queries:** Very open-ended prompts can cause the model to generate creative but incorrect information as it tries to fill in the gaps.

5. Model Size and Complexity

- **Model Size:** Larger models with more parameters can capture more nuanced patterns but are also more prone to generating complex, convincing hallucinations. The complexity of these models makes it harder to control or predict specific outputs.
- **Over-Optimization:** Fine-tuning models to perform exceptionally well on specific datasets can sometimes lead to overfitting, where the model generates outputs that are too closely tied to the training data, including its biases and errors.

6. Evaluation and Alignment

- **Lack of Proper Evaluation:** Inadequate evaluation metrics that do not properly capture the correctness of generated content can lead to unnoticed hallucinations.
- **Alignment Issues:** Models that are not well-aligned with human values and factual accuracy standards may prioritize generating plausible-sounding text over true and accurate content.

Mitigation Strategies

1. Improved Data Curation:

- Ensure high-quality, diverse, and balanced training data.
- Regularly update the training datasets to include recent and accurate information.

2. Enhanced Prompt Design:

- Use clear, specific, and unambiguous prompts to guide the model toward generating accurate responses.
- Implement prompt engineering techniques to reduce the chances of hallucinations.

3. Fact-Checking Mechanisms:

- Integrate external knowledge bases and real-time information sources to verify the generated content.
- Implement post-processing steps to fact-check and correct the model outputs.

4. Regular Evaluation:

- Continuously evaluate the model using metrics that capture the factual accuracy of responses.
- Incorporate human feedback to identify and correct hallucinations.

5. Model Training and Alignment:

- Focus on training models that align with factual accuracy and ethical standards.
- Use reinforcement learning with human feedback (RLHF) to better align the model's outputs with human expectations and facts.

By addressing these issues and employing robust mitigation strategies, the incidence of hallucinations in LLMs can be significantly reduced, leading to more reliable and accurate model outputs.

Intrinsic Versus Extrinsic Hallucinations

The following prompt was given to Llama 3.

Please compare in-context hallucinations with extrinsic hallucinations.
Llama 3:

In-context hallucinations occur when a language model generates text that is not based on any actual information or evidence, but rather is a product of its own internal workings and biases. This type of hallucination happens within the context of the conversation or text generation, and the model may not even realize it's generating fictional information.

Examples of in-context hallucinations:

- A language model generates a fictional event or detail that is not supported by any evidence or facts.
- A model creates a character or entity that does not exist in reality.
- A model provides a false or misleading explanation for a phenomenon or concept.

In-context hallucinations can be problematic because they can:

- Spread misinformation or disinformation
- Erode trust in language models and AI systems
- Lead to confusion or misinterpretation of information

Extrinsic Hallucinations

Extrinsic hallucinations, on the other hand, occur when a language model generates text based on external information or stimuli that are not actually present in the input or context. This type of hallucination is often triggered by external factors, such as:

- Biases or assumptions in the training data
- Overfitting or memorization of specific patterns or examples
- Adversarial attacks or manipulation of the input

Examples of extrinsic hallucinations:

- A language model generates text that is influenced by biases in the training data, such as stereotypes or discriminatory language.
- A model generates text that is overly reliant on memorized patterns or examples, rather than generating novel responses.

- A model is manipulated by an adversary to generate specific types of text or responses.

Extrinsic hallucinations can be problematic because they can:

- Perpetuate biases and stereotypes
- Compromise the integrity and reliability of language models
- Enable malicious actors to manipulate the output of language models

Key differences

The key differences between in-context and extrinsic hallucinations are:

- **Source:** In-context hallucinations arise from the model's internal workings, while extrinsic hallucinations are triggered by external factors.
- **Nature:** In-context hallucinations are often fictional or made-up information, while extrinsic hallucinations are influenced by external biases, patterns, or manipulation.
- **Impact:** Both types of hallucinations can be problematic, but extrinsic hallucinations can have more severe consequences, such as perpetuating biases or enabling malicious actors.

By understanding the differences between in-context and extrinsic hallucinations, we can better address the challenges and limitations of language models and work toward developing more accurate, reliable, and trustworthy AI systems.

Hallucination Detection

The following question was posed to Llama 3.1 405B:

Do you have any recommendations regarding hallucination detection?

Llama 3.1 405B:

There are several approaches and tools available for hallucination detection in large language models (LLMs). Here are some recommendations:

1. **Semantic Entropy:** This method, proposed in the *Nature* article “Detecting Hallucinations in Large Language Models Using Semantic Entropy,” uses entropy-based uncertainty estimators to detect confabulations (arbitrary and incorrect generations) in LLMs. The method works by clustering answers with similar meanings and computing the entropy over the clusters.
2. **Pythia:** This is an advanced AI hallucination detection tool that verifies the factual accuracy and coherence of outputs from LLMs. It uses a powerful knowledge graph and interconnected information networks to analyze AI-generated content in real-time.

3. **EasyDetect:** This is a unified system for hallucination detection of LLMs, which won the second prize in the model-agnostic track of the SemEval-2024 Task 6. It utilizes prompt engineering and few-shot learning to detect hallucinations.
4. **Hallucination Detection Using NER:** This approach, proposed in the arXiv paper “Hallucination Detection: Robustly Discerning Reliable Answers in Large Language Models,” uses named entity recognition (NER) to detect hallucinations in LLMs.
5. **OPDAI:** This is a system for hallucination detection of LLMs, which achieves considerable results in the model-aware track of the SemEval-2024 Task 6. It uses a combination of prompt engineering and few-shot learning to detect hallucinations.

These are just a few examples of the many approaches and tools available for hallucination detection in LLMs. The choice of approach will depend on the specific use case and requirements.

In general, it is recommended to use a combination of methods to detect hallucinations, as different methods may be more effective for different types of hallucinations. Additionally, it is important to continuously evaluate and update the detection methods to ensure they remain effective as LLMs evolve.

In terms of future work, there are several areas that require further research, including:

1. **Improving the robustness of hallucination detection methods:** Current methods may not be robust to all types of hallucinations, and further research is needed to develop more effective methods.
2. **Developing methods for detecting hallucinations in multimodal LLMs:** As LLMs become increasingly multimodal, there is a need for methods that can detect hallucinations in multiple modalities, such as text, images, and audio.
3. **Investigating the relationship between hallucinations and other AI safety issues:** Hallucinations are just one of many AI safety issues, and further research is needed to understand the relationship between hallucinations and other issues, such as bias and fairness.

Overall, hallucination detection is an important area of research that requires continued attention and investment to ensure the safe and reliable deployment of LLMs.

Model Calibration

The following prompt was given to GPT-4o.

Please explain model calibration for LLMs.
GPT-4o:

Model calibration for LLMs is a crucial aspect of ensuring the reliability and trustworthiness of their predictions. Calibration refers to the process of adjusting the model's output probabilities to better reflect the true correctness likelihood of its predictions. In other words, a well-calibrated model is one that is accurate not only in its predictions but also in its confidence about those predictions.

Why Calibration Matters

LLMs are often overconfident in their predictions, which can lead to issues in real-world applications where accuracy and reliability are paramount. For instance, in a medical diagnosis scenario, an overconfident but incorrect diagnosis can have severe consequences. Calibration helps to address this by aligning the model's confidence with its actual performance, enabling more informed decision-making.

Calibration Techniques

1. **Temperature Scaling:** This is a simple and widely used method. It involves adjusting the temperature parameter (τ) in the softmax function that generates the output probabilities. The temperature parameter controls the entropy of the output distribution. Lowering the temperature makes the model more confident, while increasing it makes the model less confident. The optimal temperature is typically found through a validation set.
2. **Vector Scaling:** This method involves scaling the output logits by a vector before applying the softmax function. This allows for more fine-grained control over the calibration process, as different classes can be scaled differently.
3. **Histogram Binning:** This method involves dividing the output probabilities into bins and then adjusting the probabilities within each bin to match the empirical accuracy of the model within that bin.
4. **Isotonic Regression:** This is a non-parametric approach that involves fitting a monotonically increasing function to the model's output probabilities. The goal is to ensure that the model's confidence is monotonically related to its accuracy.
5. **Bayesian Methods:** These involve modeling the uncertainty in the model's parameters and outputs using Bayesian inference. This can provide a more principled approach to calibration, but it can be computationally expensive.

Challenges and Limitations

1. **Data Quality:** Calibration requires high-quality data that accurately reflects the true distribution of the problem. Poor data quality can lead to poorly calibrated models.

2. **Overfitting:** Calibration techniques can sometimes overfit to the validation set, leading to poor performance on unseen data.
3. **Computational Cost:** Some calibration techniques, especially Bayesian methods, can be computationally expensive.

Best Practices

1. **Use a Representative Validation Set:** Ensure that the validation set used for calibration is representative of the test data distribution.
2. **Monitor Calibration Metrics:** Regularly monitor calibration metrics such as expected calibration error (ECE) or Brier score to ensure the model remains well-calibrated.
3. **Regularly Update Calibration:** As the model is updated or fine-tuned, recalibrate to ensure that the model's confidence remains aligned with its accuracy.

Conclusion

Model calibration is a critical step in the development and deployment of LLMs. By ensuring that the model's confidence is aligned with its accuracy, calibration can significantly improve the reliability and trustworthiness of LLMs in real-world applications. It's important to be aware of the challenges and limitations of calibration techniques and to follow best practices to achieve optimal results.

KAPLAN AND UNDERTRAINED MODELS

Kaplan et al provide (empirical) power laws regarding the performance of language models, which they assert depends on the following:

- model size
- dataset size
- amount of compute for training

In their 2020 article “Scaling Laws for Neural Language Models,” Kaplan et al. assert that changing the network width or depth have minimal effects. They also claim that optimal training of very large models involves a relatively modest amount of data. The paper with the relevant details is accessible here: <https://arxiv.org/abs/2001.08361>.

Alternatively, Chinchilla is a 70B LLM that was trained on a dataset that is much larger than the size that is recommended by Kaplan et al. In fact, Chinchilla achieved state-of-the-art (SOTA) status has surpassed the performance of the following LLMs, all of which are between two and seven times larger than Chinchilla:

- Gopher (280B)
- GPT-3 (175B)
- J1-Jumbo (178B)

- LaMDA (137B)
- MT-NLG (530B)

Obviously an LLM that exceeds 1 trillion parameters faces a significant challenge in order to create datasets of the recommended size as described in the paper from the authors of Chinchilla. One interesting possibility involves automatic speech recognition (ASR) that might enable the generation of datasets that are larger than 10 trillion tokens by transcribing audio to text.

SUCCESS STORIES IN GENERATIVE AI

The following prompt was given to ChatGPT to describe success stories in generative AI.

Please provide a detailed list of success stories in generative AI.
ChatGPT:

Generative AI has led to numerous success stories across a wide array of industries, showcasing the technology's versatility and transformative potential. Here are some detailed examples illustrating how generative AI has been successfully applied:

1. Art and Design

- **OpenAI's DALL-E:** This AI model can generate novel images from textual descriptions, combining concepts in unique ways that are often surprising and creative. It has been used by artists and designers to spark creativity and generate ideas that are then refined into final artworks.
- **Google's DeepDream:** Originally developed to help scientists and engineers see what a deep neural network is seeing when it looks at a given image, DeepDream has become a popular tool in the art world for transforming photographs into surreal and imaginative pieces.

2. Media and Entertainment

- **Warner Music's Signing of Endel:** Endel, a startup that uses AI to create personalized sound environments, signed a deal with Warner Music to produce twenty albums. The AI analyzes factors like time of day, weather, and heart rate to create custom soundscapes that improve focus, relaxation, and sleep.
- **Scriptwriting by AI in "Sunspring":** In 2016, an AI named Benjamin wrote the screenplay for "Sunspring," a short sci-fi film. The script was bizarre yet intriguing, demonstrating the potential of AI in generating creative content for the film industry.

3. Healthcare

- **Drug Discovery:** Companies like Atomwise use AI to predict how different chemicals might interact with the body, speeding up the discovery process for new medications. Atomwise's AI has been used

to predict treatments for multiple diseases, potentially reducing development times and costs significantly.

- **Biomarker Development:** Deep Genomics uses generative AI to identify biomarkers and new molecular entities for genetic disorders. Their platform has successfully identified several viable targets for genetic diseases previously deemed undruggable.

4. Technology and Engineering

- **AI-Generated Code by GitHub Copilot:** Developed by GitHub and OpenAI, Copilot suggests whole lines or blocks of code as developers write, effectively providing pair programming support. It's trained on a dataset of public source code and has improved productivity for many developers.
- **Autodesk's Dreamcatcher:** A project aimed at creating a generative design system that uses AI to create thousands of design alternatives based on specific goals and constraints set by the user. It's particularly useful in automotive, manufacturing, and architecture industries.

5. Business and Marketing

- **Persado:** Persado uses generative AI to create marketing language that resonates more effectively with consumers. The AI analyzes data from previous campaigns to determine the words and phrases that are most likely to engage specific audiences, leading to improved conversion rates and customer engagement.
- **Jasper (formerly Jarvis):** An AI content platform that helps businesses create high-quality content for blogs, Web sites, and social media posts quickly and efficiently, significantly reducing the time and cost associated with content creation.

6. Fashion

- **Stitch Fix's Hybrid Design:** Combining human designers with AI tools, Stitch Fix uses AI to analyze trends and customer preferences to help design clothing that is more likely to appeal to its customers, improving satisfaction and reducing returns.

7. Education

- **Duolingo:** While primarily known for its language learning apps, Duolingo uses generative AI to create new practice sentences in different languages. This not only keeps the learning material fresh and engaging but also ensures a wide variety of examples are covered.

These success stories highlight the broad applicability of generative AI, driving innovations that not only enhance business efficiency but also contribute to creative industries and complex problem-solving in fields like healthcare and engineering.

REAL-WORLD USE CASES FOR GENERATIVE AI

The following prompt was given to Cohere to describe real-world use cases for Generative AI and their impact.

Please describe real-world use cases for generative AI, along with their potential impact.

Cohere:

Generative artificial intelligence (AI) has the potential to revolutionize various industries and solve complex problems. Here's an improved version of the description, including additional use cases and their potential impact:

Healthcare:

Drug Discovery and Design: Generative AI can be used to identify potential drug candidates by generating molecular structures with desired properties. This accelerates the drug discovery process, leading to faster development of treatments for diseases.

Medical Image Analysis: Generative models can assist in detecting and diagnosing medical conditions from medical images, such as MRI scans or X-rays. This helps healthcare professionals in making more accurate diagnoses and improving patient care.

Personalized Medicine: Generative AI can analyze patient data to provide personalized treatment plans and recommendations. By considering individual genetic variations and health histories, generative AI enables more precise and effective healthcare.

Climate and Environment:

Climate Modeling and Prediction: Generative AI models can analyze vast amounts of climate data to predict and model complex climate patterns. This aids in understanding climate change, extreme weather events, and the development of mitigation strategies.

Sustainable Resource Management: Generative AI can optimize the use of natural resources, such as water or energy, by generating efficient allocation plans. This helps reduce waste, improve sustainability, and preserve the environment.

Biodiversity Conservation: Generative models can assist in identifying and classifying plant and animal species, supporting biodiversity conservation efforts. They can also generate habitat restoration plans to promote ecological balance.

Creative Industries:

Content Generation: Generative AI can create text, images, or even music with remarkable creativity and originality. This technology can assist writers, artists, musicians, and content creators in producing unique and engaging content.

Art and Design: Generative AI models can generate aesthetically pleasing artwork, designs, or architectural concepts. They can inspire human creators and lead to innovative artistic styles and trends.

Video and Game Development: Generative AI can be used to create realistic animations, generate game environments, or develop nonplayer characters, enhancing the immersive experience for gamers.

Education:

Personalized Learning: Generative AI can adapt educational content to individual student needs, providing customized lessons, practice problems, and feedback. This personalized approach improves learning outcomes and engagement.

Language Learning: Generative models can facilitate language acquisition by generating interactive and contextualized language-learning materials. They can provide tailored exercises, conversations, and feedback to students.

Virtual Tutoring: Generative AI can power virtual tutors that offer personalized guidance and explanations to students, supplementing traditional teaching methods.

Finance and Economics:

Fraud Detection and Prevention: Generative AI models can analyze patterns in financial data to detect fraudulent activities, such as credit card fraud or money laundering. This helps protect individuals and organizations from financial losses.

Investment Strategies: Generative AI can generate insights and identify patterns in financial markets, aiding in investment decision-making. It can analyze vast amounts of data to predict market trends and optimize investment portfolios.

Risk Assessment and Management: Generative models can assess and manage financial risks, such as credit risk or market risk, by generating probabilistic scenarios and providing risk mitigation strategies.

Transportation and Logistics:

Autonomous Vehicles: Generative AI can enhance the perception and decision-making capabilities of self-driving cars, improving their ability to recognize objects, predict trajectories, and make safe driving decisions.

Route Optimization: Generative models can optimize delivery routes for logistics and transportation companies, reducing travel distances, fuel consumption, and delivery times.

Traffic Flow Optimization: Generative AI can analyze traffic patterns and suggest improvements to reduce congestion and optimize traffic flow in urban areas.

Social Impact and Humanitarian Efforts:

Disaster Response: Generative AI can assist in disaster relief efforts by analyzing damage, identifying affected areas, and coordinating rescue and relief operations.

Language Translation and Preservation: Generative models can facilitate cross-lingual communication and help preserve endangered languages, promoting cultural exchange and understanding.

Social Good Initiatives: Generative AI can be leveraged to address social issues, such as generating awareness campaigns for humanitarian causes or providing personalized recommendations for social services.

Potential Impact:

The real-world use cases of generative AI have the potential to bring about significant positive changes. They can enhance human creativity, improve decision-making, increase efficiency, and solve complex problems in various domains. Generative AI can drive innovation, foster economic growth, and improve the overall quality of life for people worldwide.

It is important to approach the development and deployment of generative AI with caution and ethical considerations. Addressing biases, ensuring privacy and security, and establishing governance frameworks are crucial to mitigate potential risks and ensure that the benefits of generative AI are widely accessible and beneficial to society.

DEEPMIND

DeepMind has made significant contributions to AI, which includes the creation of various AI systems. DeepMind was established in 2010 and became a subsidiary of Google in 2014, and its home page is here: <https://deepmind.com/>

DeepMind created the 280GB language model Gopher that significantly outperforms its competitors, including GPT-3, J1-Jumbo, and MT-NLG. DeepMind also developed AlphaFold that solved a protein folding task in thirty minutes that had eluded researchers for ten years. Moreover, DeepMind made AlphaFold available for free for everyone in July 2021. DeepMind has made significant contributions in the development of world caliber AI game systems, some of which are discussed in the next section.

DeepMind and Games

DeepMind is the force behind the AI systems StarCraft and AlphaGo that defeated the best human players in Go (which is considerably more difficult than chess). These games provide “perfect information,” whereas games with “imperfect information” (such as Poker) have posed a challenge for ML models.

AlphaGo Zero (the successor of AlphaGo) mastered the game through self-play in less time and with less computing power. AlphaGo Zero exhibited extraordinary performance by defeating AlphaGo 100–0. Another powerful

system is AlphaZero that also used a self-play technique learned to play Go, chess, and shogi, and also achieved SOTA performance results.

By way of comparison, ML models that use tree search are well-suited for games with perfect information. By contrast, games with imperfect information (such as Poker) involve hidden information that can be leveraged to devise counter strategies to counteract the strategies of opponents. In particular, AlphaStar is capable of playing against the best players of StarCraft II, and also became the first AI to achieve SOTA results in a game that requires “strategic capability in an imperfect information world.”

Player of Games (PoG)

The DeepMind team at Google devised the general-purpose PoG (player of games) algorithm that is based on the following techniques:

- CFR (counterfactual regret minimization)
- CVPN (counterfactual value-and-policy network)
- GT-CFT (growing tree CFR)
- CVPN

The counterfactual value-and-policy network (CVPN) is a neural network that calculates the counterfactuals for each state belief in the game. This is key to evaluating the different variants of the game at any given time.

Growing tree CFR (GT-CFR) is a variation of CFR that is optimized for game-trees trees that grow over time. GT-CFR is based on two fundamental phases, which is discussed in more detail here:

<https://medium.com/syncedreview/deepminds-pog-excels-in-perfect-and-imperfect-information-games-advancing-research-on-general-9dbad5c04221>

OPENAI

OpenAI is an AI research company that has made significant contributions to AI, including DALL-E and ChatGPT, and its home page is here: <https://openai.com/api/>

OpenAI was founded in San Francisco by Elon Musk and Sam Altman (and others), and one of its stated goals is to develop AI that benefits humanity. Given Microsoft’s massive investments in and deep alliance with the organization, OpenAI might be viewed as an arm of Microsoft. OpenAI is the creator of the GPT-x series of LLMs as well as ChatGPT that was made available on November 11, 2022.

OpenAI made GPT-3 commercially available via API for use across applications, charging on a per-word basis. GPT-3 was announced in July 2020 and was available through a beta program. In November 2021 OpenAI made GPT-3 open to everyone, and more details are accessible here:

<https://openai.com/blog/api-no-waitlist/>

In addition, OpenAI developed DALL-E that generates images from text. OpenAI initially did not permit users to upload images that contained realistic faces. Later (Q4/2022) OpenAI changed its policy to allow users to upload faces into its online system. Check the OpenAI Web page (<https://openai.com/>) for more details. Incidentally, diffusion models have superseded the benchmarks of DALL-E.

OpenAI has also released a public beta of Embeddings, which is a data format that is suitable for various types of tasks with machine learning, as described here:

<https://beta.openai.com/docs/guides/embeddings>

OpenAI is the creator of Codex that provides a set of models that were trained on NLP. The initial release of Codex was in private beta, and more information is accessible here: <https://beta.openai.com/docs/engines/instruct-series-beta>

OpenAI provides four models that are collectively called their Instruct models, which support the ability of GPT-3 to generate natural language. These models will be deprecated in early January 2024 and replaced with updated versions of GPT-3, ChatGPT, and GPT-4.

To learn more about the features and services that OpenAI offers, navigate to the following link: <https://platform.openai.com/overview>

COHERE

Cohere is a start-up and a competitor of OpenAI, and its home page is here: <https://cohere.ai/>

Cohere develops cutting-edge NLP technology that is commercially available for multiple industries. Cohere is focused on models that perform textual analysis instead of models for text generation (such as GPT-based models). The founding team of Cohere is impressive: CEO Aidan Gomez is one of the co-inventors of the transformer architecture, and CTO Nick Frosst is a protege of Geoff Hinton.

HUGGING FACE

Hugging Face is a popular community-based repository for open-source NLP technology, and its home page is here: <https://github.com/huggingface>

Unlike OpenAI or Cohere, Hugging Face does not build its own NLP models. Instead, Hugging Face is a platform that manages a plethora of open-source NLP models that customers can fine-tune and then deploy those fine-tuned models. Indeed, Hugging Face has become the eminent location for people to collaborate on NLP models, and sometimes described as “GitHub for machine learning and NLP.”

Hugging Face Libraries

Hugging Face provides three important libraries: datasets, tokenizers, and transformers. The Accelerate library supports PyTorch models. The datasets library provides an assortment of libraries for NLP. The tokenizers library enables users to convert text data to numeric values.

Perhaps the most impressive library is the transformers library that provides an enormous set of pretrained BERT-based models in order to perform a wide variety of NLP tasks. The Github repository is here: <https://github.com/huggingface/transformers>

Hugging Face Model Hub

Hugging Face provides a model hub that provides a plethora of models that are accessible online. Moreover, the Web site supports online testing of its models, which includes the following tasks:

- masked word completion with BERT
- name entity recognition with Electra
- natural language inference with RoBERTa
- question answering with DistilBERT
- summarization with BART
- text generation with GPT-2
- translation with T5

Navigate to the following link to see the text generation capabilities of “write with transformer”: <https://transformer.huggingface.co>

META AI

Meta AI is an AI assistant that is built with Llama 3. Meta AI can be used with multiple products (such as WhatsApp and Messenger) from Meta, and it is also available on mobile phones. The following prompt was given to Meta AI:

Please provide a detailed description of MetaAI.
Meta AI

Meta AI: Your Friendly AI Assistant

I'm Meta AI, an AI assistant developed by Meta Platforms Inc. My primary function is to understand and respond to human input in a helpful and engaging manner. I can answer questions, provide information, and even generate text based on a given prompt.

Key Features:

1. **Natural Language Processing (NLP):** I'm trained on a massive dataset of text from various sources, allowing me to understand and process human language.

2. **Knowledge Base:** I have access to a vast knowledge base that I can draw upon to answer questions and provide information.
3. **Contextual Understanding:** I can understand the context of a conversation or question, enabling me to provide more accurate and relevant responses.
4. **Text Generation:** I can generate text based on a prompt, topic, or style.
5. **Conversational Interface:** I'm designed to engage in natural-sounding conversations, using a human-like tone and language.

Capabilities:

- Answering questions on various topics, from science and history to entertainment and culture
- Generating text summaries, articles, or creative writing
- Offering suggestions and ideas for creative projects
- Providing definitions and explanations for complex terms and concepts
- Conversing on various topics, from news and current events to hobbies and interests

Limitations:

- While I strive to provide accurate and helpful responses, I'm not perfect and may make mistakes or not fully understand certain contexts.
- My knowledge base is vast but not exhaustive, so I may not always have the answer to a specific question.
- I'm a machine, so while I can generate text, I don't have personal experiences or emotions like humans do.

Overall, I'm here to assist and provide helpful information, answer questions, and engage in conversation to the best of my abilities. Feel free to ask me anything!

AI21

AI21 is a company that provides proprietary large language models via API to support the applications of its customers. The current SOTA model of AI21 is called *Jurassic-1* (roughly the same size as GPT-3), and AI21 also creates its own applications on top of *Jurassic-1* and other models. The current application suite of AI21 involves tools that can augment reading and writing.

Primer is an older competitor in this space, founded two years before the invention of the transformer. The company primarily serves clients in government and defense.

ANTHROPIC AND CLAUDE 3

Anthropic was created in 2021 by former employees of OpenAI and its home page is here: <https://www.anthropic.com/>

Anthropic has significant financial support from an assortment of companies, including Google and Salesforce. As this book goes to print, Anthropic released Claude 3 as a competitor to ChatGPT, and it's available in three versions: Opus, Sonnet, and Haiku (Opus is the most powerful version).

Claude 3 Opus has a high degree of comprehension and expert level knowledge in fields such as mathematics. Opus is currently available in many countries for a monthly subscription of USD 20.

Claude 3 Sonnet is twice as fast as earlier versions of Claude (i.e., Claude 2 and Claude 2.1), along with improved reasoning capabilities, and it's well-suited for fast answers to queries. In addition, Sonnet is freely available.

Claude 3 Haiku the most cost effective as well as the fastest version of Claude 3, and it's available on a per-token basis.

Other aspects of Claude 3 to keep in mind when comparing Claude 3 with other LLMs. For example, Claude 3 is more likely to respond to prompts than earlier versions. Second, Claude 3 is less likely to respond with incorrect results and more likely to indicate that it lacks information regarding a prompt. In addition, Claude 3 currently provides a 200K context window, and is likely to support one million tokens in future iterations of Claude 3.

SUMMARY

This chapter started with an introduction to generative AI and several components of generative AI, including LLMs such as ChatGPT, Gemini, Claude 3, Meta AI, DALLÉ-3, and so forth. Readers also learned about the difference between generative AI versus Conversational AI, AGI (Artificial General Intelligence), and several branches of machine learning, such as deep learning, natural language processing, and reinforcement learning.

Readers also acquired a basic understanding of LLMs and their role in generative AI, and in the final section of this chapter, they were introduced to creating generative AI applications, which included a rudimentary Python code sample.

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CLAUDE 3 AND PYTHON CODE

This chapter contains examples of using Claude 3 in order to perform an eclectic mix of Python tasks, from generating Python code for a simple calculator to generating code to calculate probabilities involved in throwing two or more dice. Note that Claude 3 generated all the code samples in this chapter.

The first part of this chapter starts with an example of generating a simple numeric calculator that you can extend by providing additional functionality (such as support for trigonometric functions). This section also contains Python code for handling text file and scraping Web sites.

The second part of this chapter contains Python code for creating a basic chatbot as well as performing data visualization. In addition, you will learn how to generate random numbers Pandas. The third part of this chapter shows you how to generate Fibonacci numbers using an iterative solution.

SIMPLE CALCULATOR

The following prompt was given to Claude 3 regarding a Python code sample for a calculator.

Please generate a Python code sample that performs as a simple calculator, along with an explanation of the code.

Claude 3:

I'll explain how a basic calculator works in Python and share a simple implementation.

Listing 6.1 displays the contents of `simple_calculator.py` that Claude 3 generated as a rudimentary calculator.

LISTING 6.1: simple_calculator.py

```
def calculator():
    print("Simple Calculator")
    print("Operations: +, -, *, /")

    # Get input from user
    num1 = float(input("Enter first number: "))
    operator = input("Enter operator: ")
    num2 = float(input("Enter second number: "))

    # Perform calculation based on operator
    if operator == "+":
        result = num1 + num2
    elif operator == "-":
        result = num1 - num2
    elif operator == "*":
        result = num1 * num2
    elif operator == "/":
        if num2 == 0:
            return "Error: Cannot divide by zero"
        result = num1 / num2
    else:
        return "Invalid operator"

    return f"{num1} {operator} {num2} = {result}"

# Run the calculator
if __name__ == "__main__":
    print(calculator())
```

Explanation:

This calculator:

1. Takes two numbers and an operator as input
2. Supports basic operations (+, -, *, /)
3. Handles division by zero errors
4. Returns the formatted result

Now launch the code in Listing 6.1, and calculate the product of 9 and 23, as shown below:

```
Simple Calculator
Operations: +, -, *, /
Enter first number: 9
Enter operator: *
Enter second number: 23
9.0 * 23.0 = 207.0
```

As you can see, there is no error checking in order to ensure that *a* and *b* are both numeric values, and there is no check for the operator to ensure that it always consists of lowercase letters. Now launch the code in Listing 6.1 and you will see that its output is the value 207.

SIMPLE FILE HANDLING

The following prompt was given to Claude 3 regarding a Python code sample that performs simple file handling.

Please generate a Python code sample that performs simple file handling, along with an explanation of the code.

Claude 3:

I'll create a sample that demonstrates basic file operations in Python including reading, writing, and appending to files.

Listing 6.2 displays the contents of `file_handling.py` that Claude 3 generated in order to read content from a file and then write new content to a different file.

LISTING 6.2: `file_handling.py`

```
def demonstrate_file_operations():
    # 1. Writing to a file
    try:
        with open('sample.txt', 'w') as file:
            file.write("Line 1: Hello, this is a test
file.\n")
            file.write("Line 2: Python file handling is
simple!\n")
            print("Successfully wrote to file.")
    except IOError as e:
        print(f"Error writing to file: {e}")

    # 2. Reading from a file
    try:
        with open('sample.txt', 'r') as file:
            # Read entire file
            content = file.read()
            print("\nReading entire file:")
            print(content)

            # Read file line by line
            with open('sample.txt', 'r') as file:
                print("\nReading line by line:")
                for line in file:
                    print(line.strip()) # strip() removes
trailing newlines
    except FileNotFoundError:
        print("Error: File not found")
    except IOError as e:
        print(f"Error reading file: {e}")

    # 3. Appending to a file
```

```

try:
    with open('sample.txt', 'a') as file:
        file.write("Line 3: This line is appended!\n")
        print("\nSuccessfully appended to file.")
except IOError as e:
    print(f"Error appending to file: {e}")

# 4. Reading specific number of characters
try:
    with open('sample.txt', 'r') as file:
        print("\nReading first 10 characters:")
        print(file.read(10))
except IOError as e:
    print(f"Error reading file: {e}")

if __name__ == "__main__":
    demonstrate_file_operations()

```

Explanation:

Let me explain the key concepts in this code:

1. File Opening Modes:
 - 'w': Write mode (creates new file/overwrites existing)
 - 'r': Read mode (default)
 - 'a': Append mode (adds to end of file)
2. Using `with` Statement:
 - Automatically handles file closing
 - Ensures proper resource management
 - Recommended over manual `file.open()` and `file.close()`
3. Error Handling:
 - Uses `try/except` blocks to handle common file operations errors
 - Catches specific exceptions like `FileNotFoundError` and `IOError`
4. Key File Operations:
 - `write()`: Writes strings to file
 - `read()`: Reads entire file content
 - `readline()`: Reads one line at a time
 - `strip()`: Removes trailing newlines and whitespace
5. File Iteration:
 - Can iterate through file line by line using a `for` loop
 - Memory efficient for large files

The code demonstrates these common file operations:

1. Creating and writing to a new file
2. Reading an entire file at once
3. Reading a file line by line

4. Appending to an existing file
5. Reading specific number of characters

Now launch the code in Listing 6.2 and notice a newly created text file called `sample.txt` with the following contents:

```
Line 1: Hello, this is a test file.
Line 2: Python file handling is simple!
Line 3: This line is appended!
```

SIMPLE WEB SCRAPING

The following prompt was given to Claude 3 regarding a Python code sample that performs basic Web scraping.

Please generate a Python code sample that performs basic Web scraping, along with an explanation of the code.

Claude 3:

I'll create a Python script that demonstrates basic web scraping using the ``requests`` and ``BeautifulSoup`` libraries to extract information from a webpage.

Listing 6.3 displays the contents of `web_scraping.py` that Claude 3 generated to perform Web scraping. Note: make sure you have installed the libraries `BeautifulSoup` and `requests`, which you can perform from the command line as follows:

```
pip3 install bs4
pip3 install requests
```

In addition, you need to specify a valid URL for the value of the variable `url`.

LISTING 6.3: `web_scraping.py`

```
import requests
from bs4 import BeautifulSoup
import csv
from datetime import datetime

def web_scraper():
    # Headers to mimic a browser request
    headers = {
        'User-Agent': 'Mozilla/5.0 (Windows NT 10.0;
Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/91.0.4472.124 Safari/537.36'
    }

    def scrape_website(url):
        try:
            # Send GET request to the URL
```

```

        print(f"Fetching data from: {url}")
        response = requests.get(url, headers=headers)
        response.raise_for_status() # Raise exception
for bad status codes

        # Parse the HTML content
        soup = BeautifulSoup(response.content, 'html.
parser')

        return soup

    except requests.RequestException as e:
        print(f"Error fetching the webpage: {e}")
        return None

def save_to_csv(data, filename):
    try:
        with open(filename, 'w', newline='',
encoding='utf-8') as file:
            writer = csv.DictWriter(file,
fieldnames=data[0].keys())
            writer.writeheader()
            writer.writerows(data)
            print(f"Data successfully saved to {filename}")
    except IOError as e:
        print(f"Error saving to CSV: {e}")

def extract_information(soup):
    if not soup:
        return []

    # Example: Extract all article titles and links
    articles = []
    # This is an example for a generic news website
    # You'll need to modify the selectors based on the
actual website
    article_elements = soup.find_all('article')

    for article in article_elements:
        try:
            title_element = article.find('h2')
            link_element = article.find('a')
            date_element = article.find('time')

            if title_element and link_element:
                articles.append({
                    'title': title_element.text.strip(),
                    'link': link_element.get('href', ''),
                    'date': date_element.text.strip()
if date_element else 'N/A',
                    'scraped_at': datetime.now().
strftime("%Y-%m-%d %H:%M:%S")

```

```

        })
    except AttributeError as e:
        print(f"Error extracting article data: {e}")
        continue

    return articles

# Example usage
url = "https://example.com" # Replace with actual
website URL
soup = scrape_website(url)

if soup:
    # Extract the data
    articles = extract_information(soup)

    # Print results
    print(f"\nFound {len(articles)} articles:")
    for article in articles:
        print(f"\nTitle: {article['title']}")
        print(f"Link: {article['link']}")
        print(f>Date: {article['date']}")

    # Save to CSV
    if articles:
        save_to_csv(articles, f'articles_{datetime.
now().strftime("%Y%m%d")}.csv')

if __name__ == "__main__":
    web_scraper()

```

Let me explain the key components of this web scraper:

1. Required Libraries:
 - `requests`: For making HTTP requests to websites
 - `BeautifulSoup`: For parsing HTML content
 - `csv`: For saving data to CSV files
 - `datetime`: For timestamps
2. Main Components:
 - `scrape\_website()`: Handles the HTTP request and initial HTML parsing
 - `extract\_information()`: Extracts specific data from the parsed HTML
 - `save\_to\_csv()`: Saves the scraped data to a CSV file
3. Key Features:
 - Uses headers to mimic a browser request (helps avoid blocking)
 - Includes error handling for network requests and parsing

- Organizes data into a structured format
 - Saves results to a timestamped CSV file
4. Important Concepts:
- HTML parsing with BeautifulSoup
 - CSS selector usage for finding elements
 - Error handling for robust scraping
 - Data storage and export
5. Best Practices Implemented:
- Uses proper headers to identify the scraper
 - Includes error handling throughout
 - Structured data extraction
 - Clean data storage solution

Remember to:

1. Check the website's robots.txt file for scraping permissions
2. Include proper delays between requests for larger scraping jobs
3. Modify the CSS selectors based on the target website's structure
4. Handle rate limiting and blocking
5. Follow the website's terms of service

Would you like me to explain any specific part in more detail or add additional features like:

- Rate limiting
- Proxy support
- More complex data extraction
- Different output formats?

Note: Make sure to respect websites' terms of service and robots.txt files when scraping, and implement appropriate delays between requests to avoid overwhelming servers.

Replace the fictitious URL with a valid URL, then launch the code in Listing 6.3, and check the generated output.

BASIC CHAT BOT

The following prompt was given to Claude 3 regarding a Python code sample that provides simple chatbot functionality.

Please generate a Python code sample that performs basic chatbot functionality, along with an explanation of the code.

Claude 3:

I'll create a simple chatbot that uses pattern matching and predefined responses. This example will demonstrate basic

chatbot functionality with the ability to understand simple queries and maintain context.

Listing 6.4 displays the contents of `basic_chatbot.py` that Claude 3 generated to perform chatbot functionality.

LISTING 6.4: `basic_chatbot.py`

```
import re
from datetime import datetime
import random

class SimpleChatbot:
    def __init__(self):
        # Initialize bot's name and basic patterns
        self.name = "PyBot"
        self.user_name = None

        # Store conversation context
        self.context = {
            "last_topic": None,
            "user_mood": None,
            "conversation_started": datetime.now()
        }

        # Define response patterns and their corresponding
        answers
        self.patterns = {
            r'hi|hello|hey': self._greeting_response,
            r'how are you': self._mood_response,
            r'my name is ([\w\s]+)': self._remember_name,
            r'what is your name': self._tell_name,
            r'what time': self._tell_time,
            r'bye|goodbye': self._farewell_response,
            r'help|what can you do': self._help_response,
            r'weather|temperature': self._weather_response,
            r'tell me a joke': self._joke_response,
            r'thank you|thanks': self._gratitude_response
        }

        # Different response variations for more natural
        conversation
        self.responses = {
            "greeting": [
                "Hello! How can I help you today?",
                "Hi there! Nice to meet you!",
                "Hey! How's your day going?"
            ],
            "unknown": [
                "I'm not sure I understand. Could you
                rephrase that?",
```

```

        "I'm still learning! Could you try asking
in a different way?",
        "I didn't quite catch that. Can you explain
more?"
    ],
    "jokes": [
        "Why don't scientists trust atoms? Because
they make up everything!",
        "What did the Python say when it was
confused? Print('Help')",
        "Why did the programmer quit his job?
Because he didn't get arrays!"
    ]
}

def _greeting_response(self, message):
    if self.user_name:
        return f"{random.choice(self.
responses['greeting'])} Nice to see you again, {self.user_
name}!"
    return random.choice(self.responses['greeting'])

def _mood_response(self, message):
    return "I'm functioning well, thank you for asking!
How about you?"

def _remember_name(self, message):
    # Extract name using regex
    match = re.search(r'my name is ([\w\s]+)', message.
lower())
    if match:
        self.user_name = match.group(1).strip().title()
        return f"Nice to meet you, {self.user_name}!"
    return "I didn't catch your name. Could you please
repeat it?"

def _tell_name(self, message):
    return f"My name is {self.name}! I'm a simple
chatbot."

def _tell_time(self, message):
    current_time = datetime.now().strftime("%H:%M")
    return f"The current time is {current_time}"

def _farewell_response(self, message):
    if self.user_name:
        return f"Goodbye, {self.user_name}! Have a
great day!"
    return "Goodbye! Have a great day!"

```

```

    def _help_response(self, message):
        return """I can help you with:
1. Greeting and basic conversation
2. Telling the time
3. Remembering your name
4. Telling jokes
5. Responding to thanks
Just ask me naturally!"""

    def _weather_response(self, message):
        return "I'm sorry, I don't have access to real
weather data, but I can recommend a weather service!"

    def _joke_response(self, message):
        return random.choice(self.responses["jokes"])

    def _gratitude_response(self, message):
        return "You're welcome! Is there anything else I
can help you with?"

    def _update_context(self, message):
        # Simple sentiment analysis
        positive_words = ['happy', 'good', 'great',
'awesome', 'excellent']
        negative_words = ['sad', 'bad', 'terrible',
'awful', 'unhappy']

        message_lower = message.lower()

        for word in positive_words:
            if word in message_lower:
                self.context["user_mood"] = "positive"
                break
        for word in negative_words:
            if word in message_lower:
                self.context["user_mood"] = "negative"
                break

    def respond(self, message):
        # Update context with new message
        self._update_context(message)

        # Check each pattern for a match
        for pattern, response_func in self.patterns.
items():
            if re.search(pattern, message.lower()):
                return response_func(message)

        # If no pattern matches, return default response
        return random.choice(self.responses["unknown"])

```

```

def main():
    # Initialize chatbot
    chatbot = SimpleChatbot()

    print(f"{chatbot.name}: Hello! I'm {chatbot.name}.
    What's your name?")

    # Main conversation loop
    while True:
        # Get user input
        user_input = input("You: ").strip()

        # Check for exit condition
        if user_input.lower() in ['quit', 'exit']:
            print(f"{chatbot.name}: Goodbye! Have a great
day!")
            break

        # Get and print chatbot response
        response = chatbot.respond(user_input)
        print(f"{chatbot.name}: {response}")

if __name__ == "__main__":
    main()

```

This chatbot implementation includes several key features:

1. Pattern Matching:
 - Uses regular expressions to match user input patterns
 - Supports various types of queries and responses
2. Context Management:
 - Stores user name and conversation context
 - Tracks user mood through basic sentiment analysis
 - Maintains conversation history
3. Natural Responses:
 - Uses variety in responses to seem more natural
 - Includes random selection from multiple response options
 - Personalizes responses using stored context
4. Basic Features:
 - Greeting and farewell responses
 - Name remembering
 - Time telling
 - Joke telling
 - Help information
 - Basic weather response (placeholder)
5. Main Components:
 - `SimpleChatbot` class with various helper methods

- Pattern-response mapping
- Context updating mechanism
- Main conversation loop

6. Best Practices:

- Organized code structure
- Error handling
- Modular design for easy expansion
- Clear response categorization

To use the chatbot:

1. Run the script
2. Enter messages naturally
3. Type 'quit' or 'exit' to end the conversation

Some example interactions:

PyBot: Hello! I'm PyBot. What's your name?

You: My name is Alice

PyBot: Nice to meet you, Alice! How can I help you today?

You: What time is it?

PyBot: The current time is 14:30

You: Tell me a joke

PyBot: Why don't scientists trust atoms? Because they make up everything!

Would you like me to explain any specific part in more detail or add more advanced features like:

- Natural Language Processing (NLP) integration
- Database storage for conversations
- More complex context management
- API integrations for weather, news, etc.?

Now launch the code in Listing 6.4 and you can “interact” with the chatbot by typing your responses to its prompts.

BASIC DATA VISUALIZATION

The following prompt was given to Claude 3 regarding a Python code sample that performs basic data visualization.

Please generate a Python code sample that performs basic data visualization, along with an explanation of the code.

Claude 3:

I'll create a Python script that demonstrates various types of data visualization using popular libraries like matplotlib and seaborn.

Listing 6.5 displays the contents of `basic_visualization.py` that Claude 3 generated to perform simple data visualization. Note: For this task,

you'll need to have the matplotlib library and seaborn library installed as follows:

```
python3 -m pip install matplotlib
python3 -m pip install seaborn
```

LISTING 6.5: basic_visualization.py

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta

def create_sample_data():
    # Create sample sales data
    np.random.seed(42)
    dates = pd.date_range(start='2023-01-01', end='2023-12-31', freq='D')

    data = {
        'date': dates,
        'sales': np.random.normal(1000, 200, len(dates)),
        'category': np.random.choice(['Electronics',
        'Clothing', 'Food', 'Books'], len(dates)),
        'customer_satisfaction': np.random.normal(4.0, 0.5,
        len(dates)).clip(1, 5),
        'region': np.random.choice(['North', 'South',
        'East', 'West'], len(dates))
    }

    return pd.DataFrame(data)

def create_visualizations(df):
    # Set the style for all plots
    plt.style.use('seaborn')

    # 1. Line Plot - Daily Sales Trend
    plt.figure(figsize=(12, 6))
    plt.plot(df['date'], df['sales'], linewidth=2)
    plt.title('Daily Sales Trend Over Time')
    plt.xlabel('Date')
    plt.ylabel('Sales Amount')
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.savefig('sales_trend.png')
    plt.close()

    # 2. Bar Plot - Average Sales by Category
    plt.figure(figsize=(10, 6))
    category_avg = df.groupby('category')['sales'].mean().
    sort_values(ascending=False)
```

```

category_avg.plot(kind='bar')
plt.title('Average Sales by Category')
plt.xlabel('Category')
plt.ylabel('Average Sales')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('category_sales.png')
plt.close()

# 3. Box Plot - Sales Distribution by Region
plt.figure(figsize=(10, 6))
sns.boxplot(x='region', y='sales', data=df)
plt.title('Sales Distribution by Region')
plt.xlabel('Region')
plt.ylabel('Sales')
plt.tight_layout()
plt.savefig('region_distribution.png')
plt.close()

# 4. Histogram - Customer Satisfaction Distribution
plt.figure(figsize=(10, 6))
plt.hist(df['customer_satisfaction'], bins=20,
edgecolor='black')
plt.title('Distribution of Customer Satisfaction
Scores')
plt.xlabel('Satisfaction Score')
plt.ylabel('Frequency')
plt.tight_layout()
plt.savefig('satisfaction_distribution.png')
plt.close()

# 5. Heatmap - Correlation Matrix
plt.figure(figsize=(8, 6))
numeric_cols = df.select_dtypes(include=[np.number]).
columns
correlation = df[numeric_cols].corr()
sns.heatmap(correlation, annot=True, cmap='coolwarm',
center=0)
plt.title('Correlation Matrix of Numeric Variables')
plt.tight_layout()
plt.savefig('correlation_matrix.png')
plt.close()

# 6. Scatter Plot - Sales vs. Customer Satisfaction
plt.figure(figsize=(10, 6))
plt.scatter(df['customer_satisfaction'], df['sales'],
alpha=0.5)
plt.title('Sales vs. Customer Satisfaction')
plt.xlabel('Customer Satisfaction')
plt.ylabel('Sales')
plt.tight_layout()

```

```

plt.savefig('sales_satisfaction_scatter.png')
plt.close()

# 7. Pie Chart - Sales Distribution by Category
plt.figure(figsize=(10, 8))
category_totals = df.groupby('category')['sales'].
sum()
plt.pie(category_totals, labels=category_totals.index,
autopct='%1.1f%%')
plt.title('Sales Distribution by Category')
plt.axis('equal')
plt.tight_layout()
plt.savefig('category_distribution_pie.png')
plt.close()

def generate_summary_statistics(df):
    """Generate summary statistics for the dataset"""
    summary = {
        'total_sales': df['sales'].sum(),
        'average_daily_sales': df['sales'].mean(),
        'highest_sales_day': df.loc[df['sales'].idxmax(),
'date'],
        'lowest_sales_day': df.loc[df['sales'].idxmin(),
'date'],
        'average_satisfaction': df['customer_
satisfaction'].mean(),
        'total_days': len(df),
        'sales_by_region': df.groupby('region')['sales'].
sum().to_dict(),
        'sales_by_category': df.groupby('category')
['sales'].sum().to_dict()
    }
    return summary

def main():
    # Create sample data
    print("Generating sample data...")
    df = create_sample_data()

    # Create visualizations
    print("Creating visualizations...")
    create_visualizations(df)

    # Generate and print summary statistics
    print("\nSummary Statistics:")
    summary = generate_summary_statistics(df)
    for key, value in summary.items():
        print(f"{key.replace('_', ' ').title()}: {value}")

    print("\nVisualization files have been saved in the
current directory.")

```



```
if __name__ == "__main__":
    main()
```

This code demonstrates various data visualization techniques. Let me explain the key components:

1. Required Libraries:
 - numpy: For numerical operations
 - pandas: For data manipulation
 - matplotlib: For basic plotting
 - seaborn: For enhanced visualizations
2. Main Functions:
 - create_sample_data(): Generates sample sales data
 - create_visualizations(): Creates different types of plots
 - generate_summary_statistics(): Calculates key metrics
3. Types of Visualizations:
 - Line Plot: Shows trends over time
 - Bar Plot: Compares categories
 - Box Plot: Shows distribution and outliers
 - Histogram: Shows distribution of values
 - Heatmap: Shows correlations
 - Scatter Plot: Shows relationships between variables
 - Pie Chart: Shows proportions
4. Visualization Best Practices:
 - Consistent styling using seaborn
 - Clear titles and labels
 - Appropriate figure sizes
 - Proper rotation of labels
 - Saving plots to files
5. Key Features:
 - Automatic data generation
 - Multiple visualization types
 - Summary statistics
 - File saving functionality

The code creates these visualizations:

1. Daily Sales Trend: Line plot showing sales over time
2. Category Analysis: Bar plot of average sales by category
3. Regional Analysis: Box plot showing sales distribution by region
4. Customer Satisfaction: Histogram of satisfaction scores
5. Correlation Analysis: Heatmap showing relationships between variables
6. Sales vs. Satisfaction: Scatter plot showing relationship
7. Category Distribution: Pie chart showing sales proportions

Would you like me to explain any specific visualization in more detail or add additional features like:

- Interactive visualizations
- Advanced statistical analysis
- Different chart types
- Custom color schemes
- Animation capabilities?

Now launch the code in Listing 6.5 and you will see the following output on the command line:

```
Summary Statistics:
Total Sales: 365726.0873614802
Average Daily Sales: 1001.9892804424115
Highest Sales Day: 2023-07-29 00:00:00
Lowest Sales Day: 2023-09-20 00:00:00
Average Satisfaction: 4.042200553060934
Total Days: 365
Sales By Region: {'East': 86460.83984929087, 'North':
93291.50574379494, 'South': 91825.10458801102, 'West':
94148.6371803834}
Sales By Category: {'Books': 77055.44825022355, 'Clothing':
96802.84800150104, 'Electronics': 103622.83120503808,
'Food': 88244.95990471757}
```

In addition, you will see seven new PNG files as shown below:

```
sales_trend.png
category_sales.png
region_distribution.png
satisfaction_distribution.png
correlation_matrix.png
sales_satisfaction_scatter.png
category_distribution_pie.png
```

The contents of `sales_trend.png` are displayed in Figure 6.1.

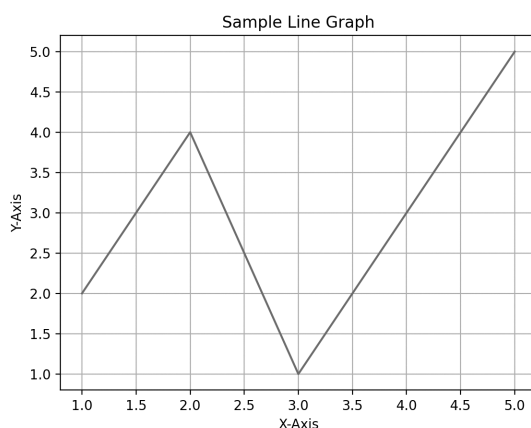


FIGURE 6.1 A Sales-Related Graph Rendered Via Matplotlib.

BASIC PANDAS

The following prompt was given to Claude 3 regarding a Python code sample that performs numerical calculations (such as the mean) on a CSV file.

Please generate a Python code sample that reads the contents of a CSV file, and then computes the mean of a column in a subset of the data.

Claude 3:

I'll create a Python script that demonstrates how to read a CSV file, filter the data based on conditions, and calculate statistics using Pandas.

Listing 6.6 displays the contents of `basic_pandas.py` that Claude 3 generated to read the contents of a CSV file, and then compute the mean of a target column.

LISTING 6.6: `basic_pandas.py`

```
import pandas as pd
import numpy as np
from typing import Optional, List
import logging

class DataAnalyzer:
    def __init__(self, file_path: str):
        """
        Initialize DataAnalyzer with the path to CSV file.

        Args:
            file_path (str): Path to the CSV file
        """
        self.file_path = file_path
        self.data = None

        # Set up logging
        logging.basicConfig(
            level=logging.INFO,
            format='%(asctime)s - %(levelname)s -
%(message)s'
        )
        self.logger = logging.getLogger(__name__)

    def read_csv(self) -> bool:
        """
        Read the CSV file into a pandas DataFrame.

        Returns:
            bool: True if successful, False otherwise
        """
```

```

        """
        try:
            self.logger.info(f"Reading CSV file: {self.file_
path}")

            self.data = pd.read_csv(self.file_path)
            self.logger.info(f"Successfully read CSV with
{len(self.data)} rows")
            return True
        except FileNotFoundError:
            self.logger.error(f"File not found: {self.file_
path}")
            return False
        except pd.errors.EmptyDataError:
            self.logger.error("The CSV file is empty")
            return False
        except Exception as e:
            self.logger.error(f"Error reading CSV file:
{str(e)}")
            return False

    def validate_column(self, column: str) -> bool:
        """
        Validate if a column exists in the DataFrame.

        Args:
            column (str): Column name to validate

        Returns:
            bool: True if column exists, False otherwise
        """
        if self.data is None:
            self.logger.error("No data loaded. Please read
CSV file first")
            return False

        if column not in self.data.columns:
            self.logger.error(f"Column '{column}' not found
in data")
            return False

        return True

    def compute_mean(self,
                    target_column: str,
                    filter_column: Optional[str] = None,
                    filter_value: Optional[any] = None) ->
Optional[float]:
        """
        Compute mean of target_column, optionally filtered
        by filter_column == filter_value.

```

```

    Args:
        target_column (str): Column to compute mean for
        filter_column (str, optional): Column to filter
on
        filter_value (any, optional): Value to filter by

    Returns:
        float or None: Computed mean or None if error
occurs
    """
    try:
        # Validate target column
        if not self.validate_column(target_column):
            return None

        # Create filtered dataset if filter is provided
        if filter_column and filter_value:
            if not self.validate_column(filter_column):
                return None

            filtered_data = self.data[self.data[filter_
column] == filter_value]
            self.logger.info(f"Filtered data contains
{len(filtered_data)} rows")

            if filtered_data.empty:
                self.logger.warning(f"No data found for
{filter_column} = {filter_value}")
                return None
            else:
                filtered_data = self.data

        # Compute mean
        mean_value = filtered_data[target_column].mean()
        self.logger.info(f"Computed mean: {mean_
value:.2f}")

        return mean_value

    except TypeError:
        self.logger.error(f"Column '{target_column}'
contains non-numeric data")
        return None
    except Exception as e:
        self.logger.error(f"Error computing mean:
{str(e)}")
        return None

    def get_summary_statistics(self,
                               target_column: str,

```

```

        filter_column: Optional[str] =
None,
        filter_value: Optional[any] =
None) -> Optional[dict]:
    """
    Get summary statistics for target_column with
    optional filtering.

    Args:
        target_column (str): Column to compute
statistics for
        filter_column (str, optional): Column to filter
on
        filter_value (any, optional): Value to filter by

    Returns:
        dict or None: Dictionary containing summary
statistics or None if error occurs
    """
    try:
        if not self.validate_column(target_column):
            return None

        # Create filtered dataset if filter is provided
        if filter_column and filter_value:
            if not self.validate_column(filter_column):
                return None

            filtered_data = self.data[self.data[filter_
column] == filter_value]
        else:
            filtered_data = self.data

        stats = {
            'mean': filtered_data[target_column].
mean(),
            'median': filtered_data[target_column].
median(),
            'std': filtered_data[target_column].std(),
            'min': filtered_data[target_column].min(),
            'max': filtered_data[target_column].max(),
            'count': filtered_data[target_column].
count()
        }

        return stats

    except Exception as e:
        self.logger.error(f"Error computing summary
statistics: {str(e)}")
        return None

```

```

def main():
    # Example usage
    analyzer = DataAnalyzer('sample_data.csv')

    # Read the CSV file
    if not analyzer.read_csv():
        return

    # Example 1: Compute mean of a column
    mean_value = analyzer.compute_mean('salary')
    if mean_value is not None:
        print(f"\nOverall mean salary: ${mean_value:,.2f}")

    # Example 2: Compute mean with filtering
    dept_mean = analyzer.compute_mean('salary',
    'department', 'IT')
    if dept_mean is not None:
        print(f"Mean salary in IT department: ${dept_
mean:,.2f}")

    # Example 3: Get summary statistics
    stats = analyzer.get_summary_statistics('salary',
    'department', 'IT')
    if stats:
        print("\nSummary Statistics for IT Department
Salaries:")
        for stat, value in stats.items():
            print(f"{stat.title()}: ${value:,.2f}")

if __name__ == "__main__":
    # Example of how the CSV file should look:
    sample_data = """
employee_id,name,department,salary,years_experience
1,John Doe,IT,75000,5
2,Jane Smith,HR,65000,3
3,Bob Johnson,IT,85000,7
4,Alice Brown,Finance,70000,4
"""

    # Create sample CSV file
    with open('sample_data.csv', 'w') as f:
        f.write(sample_data.strip())

    main()

```

This code demonstrates several key concepts:

1. Class Structure:

- `DataAnalyzer` class encapsulates all functionality
- Methods for reading data and computing statistics
- Error handling and logging throughout

2. Key Features:
 - CSV file reading with error handling
 - Column validation
 - Flexible filtering options
 - Summary statistics calculation
 - Comprehensive logging
3. Main Functions:
 - ``read_csv()``: Reads and validates the CSV file
 - ``validate_column()``: Checks if columns exist
 - ``compute_mean()``: Calculates mean with optional filtering
 - ``get_summary_statistics()``: Provides comprehensive statistics
4. Error Handling:
 - File not found errors
 - Empty file errors
 - Invalid column errors
 - Non-numeric data errors
 - General exception handling
5. Type Hints:
 - Used throughout for better code documentation
 - Helps with IDE support and code maintenance

Usage examples:

```
```python
Create analyzer instance
analyzer = DataAnalyzer('data.csv')

Read CSV file
analyzer.read_csv()

Calculate overall mean of a column
mean_salary = analyzer.compute_mean('salary')

Calculate mean with filtering
it_salary = analyzer.compute_mean('salary', 'department',
'IT')

Get comprehensive statistics
stats = analyzer.get_summary_statistics('salary',
'department', 'IT')
```

Would you like me to explain any specific part in more detail or add additional features like:

- Data validation and cleaning
- More statistical calculations
- Data visualization
- Export functionality
- Different file format support?



Now launch the code in Listing 6.6 and you will see the following output:

```
2024-11-18 13:35:08,816 - INFO - Reading CSV file: sample_
data.csv
2024-11-18 13:35:08,821 - INFO - Successfully read CSV with
4 rows
2024-11-18 13:35:08,822 - INFO - Computed mean: 73750.00

Overall mean salary: $73,750.00
2024-11-18 13:35:08,823 - INFO - Filtered data contains 2
rows
2024-11-18 13:35:08,823 - INFO - Computed mean: 80000.00
Mean salary in IT department: $80,000.00

Summary Statistics for IT Department Salaries:
Mean: $80,000.00
Median: $80,000.00
Std: $7,071.07
Min: $75,000.00
Max: $85,000.00
Count: $2.00
```

## GENERATE RANDOM DATA

---

The following prompt was given to Claude 3 regarding a Python code sample that generates random data.

Please generate a Python code sample that generates random data, along with an explanation of the code.

Claude 3:

I'll create a Python script that demonstrates various ways to generate random data using different distributions and patterns.

Listing 6.7 displays the contents of `random_data_stats.py` that Claude 3 generated to generate a set of random numbers and then display the contents of those numbers, as well as the mean and standard deviation of those numbers.

### **LISTING 6.7: `random_data_stats.py`**

```
import numpy as np
import pandas as pd
from datetime import datetime, timedelta
from typing import List, Dict, Union, Optional
import random
import string
import logging
```

```

class RandomDataGenerator:
 def __init__(self, seed: Optional[int] = None):
 """
 Initialize the random data generator with optional
 seed.

 Args:
 seed (int, optional): Random seed for
 reproducibility
 """
 # Set up logging
 logging.basicConfig(
 level=logging.INFO,
 format='%(asctime)s - %(levelname)s -
%(message)s'
)
 self.logger = logging.getLogger(__name__)

 # Set random seed if provided
 if seed is not None:
 np.random.seed(seed)
 random.seed(seed)
 self.logger.info(f"Random seed set to: {seed}")

 def generate_numeric_data(self,
 size: int,
 distribution: str = 'normal',
 params: Dict[str, float] = None)
 -> np.ndarray:
 """
 Generate random numeric data following specified
 distribution.

 Args:
 size (int): Number of samples to generate
 distribution (str): Type of distribution
 ('normal', 'uniform', 'poisson', etc.)
 params (dict): Distribution parameters

 Returns:
 np.ndarray: Generated random data
 """
 if params is None:
 params = {}

 try:
 if distribution == 'normal':
 mean = params.get('mean', 0.0)
 std = params.get('std', 1.0)
 data = np.random.normal(mean, std, size)
 elif distribution == 'uniform':

```

```

 low = params.get('low', 0.0)
 high = params.get('high', 1.0)
 data = np.random.uniform(low, high, size)
 elif distribution == 'poisson':
 lam = params.get('lambda', 1.0)
 data = np.random.poisson(lam, size)
 else:
 raise ValueError(f"Unsupported
distribution: {distribution}")

 self.logger.info(f"Generated {size}
{distribution} distributed numbers")
 return data

except Exception as e:
 self.logger.error(f"Error generating numeric
data: {str(e)}")
 return np.array([])

def generate_categorical_data(self,
 size: int,
 categories: List[str],
 probabilities:
Optional[List[float]] = None) -> List[str]:
 """
 Generate random categorical data.

 Args:
 size (int): Number of samples to generate
 categories (list): List of possible
categories
 probabilities (list, optional): Probability
weights for categories

 Returns:
 list: Generated categorical data
 """
 try:
 data = np.random.choice(categories, size=size,
p=probabilities)
 self.logger.info(f"Generated {size} categorical
values")

 return data.tolist()
 except Exception as e:
 self.logger.error(f"Error generating
categorical data: {str(e)}")
 return []

def generate_datetime_data(self,
 size: int,

```

```

 start_date: str = '2023-01-
01',
 end_date: str = '2023-12-31')
-> List[datetime]:
 """
 Generate random datetime data within a specified
 range.

 Args:
 size (int): Number of dates to generate
 start_date (str): Start date in 'YYYY-MM-DD'
format
 end_date (str): End date in 'YYYY-MM-DD' format

 Returns:
 list: Generated datetime data
 """
 try:
 start = datetime.strptime(start_date, '%Y-%m-
%d')
 end = datetime.strptime(end_date, '%Y-%m-%d')

 # Calculate date range in days
 date_range = (end - start).days

 # Generate random days offset
 random_days = np.random.randint(0, date_range,
size=size)

 # Create datetime objects
 dates = [start + timedelta(days=int(days)) for
days in random_days]
 dates.sort() # Sort dates chronologically

 self.logger.info(f"Generated {size} datetime
values")
 return dates

 except Exception as e:
 self.logger.error(f"Error generating datetime
data: {str(e)}")
 return []

 def generate_text_data(self,
 size: int,
 min_words: int = 5,
 max_words: int = 10,
 word_length: int = 5) ->
List[str]:
 """
 Generate random text data.

```

```

 Args:
 size (int): Number of texts to generate
 min_words (int): Minimum words per text
 max_words (int): Maximum words per text
 word_length (int): Average word length

 Returns:
 list: Generated text data
 """
 try:
 texts = []
 for _ in range(size):
 # Generate random number of words
 num_words = random.randint(min_words, max_
words)

 # Generate random words
 words = [''.join(random.choices(string.
ascii_lowercase, k=word_length))
 for _ in range(num_words)]

 texts.append(' '.join(words))

 self.logger.info(f"Generated {size} text
samples")
 return texts

 except Exception as e:
 self.logger.error(f"Error generating text data:
{str(e)}")
 return []

 def generate_dataset(self,
 size: int,
 schema: Dict[str, Dict]) ->
pd.DataFrame:
 """
 Generate a complete dataset based on provided
 schema.

 Args:
 size (int): Number of rows to generate
 schema (dict): Data schema specification

 Returns:
 pd.DataFrame: Generated dataset
 """
 try:
 data = {}

 for column, specs in schema.items():

```

```

 data_type = specs.get('type', 'numeric')

 if data_type == 'numeric':
 data[column] = self.generate_numeric_
data(
 size,
 distribution=specs.
get('distribution', 'normal'),
 params=specs.get('params', {})
)
 elif data_type == 'categorical':
 data[column] = self.generate_
categorical_data(
 size,
 categories=specs.get('categories',
[[]],
 probabilities=specs.
get('probabilities', None)
)
 elif data_type == 'datetime':
 data[column] = self.generate_datetime_
data(
 size,
 start_date=specs.get('start_date',
'2023-01-01'),
 end_date=specs.get('end_date',
'2023-12-31')
)
 elif data_type == 'text':
 data[column] = self.generate_text_data(
 size,
 min_words=specs.get('min_words',
5),
 max_words=specs.get('max_words',
10),
 word_length=specs.get('word_
length', 5)
)

 df = pd.DataFrame(data)
 self.logger.info(f"Generated dataset with
{size} rows and {len(schema)} columns")
 return df

 except Exception as e:
 self.logger.error(f"Error generating dataset:
{str(e)}")
 return pd.DataFrame()

def main():
 # Initialize generator with seed for reproducibility

```

```

generator = RandomDataGenerator(seed=42)

Example schema for a sample dataset
schema = {
 'id': {
 'type': 'numeric',
 'distribution': 'uniform',
 'params': {'low': 1000, 'high': 9999}
 },
 'date': {
 'type': 'datetime',
 'start_date': '2023-01-01',
 'end_date': '2023-12-31'
 },
 'category': {
 'type': 'categorical',
 'categories': ['A', 'B', 'C'],
 'probabilities': [0.4, 0.3, 0.3]
 },
 'value': {
 'type': 'numeric',
 'distribution': 'normal',
 'params': {'mean': 100, 'std': 15}
 },
 'description': {
 'type': 'text',
 'min_words': 3,
 'max_words': 7
 }
}

Generate dataset
df = generator.generate_dataset(size=10, schema=schema)

Display results
print("\nGenerated Dataset:")
print(df)

Save to CSV
df.to_csv('random_data.csv', index=False)
print("\nData saved to 'random_data.csv'")

if __name__ == "__main__":
 main()

```

This code demonstrates several key features:

1. Supported Data Types:
  - Numeric (various distributions)
  - Categorical

- Datetime
- Text

## 2. Key Features:

- Configurable random seed
- Multiple distribution types
- Comprehensive error handling
- Logging functionality
- Flexible schema definition

## 3. Main Methods:

- ``generate_numeric_data()``: Creates numeric data with different distributions
- ``generate_categorical_data()``: Creates categorical data with optional probabilities
- ``generate_datetime_data()``: Creates datetime data within a range
- ``generate_text_data()``: Creates random text data
- ``generate_dataset()``: Creates complete datasets based on schema

## 4. Supported Distributions:

- Normal (Gaussian)
- Uniform
- Poisson

## 5. Schema-based Generation:

- Flexible schema definition
- Multiple column types
- Customizable parameters
- DataFrame output

## Usage examples:

```
generator = RandomDataGenerator(seed=42)
```

```
Generate numeric data
```

```
numbers = generator.generate_numeric_data(
 size=1000,
 distribution='normal',
 params={'mean': 0, 'std': 1}
)
```

```
Generate categorical data
```

```
categories = generator.generate_categorical_data(
 size=1000,
 categories=['A', 'B', 'C'],
 probabilities=[0.4, 0.3, 0.3]
)
```



```
Generate complete dataset
df = generator.generate_dataset(size=1000, schema=schema)
```

Would you like me to explain any specific part in more detail or add additional features like:

- More distribution types
- Data validation
- Correlation between columns
- Export to different formats
- Data visualization capabilities?

Now launch the code in Listing 6.7 and you will see the following type of output (your output will be slightly different), the second part of which is stored in the newly generated file `random_data.csv`.

```
2024-11-18 13:43:08,889 - INFO - Random seed set to: 42
2024-11-18 13:43:08,889 - INFO - Generated 10 uniform
distributed numbers
2024-11-18 13:43:08,889 - INFO - Generated 10 datetime
values
2024-11-18 13:43:08,891 - INFO - Generated 10 categorical
values
2024-11-18 13:43:08,891 - INFO - Generated 10 normal
distributed numbers
2024-11-18 13:43:08,891 - INFO - Generated 10 text samples
2024-11-18 13:43:08,892 - INFO - Generated dataset with 10
rows and 5 columns
```

Generated Dataset:

	id	date	value	description
		category		
0	4370.486530 2023-01-22	A	99.633172	ahftr xckaf nafqo
1	9555.478043 2023-05-30	A	105.333270	lhwte khftc jjigb ldxch
2	7587.213482 2023-06-10	B	106.255167	xjebr uzwwj lvejrb srqbn gmhyr fitbv ukbxo
3	6387.327699 2023-07-11	A	112.486928	fmxqd dtotl pjzdm twder pkpmg
4	2404.011745 2023-09-15	A	95.599013	wasrn gqcll ywgne xwhqp dtoun alayw vhbwy
5	2403.794689 2023-10-04	C	99.552421	tvznd ryenp zytrs
6	1522.694426 2023-10-21	A	101.426887	zqlnd fipff bqfxw bgrfd yomuu ecllm
7	8794.719136 2023-11-05	A	109.969815	rzcki wgelk hgylw
8	6409.433991 2023-11-10	B	97.896723	cqoac tgkmw xeaky uhssu rmefb tbibm xnbw
9	7371.945128 2023-12-10	A	99.502099	prgdx gpqkp nyfsg

Data saved to 'random\_data.csv'

**RECURSION: FIBONACCI NUMBERS**

---

The following prompt was given to Claude 3 regarding a Python code sample that generates Fibonacci numbers.

Please generate a Python code sample that generates Fibonacci numbers, along with an explanation of the code.

Claude 3:

Let me explain how to generate Fibonacci numbers with a clear code example.

Here's a Python implementation that generates Fibonacci numbers:

Listing 6.8 displays the contents of `fibonacci.py` that shows you how to compute the *n*-th Fibonacci number using recursion.

**LISTING 6.8: fibonacci.py**

```
def fibonacci(n):
 """
 Generate the first n Fibonacci numbers.
 Returns a list of integers.
 """
 fib_sequence = [0, 1] # Initialize with first two
 numbers

 for i in range(2, n):
 # Each number is the sum of the two preceding
 numbers
 next_number = fib_sequence[i-1] + fib_sequence[i-2]
 fib_sequence.append(next_number)

 return fib_sequence

Example usage
first_10 = fibonacci(10)
print(first_10) # Output: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

Let's break down how this works:

1. The function takes a parameter `n` which specifies how many Fibonacci numbers to generate
2. We start with the first two numbers `[0, 1]` in the sequence
3. For each subsequent number:
  - We take the last two numbers in the sequence
  - Add them together to get the next number
  - Append this new number to our sequence
4. Finally, we return the complete sequence

The Fibonacci sequence has many interesting properties and appears frequently in nature and mathematics. Would you like me to show you any variations of this implementation, such as using recursion or generating numbers up to a specific value?

Now launch the code in Listing 6.8 and you will see the following output:

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

## SUMMARY

---

This chapter contains Python-based solutions generated by Claude 3 to a variety of tasks. The first set of tasks involved tasks such as creating a simple calculator, Web scraping, and file-based operations.

Then you learned about data visualization using a popular open source Python-based library called `Matplotlib`. In addition, you learned how to calculate Fibonacci numbers using an iterative solution.

Congratulations on reaching the final chapter of this journey into Python and Claude 3! Completing this book is a testament to your dedication, curiosity, and drive to expand your technical skills. You've delved into powerful concepts, tackled hands-on examples, and equipped yourself with knowledge to create and innovate in meaningful ways. As you move forward, remember that the real power of programming lies in continuous learning and application. Celebrate this achievement—you've earned it! Here's to the many projects, discoveries, and breakthroughs that lie ahead in your Python and Claude 3 adventures.

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